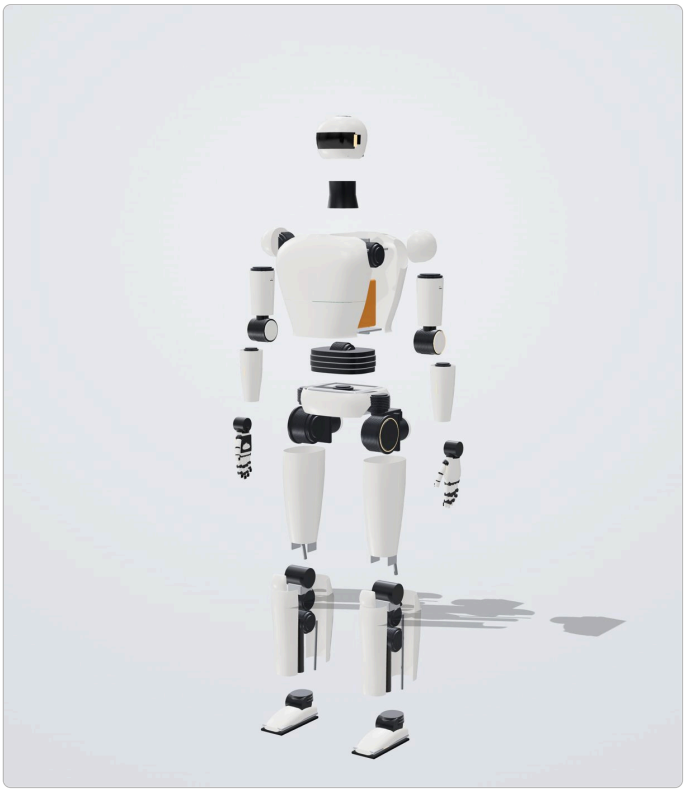
▼ PHASES AND APPENDICES

P00	Workshop, safety and documentation setup
P01	Test infrastructure: gantry, dynamometer, coolant cart, e-stop chain, bench power
P02	Electronics bring-up: RT computer, EtherCAT, drive carrier, 150 A GaN drive, safety MCU
P03	KA-90P and KA-110P planetary actuators (first articles, then series)
P04	KH arm and neck actuators: KH-C25 and KH-C20 cycloids, KH-14 strain-wave
P05	KM-130x50-LC motors and KA-130K / KA-130H actuators
P06	Two hot-swap 16S4P battery packs with in-pack BMS and FET switch
P07	Legs: hip, thigh with knee four-bar, shank with ankle push-rods, foot, leg coolant lines
P08	Pelvis, waist, torso frame, battery bay, power distribution and compute
P09	Arms: shoulders, upper arms, elbows, forearms, wrists
P10	Hands Mk II (13 actuated DOF each, palm camera)
P11	Neck, head, cameras, LiDAR, calibration
P12	Harness, shells, final leak check and final assembly
P13	Software bring-up and commissioning hand-off
App. A	Testing and commissioning gates
App. B	Manufacturing package
App. C	Optional: scratch-built motors and actuators

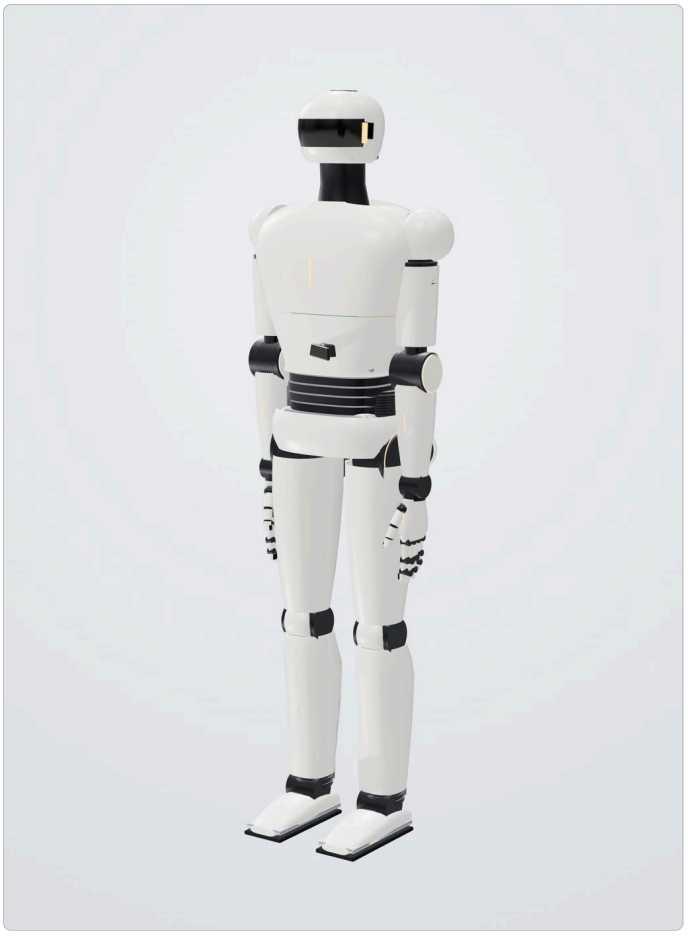
FULL-SIZE HUMANOID · MK II · 1.75 M · 102.5 KG ESTIMATED (112.7 KG DESIGN MASS) · 16S LI-ION, TWO HOT-SWAP PACKS

KESTREL K1 build tutorial

Every phase from an empty workshop to a commissioned robot. In each phase: visual reference, parts with BOM IDs, tools, manufacturing, assembly with torques, electrical work, and the bench test that has to pass before you move on. 37,932 words, 45 figures and drawing sheets.



RENDER Exploded assembly



RENDER Fully assembled

This is the step-by-step build guide for the **Mk II** revision. **All phases plus the commissioning, manufacturing and scratch-build appendices in one illustrated file: KESTREL-K1-build-tutorial.pdf** (regenerate it with `make_handbook.py`). Work through the phases **in order**. Each phase ends in a test gate (docs/11). Do not start a phase until the previous gate has passed and been logged.

Before you order metal, clear the design holds in docs/09 §11. Several detailed parts inherited from K1 (thigh plate hip flange, knee hub, KA-130 bearing seat, hollow bores) cannot carry the Mk II loads as modelled.

Mk II at a glance (numbers from `docs/generated/`)

Item	Mk II value	Replaces (K1)
Design mass / estimate	**112.7 kg** design (102.5 kg estimate × 1.10)	92.6 kg
Work loads	**30 kg sustained two-handed, 50 kg momentary lift**, 10 kg per hand at reach, 15 kg one-handed	20 kg squat, 5 kg per hand
Knee / hip pitch	KA-130K 17:1 (493 / 295 N·m) and KA-130H 14:1 (406 / 243 N·m), **liquid-cooled KM-130x50-LC**, **150 A GaN drives**	KM-130x45/x40, 99 A drive, ducted air
Hip roll, waist pitch	KA-110P: **RI115-PH**, 29:1, 427 / 125 N·m, **output flange 12× M5 12.9 on PCD 100**	ILM85x26, 32:1, 10× M4 on PCD 85
Hip yaw, waist yaw, ankles	KA-90P: RI85-PH, 28:1, 129 / 44 N·m	23:1
Shoulders, elbows	**KH-C25 / KH-C20: machined two-disc cycloids 30:1** (362 / 158 N·m motor peak, firmware-limited to 160 / 105 N·m)	strain-wave CSG-25/20-50
Wrist yaw, neck	KH-14: KM-50x20 + CSG-14-50 (limited to 23 N·m)	unchanged
Wrist pitch/roll	**KW-L2**: maxon ECX FLAT 32 L brushless + **1 mm-lead** ball screw on a **24 V** forearm rail	DCX 16 L on 12 V
Hands	**13 DOA**, linkage + tendon, palm camera, ≥ 200 N power grasp, 1.15 kg	10 DOA

Item	Mk II value	Replaces (K1)
Battery	**two hot-swap 16S4P packs** (2 × 1.15 kWh, 2 × 5.9 kg), in-pack BMS, robot-side ideal-diode / tie power path with a fail-locked removal interlock	one 16S6P pack, powered-down swap
Cooling	**sealed liquid loop** (2 pumps, radiator + 2 fans, 40 mL spring compensator, dry-break QDs at the hips) for the knee/hip jackets and the Thor cold plate, plus 2 electronics blowers	blowers and ducts only

How each phase is laid out

Section	Contents
A. Visual reference	renders, cutaways, diagrams, CAD files and drawing sheets for this phase
B. Parts list	every part with its BOM ID (`data/bom.csv`, generated/bom.md; Mk II additions start at B101), quantity, specification
C. Tools & equipment	what you need on the bench
D. Manufacturing	how to make the custom parts: material, process, tolerances, finishing
E. Assembly	ordered steps with orientation, fasteners, torques, adhesives, alignment
F. Electrical	wiring, connectors, pinouts, voltages, fuses, precautions
G. Testing	the bench test that proves the assembly before it goes into the robot, with pass criteria

Mk II additions that are large enough to need their own A–G sequence are **sub-phases** inside the phase where they are built (for example P05.1).

Phase map

Phase	What you build	Gate
P00	Workshop layout, lithium and 67 V DC safety, coolant handling, documentation system	—
P01	Gantry + fall-arrest harness, dynamometer, coolant test cart, e-stop chain, bench power	—
P02	RT computer + EtherCAT master, drive carrier, 150 A GaN drive, safety MCU	G0
P03	KA-90P (first article), then KA-110P with the 12× M5 output flange	G1
P04	KH-C25 / KH-C20 cycloid actuators and KH-14; **P04.1 machined cycloid gears**	G1
P05	KM-130x50-LC motors, KA-130K / KA-130H actuators; **P05.1 liquid-cooling jackets**	G1
P06	Two hot-swap 16S4P packs; **P06.1 hot-swap battery bay**	G0 (packs, bay)
P07	Hip, thigh with knee four-bar and torque limit, shank with ankle push-rods, foot, leg coolant lines	G2
P08	Pelvis, waist, torso frame, battery bay, power distribution, compute; **P08.1 liquid-cooling loop**	G2/G3
P09	Shoulders, upper arms, elbows, forearms; **P09.1 KW-L2 wrist actuators**	G2
P10	Mk II hands: 13 actuators, linkage + tendon fingers, tactile, palm camera	G2
P11	Neck, head frame, cameras, LiDAR, calibration	G3
P12	Harness, shells, final leak check, final assembly and inspection	G3
P13	Software bring-up and progressive commissioning (hand-off to docs/11 G3–G10)	G3–G10

Why this order

1. **Rigs and safety before anything moves.** The gantry, e-stop chain, dyno and coolant cart make every later test safe and measurable.
2. **One of each actuator class first.** The first article proves the design (Kt, efficiency, thermal behaviour, backlash) before you buy more gear sets. Results feed back into `calc/kestrel/actuators.py`. For Mk II, the R10 §7 validation tests (FEMM, jacket thermal resistance, 150 A locked rotor, gear run-in, k_L, cycloid overload, KW-L2 force–speed) are part of gate G1 (docs/11 §4.1).
3. **Simplest risk first.** KA-90P (commercial motor, simplest planetary) comes before the cycloids, and both come before the custom liquid-cooled KM-130.
4. **Limbs before the body.** Each limb is tested clamped on a stand, with the coolant cart standing in for the robot's loop, before it carries the robot's weight.

Non-negotiable safety rules (repeat them to yourself every session)

- **Keyed main switch off and both packs out** before you touch any connector, shell, harness or coolant fitting. The e-stop is not an isolation: the logic bus stays live. A hot swap during the programme is done only with the robot supported and the drives in STO (P06.1).
- The bus is **up to 67.2 V DC** (above the 60 V touch-safe limit) and the packs can deliver > 1 kA into a short. Insulated tools, one hand where you can, and measure < 5 V before touching a conductor.
- No powered motion of a limb or the robot without the **e-stop in hand** and a second person present from G3 onward.
- **No unrestricted high-torque operation near people.** Reduced torque limits until the gate allows more; nobody in the cage while drives are enabled.
- No walking or payload tests without the **gantry harness** attached until gate G6 says otherwise.
- **Coolant and electronics:** fill, drain and open the loop only with both packs out; wipe spills at once; never route a fitting above a PCB.
- Charge batteries only when supervised, on a non-flammable surface, in the battery area.
- Magnetised rotors (N50UH) are handled with jigs, never fingers.
- High-torque joints are pinch hazards. Keep hands out of joint envelopes whenever drives are enabled, even in "zero torque".

P00 Workshop, safety and documentation setup

A Visual reference

- Safety architecture: `diagrams/safety_chain.svg`
- Test progression: `diagrams/test_gates.svg`

Suggested floor plan (minimum ≈ 6 × 5 m):

```
+-----------------------------------------------------------+
|  BATTERY AREA (concrete floor, metal cabinet,  |  BENCH 1: electronics / soldering   |
|  sand bin, extinguisher, smoke detector,       |  (ESD mat, fume extraction)         |
|  2-bay charging shelf, pack trolley)           |                                     |
|------------------------------------------------+-------------------------------------|
|  BENCH 2: mechanical assembly, surface plate   |  DYNO (bolted to bench, guard box)  |
|  COOLANT STATION (drip tray, spill kit, fill   |  + coolant test cart beside it      |
|  pump, labelled coolant containers)            |                                     |
|------------------------------------------------+-------------------------------------|
|  TEST CAGE 4 × 3 m under the 2 t gantry (mesh panels 1.2 m, foam floor tiles)        |
|  operator position outside the cage, e-stop on a lanyard                              |
+-----------------------------------------------------------+
```

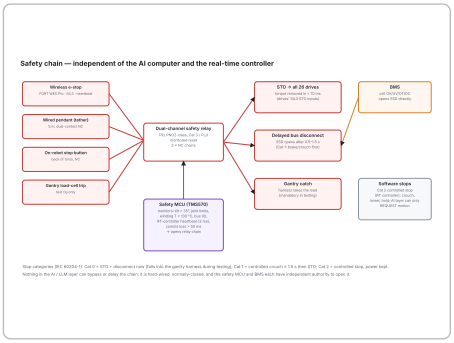


DIAGRAM Safety architecture



DIAGRAM Test progression

B Parts list

BOM	Item	Qty
—	Metal flammables cabinet for cells/packs, smoke detector, Class D or large AVD/water-mist extinguisher, fire blanket, sand bin, steel pack-quarantine box with a lid	1 set
—	Insulated hand tools rated 1000 V (IEC 60900): screwdrivers, hex keys, spanners, side cutters; insulating mat; Class 0 insulating gloves for pack work	1 set
—	CAT III 1000 V multimeter, clamp meter (DC), non-contact thermometer	1 each
—	ESD mat + wrist straps, fume extractor	1 set
—	Coolant station: drip tray, absorbent pads and spill kit, nitrile gloves, eye protection, labelled HDPE containers for coolant (propylene glycol mix) and waste	1 set
—	Labels, label printer, parts bins	—
B083	Gantry + harness (built in P01)	1

C Tools & equipment

See docs/12 §4 (instruments, hand tools, metrology). Mk II adds the insulated tool set, the coolant station and a pack trolley (two packs at 5.9 kg each are carried by their handles, never by the connector).

D Manufacturing

None.

E Setup steps

1. Set up the **battery area** at least 2 m from exits and from flammables. Put a smoke detector over it and an extinguisher by the door. Give each pack its own shelf position with a label: serial, state of charge, last check.
2. Fit **ESD protection** at the electronics bench: drives, encoders and the Thor carrier are ESD-sensitive.
3. Set up the **coolant station** on the mechanical bench, below and away from the electronics bench. Coolant never goes on the electronics bench.
4. Start a **build log**: a git repository with `logs/YYYY-MM-DD.md`, every measurement, photos, part serial numbers, torque values and test results. Calibrations are versioned here (docs/07 §5).
5. Print the **gate checklist** (docs/11 §1) and the **design-hold list** (docs/09 §11). Sign each gate and each hold as it is cleared.
6. Agree the **two-person rule** with a helper for every powered test from G3 onward and for every pack swap.

F Electrical safety rules

- **The 16S bus is 48–67.2 V DC.** It is extra-low voltage under AS/NZS 3000 (≤ 120 V ripple-free DC) but above the 60 V touch-safe limit, and each pack can deliver ≈ 1.2 kA into a short (67 V over ≈ 56 mΩ). Treat every bus conductor as live until measured below 5 V.
- **Two packs means two sources, and the e-stop is not an isolation.** The e-stop removes torque (STO) and, after 0.5–1.5 s, opens the actuator-zone eFuses, but the logic bus (computers, pumps) stays live by design (docs/05 §8). Before working on the robot: 1. turn the **keyed main switch** off (the power-path board sheds every path); 2. pull **both** packs from the bay; 3. wait 1 minute (bus capacitors discharge through the PDU bleed resistors); 4. verify < 5 V between BUS+ and BUS− and between BUS+ and chassis. There is no mechanical service disconnect in Mk II: pulling both packs is the isolation.
- **A removed pack has dead terminals by design** (its in-pack FET switch opens when the pilot loop breaks). Verify it every time with a meter before you rely on it. A pack that shows voltage at its terminals off-robot is faulty: quarantine it.
- **DC arcs do not self-extinguish.** Never unplug a power connector under load. The pack connectors have sequenced pilot contacts so the pack switches off before the power pins separate; do not defeat them.
- Use insulated tools for all battery work. Remove rings and watches.
- Never short-circuit a cell or pack "to test it".

Coolant safety rules

- The coolant is 30–40 % propylene glycol in deionised water with inhibitor: low toxicity, but **electrically conductive**. A leak onto a powered board can short 67 V. Fill, drain and open the loop only with both packs out.
- Keep the loop's fittings below and away from electronics; never let a hose run above a PCB.
- Wipe spills at once, then rinse the area with deionised water and dry it. If coolant reaches a board, it comes out for cleaning (IPA, ultrasonic if needed) and inspection before power is reapplied.
- Glycol is slippery on the cage floor: clean the floor before any standing or walking test.

Machine-shop hazards new in Mk II

- **N50UH rotors** are stronger than K1's magnets: rotor-to-stator pull is several hundred newtons. Jigs only.
- **Titanium chips** can ignite: flood coolant, no dry grinding, clean the machine afterwards.
- **Hardened 100Cr6 parts and EDM work** are outsourced; when lapping cycloid discs, wear eye protection and use a dust extractor.

G Check

Walk through a dry run with your helper of: (a) a battery fire response, (b) an e-stop response, (c) a coolant spill on the electronics bench, (d) removing both packs and proving the bus dead. Log it.

P01 Test infrastructure: gantry, dynamometer, coolant cart, e-stop chain, bench power

A Visual reference

- Safety chain: `diagrams/safety_chain.svg`

Dynamometer layout (top view):

```
[absorber ≥ 600 N·m]──[jaw coupling]──[FUTEK TRS605 1,000 N·m]──[jaw coupling]──[ACTUATOR UNDER TEST]
       bolted to 25 mm Al tooling plate, shafts aligned to 0.05 mm TIR, polycarbonate guard box over the whole line
       reaction: actuator housing bolted to an angle plate through its base-flange pattern (docs/09 §4)
       locked-output fixture: a steel reaction arm replaces the absorber for peak, locked-rotor and overload tests
```

Coolant test cart (stands in for the robot loop during P05 and P07 tests):

```
reservoir → pump → flow sensor → T_in → P gauge → [QD] → JACKET(S) UNDER TEST → [QD] → T_out → radiator + fan → reservoir
                                 (thermocouples in tees, 0–4 bar gauge, bypass valve, drain valve, drip tray under everything)
```

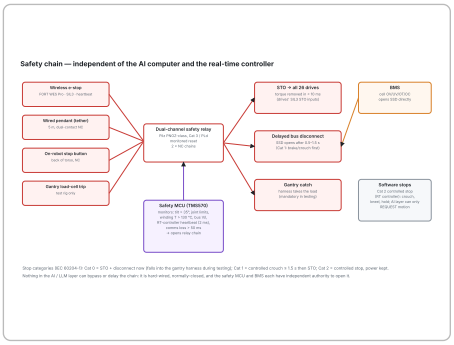


DIAGRAM Safety chain

B Parts list

BOM	Item	Qty	Spec
B083	Overhead gantry 2 t + trolley	1	≥ 2.4 m clear under the beam, beam ≥ 4 m
B083	Fall-arrest harness for the robot: chest + pelvis straps to the two lifting points, 6 mm Dyneema, shock-absorbing lanyard, chain hoist for height	1	every element rated ≥ 10 × the 112.7 kg design mass (≥ 1.13 t)
—	Lifting eyes DIN 580 **M12** (WLL 340 kg each) + key-locked steel thread inserts for 7075	2 + spare	replaces K1's M10 (WLL 230 kg); fall-arrest peaks reach ≈ 3 g
B084	FUTEK TRS605 inline torque transducer with encoder, **1,000 N·m capacity version**	1	via Metromatics (AU); covers the KA-130K 493 N·m peak with margin (the BOM line says 500 N·m: too little)
B084	Absorber for torque–speed maps: industrial servo + gearhead or a second KA-130K through a 2:1 reduction, **≥ 600 N·m** at the dyno shaft; hysteresis brake alternative ≥ 600 N·m	1	K1's ≥ 450 N·m is below the Mk II knee peak
B084	Locked-output reaction arm (steel, 1 m) + calibrated weights for the lever check	1 set	
B084	Jaw couplings rated ≥ 1,000 N·m peak, 25 mm tooling plate, angle plate, guard box	1 set	
B131	Coolant cart (loop bench): 12 V pump (same class as the robot's), reservoir, 240 mm aluminium radiator + fan, turbine flow sensor 0.1–2 L/min, 2 × K-type thermocouples in tees, 0–4 bar gauge, bypass and drain valves, 2 dry-break QD pairs (same type as the robot), EPDM hose, drip tray	1	aluminium wetted parts only
—	Hand pump with 0–6 bar gauge and shut-off for pressure tests; soap solution; bucket for submersion tests	1	
B043	FORT Wireless E-Stop Pro	1	
B044	Safety relay + wired pendant	1	
—	Bench supply ≥ 70 V / ≥ 10 A programmable, **current-limited**, + DC electronic load + 8 Ω 100 W brake resistor on a clamp board; for the 150 A tests a second supply path: one built 16S4P pack through its own fuse and disconnect (P06)	1 each	a bench supply cannot deliver the 150 A drive tests

BOM	Item	Qty	Spec
B085	Single-leg stand: vertical linear rail (HGR20 class) carrying a hip mount, base with 4 load cells, counterweight	1	sized for half of 112.7 kg plus 50 kg payload cases

C Tools

Drill press, tap set, hoist, laser level, dial indicators (shaft alignment), torque wrench, crimp tools for 8 AWG.

D Manufacturing

- **Dyno plate:** 25 mm 6061 tooling plate with an M8 grid at 50 mm pitch. Mount the angle plate with the KA-130 (PCD 125, 12× M5), **KA-110P (PCD 115, 12× M5)**, KA-90P (PCD 75, 8× M4) and KH base-flange patterns.
- **Robot lifting points:** two M12 lifting eyes, one on the pelvis bowl and one on the spine plate top rib, in key-locked steel inserts (thread engagement ≥ 1.5 d = 18 mm). Do not use wire thread inserts for lifting.
- **Leg stand:** carriage plate with the hip-yaw base pattern. The rail lets the hip move vertically only; a pneumatic counterweight (or bungee) offsets 70 % of the body weight.
- **Coolant cart:** mount everything on a steel tray with a 20 mm lip so a burst hose stays in the tray.

E Assembly

1. Install the gantry per its manual. **Load-test it at 1.25 × the 112.7 kg design mass (141 kg)** with sand bags for 10 minutes.
2. Assemble the dyno line. Align the shafts to ≤ 0.05 mm TIR and ≤ 0.1° angular with dial indicators on the coupling hubs.
3. Wire the **e-stop chain** on a test board: FORT receiver + pendant + panel button in series → safety relay (dual channel) → two outputs: (a) STO pair to the drive under test, (b) enable input of the bench disconnect.
4. Add a **brake-resistor clamp** (comparator at 70 V on / 68 V off driving a 100 V MOSFET) across the bench bus. It absorbs regeneration when the supply cannot sink current.
5. Build and **pressure-test the coolant cart** on its own (2 bar, 15 min, no leak) before connecting any actuator.

F Electrical

- Dyno bus: bench supply (or test pack) → 30 A fuse (bench supply) / 125 A pack fuse (pack) → disconnect (enable = safety relay) → bus with clamp → drive under test + absorber drive. 8 AWG for the pack path.
- Torque transducer → its amplifier → USB/analog → a logging PC (≥ 1 kHz). Log actuator current, speed, winding and coolant temperatures, flow and torque together.
- Coolant-cart pump and fan on a separate 12 V supply, so the coolant keeps flowing when the actuator is e-stopped.

G Testing

Test	Pass
E-stop: press each device while a test motor spins at 50 % speed	torque removed < 10 ms (scope the STO line and the phase current); latched until a deliberate reset
Wireless e-stop out of range / battery removed	stop within the configured heartbeat timeout; measure and log it (not published, R2)
Dyno torque calibration: 1 m lever arm + calibrated weights at 10, 100, 300 N·m	transducer within ±0.5 % FS
Locked-output arm: 500 N·m static on the arm	no visible deflection of the angle plate (< 0.1 mm at the arm root)
Gantry harness drop test with a **113 kg** sand dummy from 0.3 m slack	arrest without the dummy touching the floor; lanyard deploys as rated
Coolant cart: flow vs bypass setting; thermocouples in a common bath	0.2–1.0 L/min adjustable; thermocouples agree within 0.3 °C

P02 Electronics bring-up: RT computer, EtherCAT, drive carrier, 150 A GaN drive, safety MCU

Gate G0

A Visual reference

- Comms: `diagrams/comms_topology.svg`
- Power: `diagrams/power_architecture.svg`
- Software layers: `diagrams/software_stack.svg`

The two diagrams still show the K1 power path. Mk II (docs/05): two packs through the KES-PPM power path (P06.1), a logic bus and an actuator bus, and **28 EtherCAT slaves** (22 Everest + 4 HC2 + 2 hand controllers; each hand controller drives its wrist's KES-WRIST board, so the KW-L2 adds no slave).

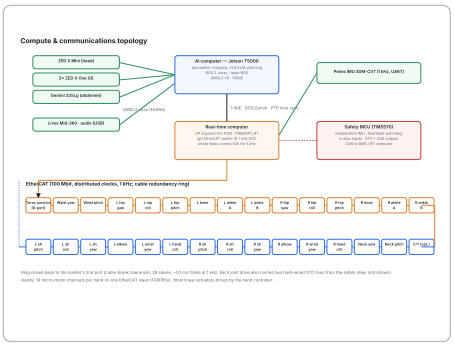


DIAGRAM Comms

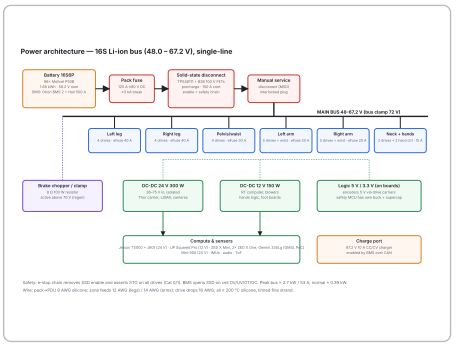


DIAGRAM Power

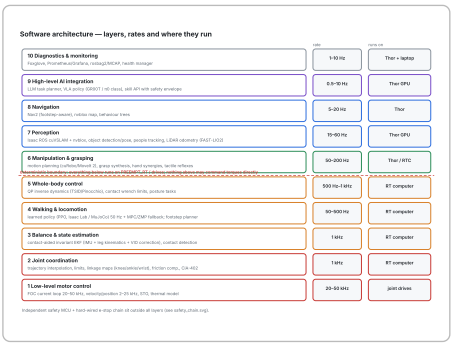


DIAGRAM Software layers

B Parts list

BOM	Item	Qty (now)
B041	UP Squared Pro 7000 (i3-N305, 2× i226-IT)	1
B026	Novanta Everest S NET drive (8–80 V, 45 A cont / 60 A 1 s)	1 (first article)
B109	EPC9186HC2 GaN motor-drive evaluation board (HC2 bring-up)	1
B025	HC2: 150 A pk / 75 A cont, 100 V-class GaN EtherCAT drive (custom: F28P65x with on-chip EtherCAT, 2 × BiSS-C, 2-channel hardware STO, 12 V aux input)	1 (first article)
B027	Drive carrier PCB (Everest)	3 (spin 1)
B042	Safety MCU board (start on the TI LAUNCHXL2-570LC43)	1
B004	CubeMars RI85-PH frameless motor (test motor in a printed fixture)	1
B017	AksIM-2 encoder + ring	1
B078/B079	EtherCAT cable (igus chainflex CF Robot), Harwin Gecko-MT 10-pin (J-LINK), XT30/XT60, JST GH	—

C Tools

Oscilloscope (4-ch, with a 100 V differential probe and a current probe ≥ 150 A), bench supply ≥ 70 V current-limited, DC load, ESD bench, hot-air station, microscope, Ethernet tap or spare NIC for Wireshark (EtherCAT dissector), IR camera.

D Manufacturing: drive carrier PCB (custom)

Specification (docs/05 §6): - 4-layer, 2 oz copper on power layers, ENIG finish; - the drive module footprint per the Novanta integration manual; - 2 × 100 µF 100 V polymer + 1 µF X7R at the bus input; SMCJ75A TVS; - EtherCAT magnetics (per the drive's reference design) to two J-LINK connectors; - 2 × RS-422 receivers (THVD1450 class) converting BiSS-C MA/SLO to the 3.3 V single-ended inputs on Everest [R2]; - 5 V / 300 mA LDO for the encoder; - NTC dividers; STO input conditioning; an ADS1262 footprint (populated on KH-C25 and KH-C20 for the torque flexure).

HC2 drive (150 A, B025). No purchasable 100 V, 150 A pk EtherCAT drive of this size is identified (docs/09 §11 H8), so the HC2 is custom (docs/05 §6.1, docs/13 §7): F28P65x

with on-chip EtherCAT, three 100 V GaN half-bridges on an EPC9186HC-class layout at 40 kHz, 2 × BiSS-C, 2-channel hardware STO (channel A cuts the high-side gate supply, channel B disables the PWM buffers, both read back), 12 V aux input, ≥ 16 × 4.7 µF + 4 × 100 µF 100 V DC-link at the bridge. Start on the EPC9186HC2 evaluation board (B109). Its stage is rated ≤ 76 V input: set the over-voltage trip at 74 V and keep the brake chopper at 70 V on / 68 V off. Firmware is the long pole; plan two board spins (B110).

Order 3 Everest carriers first, assembled by JLCPCB/PCBWay (BOM + centroid from your PCB tool).

E Assembly / setup steps

1. **RT computer OS:** Ubuntu 24.04 LTS with a PREEMPT_RT kernel (6.8+ RT). Boot parameters: `isolcpus=2,3 nohz_full=2,3 rcu_nocbs=2,3 irqaffinity=0,1`. Disable C-states deeper than C1 in the BIOS.
2. Build **IgH EtherLab 1.6** with the native `igc` driver for the i226 (`--enable-igc`). Bind port 1 (EtherCAT) to the igc driver; port 2 later closes the ring.
3. Run `cyclictest -m -p95 -t2 -a2,3 -i1000 -D 1h`. **Target: max latency < 50 µs.**
4. Drive configuration via the vendor tool (Novanta MotionLab for Everest; the HC2 drive's own tool or your firmware's CiA-402 object dictionary): - Set the motor parameters from the datasheet (RI85-PH: 14 pole pairs, Kt 0.13 N·m/A, R_LL 0.1385 Ω [ds R1]). - Encoder BiSS-C, 19 bit; current limit 5 A to start; I²t on; STO enabled. - **Store current limits in the drive's non-volatile parameters.** Mk II relies on drive-level limits on the cycloid and strain-wave joints (docs/09 §5.3), independent of the RT software.
5. Commutation: run auto-phasing with the rotor free. Check the electrical angle offset against a manual alignment.
6. Tune the current loop (vendor autotune), then verify with a step response.
7. **HC2 first light** on the RI85-PH test motor at 48 V and ≤ 10 A: EtherCAT OP, commutation, current loop, STO. The 150 A tests come later on the dyno with a KM-130 motor (P05 §G).
8. **Safety MCU:** flash the monitor firmware (heartbeat watchdog on CAN from the RT computer, e-stop input, two relay-driver outputs, IMU tilt; Mk II adds the HC2 leak inputs and the coolant > 75 °C trip, docs/05 §8). Unit-test each trip path.

F Electrical

- Bench bus 48 V from the supply (current limit 3 A) → disconnect (enable = safety relay) → Everest carrier XT30 (16 AWG) or HC2 XT60H (12 AWG). HC2 12 V aux from a separate bench supply (Gecko-MT 2-pin, 20 AWG).
- **HC2 stop function is uncertified** (custom drive). From the first HC2 power-up until its STO is validated (docs/11 §5) and through commissioning, the safety function for every HC2 is the external chain: safety relay → a DC contactor (Gigavac GX11 class, ≥ 100 A, ≥ 80 V DC break rating) in the HC2's DC feed, opened by the relay's instantaneous contacts, plus the HC2 STO inputs. On the bench the contactor sits in the bench-bus feed; on the robot, one per leg zone until gate G6.
- STO pair from the safety relay → J-LINK pins 5/6 (STO_A/STO_B), 7 (STO_COM). The signal level at the drive is **to be confirmed from the Everest S and HC2 datasheets** (docs/05 §9).
- Encoder: JST GH 6-pin (5 V, GND, MA+, MA−, SLO+, SLO−), shielded, shield to the carrier chassis pin.
- First power-up procedure for every new board: 1. Inspect the board under a microscope. 2. Check the bus for shorts (> 10 kΩ). 3. Power at 12 V with a 0.2 A limit and check rails. 4. Raise to 48 V with a 0.5 A limit. 5. Only then raise the limit to 3 A.

G Testing (gate G0)

Test	Pass
EtherCAT: master sees the drives, OP state, 1 kHz with distributed clocks for 1 h	0 lost frames; DC sync error < 1 µs
Cyclic synchronous torque mode: ±0.5 N·m square wave on the free motor (Everest and HC2)	current tracks with < 1 ms rise; no faults
STO from each e-stop device while the motor runs	phase current → 0 in < 10 ms
HC2: e-stop with the external contactor in circuit	contactor opens and the HC2 DC bus collapses within the relay's time; HC2 STO read-back consistent; the motor coasts or brakes, never drives
Drive-level current limit: command 2 × the stored limit over EtherCAT	the drive clamps at the stored limit ±5 % (proves the limit does not depend on the RT software)

Test	Pass
Heartbeat loss (kill the RT process)	safety MCU opens the relay within 5 ms of the 2 ms timeout
Over-voltage, one drive at a time: clamp disconnected, dummy load, raise the supply in 1 V steps to 1 V above the drive's configured trip (HC2 74 V, Everest 78 V)	each drive trips at its threshold ±1 V without damage; with the clamp connected the bus stays ≤ 72 V during a regen burst
Thermal: NTC reading matches a thermocouple	±2 °C

P03 KA-90P and KA-110P planetary actuators (first articles, then series)

Gate G1

Mk II values (`docs/generated/actuator_family.md`):

Class	Joints (units)	Motor	Ratio	Peak / continuous out	No-load speed	Mass	Drive
KA-90P	hip yaw ×2, waist yaw, ankle ×4 (7)	CubeMars RI85-PH	28:1	129 / 44 N·m	13.1 rad/s	1.41 kg	Everest S NET
KA-110P	hip roll ×2, waist pitch (3)	CubeMars **RI115-PH**	29:1	**427 / 125 N·m**	5.7 rad/s	2.96 kg	Everest S NET

KA-110P is no longer a stretched KA-90P: it has a Ø115 motor, an Ø≈127 mm housing, a 4-planet output stage and the **12× M5 12.9 output flange on PCD 100** (K1's 10× M4 flange would slip at 440 N·m against the 379 N·m waist-pitch momentary torque).

A Visual reference

- Architecture of the KA family (same layout as the KA-130K cutaway, smaller): `renders/18_KA130K_cutaway.png` , `renders/19_KA130K_exploded.png`
- Envelope plots: `docs/img/envelope_KA-90P.png` , `envelope_KA-110P.png`
- Gear tables, fits and flange standards: docs/09 §2, §4, §5.1
- Section (axial stack, output at top):

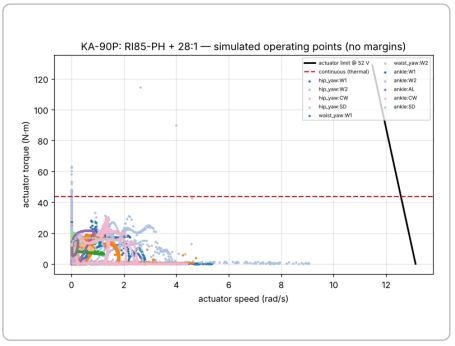
```
output flange (KA-90P 8× M4 PCD 75 | KA-110P 12× M5 PCD 100) — radial lip seal
crossed roller THK RB6013 — stage-2 carrier (output) + output AksIM-2 ring
front housing with stage-2 ring · 3 planets · stage-2 sun
stage-1 carrier → stage-2 sun ; stage-1 ring + front rotor bearing (KA-110P: inside the Ø74 rotor bore; KA-90P: in front of the motor) · 3 plan
motor housing: stator bonded in the bore —— rotor (magnets on hub) —— rear rotor bearing in the housing wall
motor encoder (KA-90P MT6835 | KA-110P AksIM-2) · drive carrier PCB · rear cap —— no through-bore with Plan A gears (docs/09 §11 H2)
```



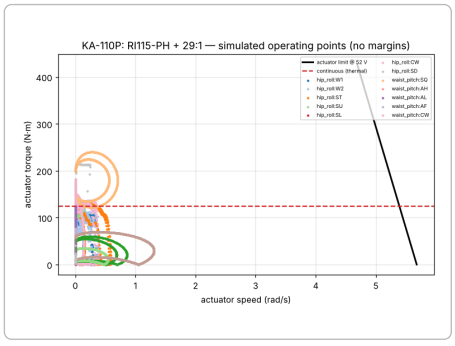
RENDER Architecture of the KA family (same layout as the KA-130K cutaway, smaller)



RENDER Architecture of the KA family (same layout as the KA-130K cutaway, smaller)



PLOT Envelope plots



PLOT Envelope plots

B Parts list (per actuator)

BOM	Item	KA-90P	KA-110P
B004 / B003	Frameless motor kit: CubeMars RI85-PH KV85 / RI115-PH KV40	1	1
B009 / B008	Planetary set 28:1 / 29:1 per docs/09 §5.1 Plan A (KA-90P: z15/3 × z27/z69 m0.6 + z18/3 × z27/z72 m0.8; KA-110P: z15/3 × z30/z75 m0.6 + z24/4 × z34/z92 m1.0), planet pins 100Cr6, INA needle cages	1	1
B014	Crossed roller THK RB6013 (60 × 90 × 13)	1	1
B016	Rotor bearings SKF 61806-2RZ (30 × 42 × 7) / 61808-2RZ (40 × 52 × 7) + wave spring	2 + 1	2 + 1
B017	AksIM-2 output encoder (ring on carrier, readhead on housing)	1	1
B018 / B017	Motor encoder: MT6835 + magnet (KA-90P) / AksIM-2 (KA-110P)	1	1
B020	NTC TDK B57861S0103F045	2	2
B022	Housing set: rear cap, motor housing, intermediate plate, front housing, rotor hub, output flange (7075-T6, docs/09 §3)	1	1
B026 + B027	Everest S NET drive + carrier PCB	1	1
B081	Output flange screws ISO 4762 12.9	8× M4 × 12	**12× M5 × 14**
B081	Base flange screws 12.9	8× M4	12× M5
B081	Dowels ISO 8734 m6, 2 per flange	Ø4 × 10	**Ø6 × 12**
B081	HeliCoil Plus stainless inserts, 2 × d long	M4 × 8	M5 × 10
B081	Housing screws 8.8 (closures), O-rings EPDM/FKM 70 (front, rear), output radial lip seal (double lip with dust lip, FKM) sized to the Ø60 / Ø80 spigot, ePTFE pressure-equalising vent (IP67, docs/03 §12)	set	set
—	Grease Klüber Isoflex Topas NB 52 ≈ 8 g / ≈ 14 g; thermally conductive potting epoxy ≥ 1.5 W/m·K Class H ≈ 10 / 20 g; Loctite 243, 638, 222	—	—

C Tools

Arbor press with bearing tools (flat drifts on the correct ring only), oven or induction heater (housing to 90 °C), non-ferrous tweezers, torque screwdrivers 0.5–6 N·m and a 2–25 N·m torque wrench, dial indicator + magnetic base, feeler gauges, rotor-insertion jig (printed sleeve + guide rod), dyno (P01).

D Manufacturing

- **Housings (B022):** from the KA-130K STEP profiles, rescaled to the KA-90P / KA-110P interface table (docs/09 §4).
- 7075-T6; Type III hard anodise black with masks on bearing seats, the stator bore, bonding and earth pads.
- Machine the bearing seats **after** anodise if not masked.
- Fits (docs/09 §2): crossed-roller seat H7; rotor bearing seats k5 (hub) and K6/H6 (housing); stator bore +0.02/+0.05 over the stack; ring-gear spigots H7/js6; dowel holes H7 (press) / G7 (slip); coaxiality of all seats ≤ 0.01 mm; seat faces perpendicular within 0.005 mm.
- KA-110P output flange: 12 × M5 on PCD 100.000 ±0.02, 2 × Ø6 H7 dowel holes at 180° on the same PCD, spigot Ø80 h6. Base flange 12 × M5 on PCD 115. HeliCoil every M5 hole.
- **Gears:** order from a gear shop to docs/09 §5.1 (18CrNiMo7-6 carburised and ground, ISO 1328-1 grade 6; rings 42CrMo4 wire EDM + nitride). Request the inspection report. On arrival: tooth count, OD, over-pin dimension, hardness spot check (file test on a non-functional face), planet bore and pin diameters.

E Assembly (KA-90P first article; KA-110P identical order, larger parts)

1. **Clean** all parts (IPA). Lay them out on a lint-free mat in the order below.
2. **Stator into the motor housing.** - Heat the housing to 90 °C. - Apply the potting epoxy thinly to the bore. - Slide the stator in with the lead exit aligned to the cable slot; hold it against the shoulder while it cures (24 h at room temperature, or per the epoxy data sheet). - Pot

the end turns with the same epoxy, keeping the rotor clearance free (check with a gauge pin).

3. **NTCs:** glue two NTCs into the end turns at 180° (thin epoxy). Route the leads with the phase leads.
4. **Rear rotor bearing** into the motor-housing rear wall (outer ring press, Loctite 638), then the wave spring.
5. **Rotor hub:** press the front rotor bearing onto the hub (inner ring only, Loctite 638). Fit the stage-1 sun (shrink or press, Loctite 638) and the motor encoder magnet (KA-90P) or AksIM-2 ring (KA-110P) on the rear end.
6. **Insert the rotor** on the guide rod through the rear bearing. **Magnet safety:** the rotor is pulled hard into the stator; use the printed sleeve, never fingers. Check free rotation and cogging feel. Check the airgap with plastic shim at 4 points: all within ±0.05 mm.
7. **Intermediate plate** (stage-1 ring + front rotor-bearing seat) onto the motor housing: dowels, 8.8 screws at the housing torque (docs/09 §2), Loctite 222. This captures the rotor.
8. **Stage 1:** grease the ring and the planets. Planets on the stage-1 carrier pins with their needle cages. Mesh them into the ring and onto the sun. Turn the rotor: no tight spots.
9. **Front housing with the stage-2 ring** (spigot H7/js6, 2 dowels, Loctite 243; KA-90P 8 × M3 12.9 at 1.7 N·m, KA-110P 12 × M4 12.9 at 4.5 N·m: this joint carries ≈ 80 % of the output torque) onto the intermediate plate (O-ring lightly greased). Stage-2 sun on the stage-1 carrier output. Planets on the output carrier.
10. **Crossed roller:** press onto the output carrier, then into the front housing, per THK's instructions (press only the ring being fitted). Fit the outer-ring clamp ring: 8 × M4 8.8 at 2.5 N·m in a star, two passes.
11. **Output encoder:** AksIM-2 ring on the output carrier, readhead on the housing. Set the gap with the supplied shim and check the signal LED or the RLS tool.
12. **Rear close-up:** motor encoder readhead, drive carrier PCB on the rear standoffs, phases (crimped ring lugs, M3 at 0.6 N·m), encoders (JST GH), NTCs. Rear cap with O-ring.
13. **Output flange:** radial lip seal (lightly greased lip) on the spigot, ePTFE vent in the rear cap. Fit the flange on its 2 dowels.
    - KA-90P: 8 × M4 12.9 at **4.5 N·m**, Loctite 243, star pattern, two passes.
    - KA-110P: **12 × M5 12.9 at 8.5 N·m**, Loctite 243, star pattern, two passes.
    - Witness-mark every screw head with a paint pen.
14. **Label** the actuator serial and record the stator, rotor, gear and bearing serials in the log.

F Electrical

- Phases U/V/W to the drive (any order, then fix by auto-phasing). Twist the three leads, ≤ 150 mm.
- Bus: KA-90P via XT30 (16 AWG); KA-110P via XT60 (12 AWG, its peak bus current is ≈ 25 A). EtherCAT + STO via the J-LINK pair.
- First power: 48 V, 3 A supply limit, STO enabled, drive in torque mode at 10 % of the peak current.
- Drive peak current: KA-90P 38 A (the RI85-PH saturates at 5.0 N·m), KA-110P 55 A (RI115-PH 16 N·m). Stored in the drive's non-volatile parameters.

G Testing on the dyno (gate G1, per actuator)

Test	Method	Pass (KA-90P / KA-110P)
Phase resistance, insulation	4-wire milliohm; 500 V insulation tester phases → housing	R_LL within ±10 % of datasheet (0.1385 / 0.134 Ω); > 100 MΩ
Back-EMF constant	back-drive at 30 rpm output, scope a line-line voltage	Ke within ±5 % of datasheet
Gear run-in (R10 §7 item 4)	30 min each direction at 25 % peak torque and 50 % speed, then 30 min at 50 % torque; drain, inspect and replace the grease	no metal flakes, temperature rise steady, noise does not grow
Kt at the output	current steps 10–100 % peak vs transducer torque	slope = Kt·N·η = 3.35 / 7.76 N·m/A within ±8 %
Efficiency map	speed × torque grid, forward and back-driven	η ≥ 0.88 at rated (model 0.92; feed the measurement into `ETA`)
Peak torque	2 s pulses at the stored current limit	≥ 129 / 427 N·m without slip, noise or witness-mark movement

Test	Method	Pass (KA-90P / KA-110P)
Backlash, stiffness	lock the output, ±torque sweep, read both encoders	backlash ≤ 10 arcmin; record torsional stiffness
Back-drive torque	turn the output by hand through the transducer (drive off)	record (QDD target < 2 / < 4 N·m)
Thermal step	hold the continuous torque (44 / 125 N·m) at 1 rad/s until steady	winding ≤ 120 °C at 25 °C ambient; record the time constant
Endurance	10^5 cycles of the walking torque profile (from `data/scenario_series.npz`)	no loss of torque, backlash growth < 2 arcmin, grease clean
Flange audit after the peak and endurance tests	re-check the witness marks and re-torque check (no rotation at the set torque)	no movement
E-stop	STO at full speed	stops; no damage

Record every result in `logs/actuators/<serial>.md`. **Only after the first article of each class passes, order and build the rest of that class.**

P04 KH arm and neck actuators: KH-C25 and KH-C20 cycloids, KH-14 strain-wave

Gate G1

Mk II values (`docs/generated/actuator_family.md`, docs/09 §5.3):

Class	Joints (units)	Motor	Gear	Motor-limited peak	Gear limit (drive current limit)	Continuous	No-load speed	Mass
KH-25 (KH-C25)	shoulder pitch, shoulder roll (4)	KM-115x32-N50 (Ø115, scratch-built)	C25 two-disc cycloid 30:1	362 N·m	**160 N·m repeated (21 A)**, 250 N·m momentary target	119 N·m motor / 100 N·m gear average	6.3 rad/s	2.60 kg
KH-20 (KH-C20)	shoulder yaw, elbow (4)	KM-85x26-N50-4t	C20 two-disc cycloid 30:1	158 N·m	**105 N·m repeated (18.8 A)**, 160 N·m momentary	44 N·m	8.6 rad/s	1.63 kg
KH-14	wrist yaw ×2, neck yaw, neck pitch (4)	KM-50x20	CSG-14-50 strain-wave	34 N·m	**23 N·m repeated (4.3 A)**	9 N·m	9.0 rad/s	0.68 kg

Every KH motor can out-torque its gear. The drive peak-current limits above are stored in the drive's non-volatile parameters (P02 §E step 4) and verified on the dyno before the actuator goes into an arm. A software bug in the RT layer must not be able to break a gear.

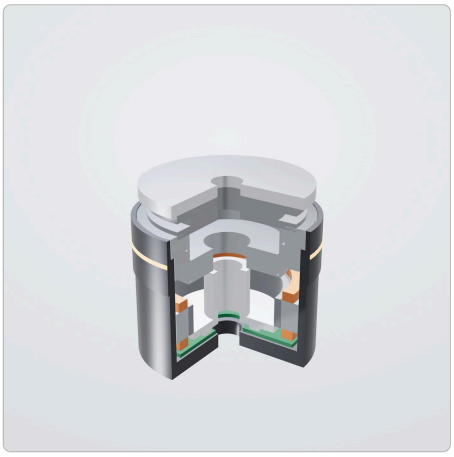
Build the cycloid gear sets first: **P04.1** (below).

A Visual reference

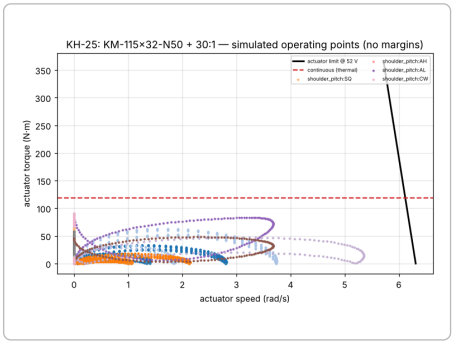
- KH-25 cutaway (K1 strain-wave version; the motor, encoder, flexure and output bearing positions still apply): `renders/20_KH25_cutaway.png`
- STEP profiles: `cad/out/step/kh25__*.step` (K1), cycloid discs `cad/out/step/cycloid_KH-C25_disc.step`, `cycloid_KH-C20_disc.step`
- Envelope plots: `docs/img/envelope_KH-25.png`, `envelope_KH-20.png`, `envelope_KH-14.png`

KH-C25 axial stack (output at top):

```
output flange (8× M4 PCD 70) — torque flexure (Ti, 4 shear gauges) — output front disc (screws into the 8 output pins)
crossed roller THK RB5013 — input-shaft front journal 61803 in the output disc
disc 1 (needle cage on eccentric sleeve 1) — spacer — disc 2 (sleeve 2, 180°) — ring with 31 Ø5 pins (fixed in the housing)
output back ring (8 solid Ø8 output pins pressed in, rotating)
rotor (KM-115x32-N50) on the hollow input shaft — stator bonded in the motor housing — rotor bearings 61808
motor encoder MT6835 — drive carrier (Everest + ADS1262) — rear cap — Ø12 cable bore
```



RENDER KH-25 cutaway (K1 strain-wave version; the motor, encoder, flexure and output bearing positions still apply)



PLOT Envelope plots

B Parts list (per actuator)

BOM	Item	KH-C25	KH-C20	KH-14
B005 / B104 / B006	Motor: KM-115x32-N50 / KM-85x26-N50-4t / KM-50x20 (docs/09 §6.1, docs/13 §3)	1	1	1

BOM	Item	KH-C25	KH-C20	KH-14
B010 / B011 / B012	Gear: C25 set / C20 set (P04.1) / Leaderdrive LHSG-14-50 component set (WG, FS, CS)	1	1	1
B015	Output crossed roller THK RB5013 (50 × 80 × 13) / THK RB4010 (40 × 65 × 10) / per the CSG-14 installation drawing (THK RB3010 class)	1	1	1
B016	Rotor bearings 61808-2RZ / 61806-2RZ / per K1 drawing + wave spring	2	2	2
B016	Input-shaft front journal bearing 61803 / 61802	1	1	—
B017, B018	AksIM-2 output + MT6835 motor encoder	1 + 1	1 + 1	1 + 1
B019	Torque flexure kit: 4 × HBK XY41-3/350 shear gauges, ADS1262 on the carrier	1	1	—
B023	Housing set incl. Ti-6Al-4V flexure (KH-C25/C20)	1	1	1
B026 + B027	Everest S NET + carrier	1	1	1
B081	Output flange screws 12.9	8× M4	**8× M4** (was M3)	6× M3
B081	Output-pin screws 12.9 (front disc into the pin ends)	8× M4	6× M3	—
B081	Dowels ISO 8734 m6 (flange, ring)	Ø4	Ø4	Ø3
— (recommended, docs/04 §8; not in the mass budget)	Spring-applied flat brake ≥ 4.5 N·m at the motor (shoulder pitch only, for the 90 N·m hold)	1	—	—
—	Grease: Nabtesco VIGOGREASE RE0 or Molykote BR2 Plus (cycloid), Harmonic Grease SK-1A (KH-14); gauge adhesive (M-Bond 610 class), protective coating (M-Coat)	—	—	—

C Tools

As P03, plus a strain-gauge bonding kit (clamps, Teflon tape, curing oven 120–175 °C for M-Bond 610), a fine soldering iron, a gauge installation tester, a rotor-locking pin for the static overload test, gauge pins Ø5 and Ø4.

D Manufacturing

- **Cycloid gear sets:** P04.1.
- **Motors:** KM-115x32-N50 needs a new Ø115 lamination die; KM-85x26 uses the Ø85 die. Wind, impregnate and test per docs/13 §3 (method) and §3.11 (these two variants), with the values in docs/09 §6.1. Run the FEMM check on each new design before cutting the die.
- **Torque flexure (KH-C25/C20):** Ti-6Al-4V, 4 radial spokes, sized for ≈ 200 με at the gear-limited torque (**160 N·m KH-C25, 105 N·m KH-C20**; K1 flexures were sized for 127 / 73 N·m and are too soft). Check by FEA. Gauge pads ground flat, Ra ≤ 0.8.
- **Housings:** 7075-T6, hard anodise with masked seats. Fits in docs/09 §2. Critical:
- ring-pin circle coaxial with the rotor bearing seats ≤ 0.01 mm;
- crossed-roller seat coaxial with the ring ≤ 0.01 mm;
- KH-14: circular spline and wave generator concentricity ≤ 0.01 mm, mounting-face perpendicularity ≤ 0.01 mm, axial room for the flexspline to "breathe" (vendor drawing).

E Assembly

Torque flexure (KH-C25, KH-C20): 1. Bond 4 shear gauges on the flexure spokes at ±45° as a full Wheatstone bridge (insensitive to bending and temperature when the gauges are matched). 2. Cure per the adhesive data sheet, wire to the 4-pin flexure connector and coat. 3. Measure the bridge: 350 Ω ± 0.5 %, insulation > 10 GΩ.

Motor (all): stator, NTCs, rear rotor bearing, rotor on the guide rod as P03 steps 2–6. On the cycloids the rotor hub is the hollow input shaft, with a keyway for the two eccentric sleeves.

Cycloid gear train (KH-C25 shown; KH-C20 identical with its own parts). Housing on the bench, output end up: 4. **Ring** into the front housing: spigot H7/js6, 2 dowels, 12.9 screws at the structural torque with Loctite 243. 5. **Output back ring** with its 8 output pins already pressed in (m6 pins into H7, Loctite 638) over the input shaft, pins pointing up. 6. **Disc 2:** key, eccentric sleeve 2, grease, needle cage, then disc 2 (mark up) over the sleeve and over the 8 output pins. 7. **Spacer**, then sleeve 1 on the same key (180° from sleeve 2), cage, and

disc 1 turned 180° about the axis relative to disc 2 (marks opposite). With 30 lobes the lobes coincide. 8. **Ring pins:** grease the 31 grooves; slide the pins in axially one by one, rocking the rotor so each groove meets a lobe valley. Fit the pin retaining washer. 9. **Output front disc** with the 61803 journal bearing onto the input shaft and onto the pin ends; 8 × M4 12.9 axially into the tapped pin ends at **4.5 N·m**, Loctite 243 (KH-C20: 6 × M3 at 1.7 N·m). The output pins are now supported at both ends. 10. Turn the rotor by hand through a full output turn: smooth, no tight spot, no knock. A tight spot means a mis-phased disc: stop and re-check step 7. 11. **Crossed roller** RB5013 (RB4010) onto the output front disc and into the housing; clamp ring 8 × M3 8.8 at 1.2 N·m in a star. 12. **Flexure** on the output disc (2 dowels, screws at the structural torque), then the AksIM-2 ring; readhead on the housing, gap set with its shim.

Strain-wave train (KH-14): 4. Wave generator onto the rotor hub (keyed, Loctite 638). 5. Circular spline into the front housing (dowelled, screws to the vendor torque). Flexspline cup bolted to the crossed-roller inner ring / output carrier to the vendor torque. 6. Grease the flexspline teeth and the wave-generator bearing with the vendor's quantity. Insert the wave generator at the vendor's orientation and rotate: smooth, with **no dedoidal (off-by-one-tooth) assembly**. An incorrect mesh destroys the gear: follow the vendor's check exactly.

Close-up (all): 13. Motor encoder at the rear, drive carrier on its standoffs, phases, encoder and flexure connectors, rear cap with O-ring. 14. **Output flange** on 2 dowels: KH-C25 8 × M4 12.9 at **4.5 N·m**; KH-C20 **8 × M4 12.9 at 4.5 N·m** (K1's 8 × M3 slipped at ≈ 125 N·m against the 115 N·m elbow momentary torque); KH-14 6 × M3 12.9 at 1.7 N·m. Loctite 243, witness marks. 15. Label and log serials.

F Electrical

- As P03 (XT30, 16 AWG drops; J-LINK; JST GH encoders).
- Flexure bridge → ADS1262 on the carrier (excitation 5 V, gain 32, 1 kSPS, sinc3 filter).
- **Drive peak-current limits (non-volatile):** KH-C25 21 A rms, KH-C20 18.8 A rms, KH-14 4.3 A rms, each then trimmed on the dyno so the measured output torque sits at 95–100 % of the gear's repeated rating.
- Shoulder-pitch brake (if fitted): 24 V coil from a torso 24 V supply through a low-side switch on the carrier; released only while the drive is enabled and healthy; it engages on STO.

G Testing (gate G1)

Test	Pass (KH-C25 / KH-C20 / KH-14)
Phase resistance, insulation, back-EMF (motor model values, docs/09 §6.1)	R_LL ±10 %, Ke ±7 %, > 100 MΩ at 500 V
Cycloid run-in: 1 h at 25 % then 1 h at 50 % of the repeated rating, both directions, 50 % speed; change the grease	no metal in the grease, steady temperature, no growing noise
Kt at the output: slope = Kt·N·η (6.43 / 4.74 / 3.98 N·m/A at η 0.85 / 0.85 / 0.75)	within ±10 %; record measured η and feed it into `ETA`
Drive current limit commanded at 2 × over EtherCAT	output torque 152–160 / 100–105 / 22–23 N·m
Cycloid static overload (R10 §7 item 6): rotor locked with the locking pin, output loaded through the 1 m arm to 250 / 160 N·m for 1 s, 3 times each direction	no permanent lost-motion change > 1 arcmin; no pin or lobe marks on teardown of the first article
Torsional stiffness (from the overload sweep)	record; feed into the arm model
Backlash / lost motion (dual encoder)	≤ 7 arcmin cycloid; ≤ 1 arcmin strain-wave
Efficiency vs speed at 50 % and 100 % of the repeated rating	cycloid ≥ 0.80 at rated (model 0.85)
Torque flexure calibration: ±full scale with lever + weights	linearity ≤ 1 % FS, hysteresis ≤ 0.5 % FS
Collision detection: padded pendulum into the output link at 1 rad/s	flexure detects within 5 ms; controller backs off
Thermal: 100 / 44 / 9 N·m at 2 rad/s to steady state	winding ≤ 120 °C; gear housing ≤ 80 °C
Endurance: 10^5 reach cycles at the arm torque profile	lost-motion growth < 1 arcmin (cycloid) / < 0.5 arcmin (strain-wave); grease clean

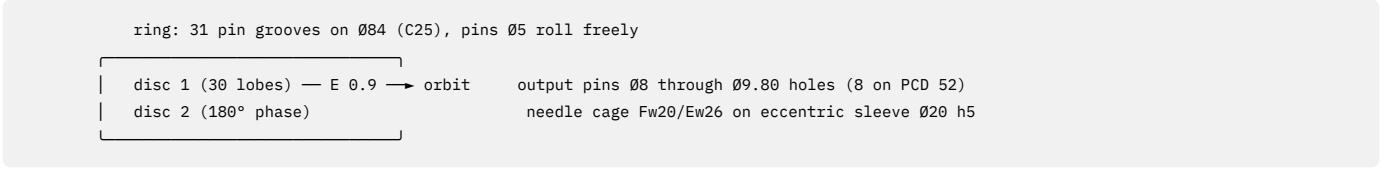
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P04.1 — Machined cycloid gears (C25 and C20)

Four C25 sets (shoulders) and four C20 sets (shoulder yaw, elbows), plus one spare disc pair of each. Specification: docs/09 §5.3. Build and test **one C25 set first** (G1 static overload above) before ordering the rest.

A Visual reference

- Profile generator: `cad/cycloid_profile.py` (writes `cad/out/cycloid_KH-C25.dxf`, `cycloid_KH-C20.dxf` and the STEP discs)
- Disc models: `cad/out/step/cycloid_KH-C25_disc.step`, `cycloid_KH-C20_disc.step`
- Equations and design background: docs/13 §4.2



B Parts list (per set)

Part	C25	C20
Discs (matched pair)	2 × 8.0 mm, 100Cr6 hardened 60–62 HRC (route A)	2 × 6.0 mm (7.0 mm per docs/13 §4.2)
Ring	42CrMo4 Q&T 28–32 HRC, nitrided, 31 grooves on Ø84.000	31 grooves on Ø68.000
Ring pins	31 × Ø5 × 19.8 needle rollers ISO 3096 G2 (or ISO 8734 Ø5 m6, 60 HRC)	31 × Ø4 × 15.8
Output pins	8 × ISO 8734 Ø8 m6, ≥ 58 HRC, ends tapped M4	6 × Ø6, ends tapped M3
Eccentric sleeves	2 × 100Cr6 60–62 HRC, Ø20 h5, E 0.900 ±0.003, keyway	2 × Ø16 h5, E 0.800
Needle cages	2 × INA K-series Fw 20 / Ew 26, width ≤ 8 mm	2 × Fw 16 / Ew 22, width ≤ 6 mm
Output back ring and front disc	17-4PH H1025 or 42CrMo4 Q&T	same
Spacer between discs	1 mm 100Cr6 shim, ground	1 mm

C Tools & equipment (outsourced items in brackets)

Python with CadQuery (to run the profile script), [wire-EDM shop with 3-skim capability], [heat treater], [CMM or optical comparator service], surface plate, micrometers, gauge pins Ø5 / Ø4 (class ZZ or better), lapping plate and fine diamond paste, ultrasonic cleaner.

D Manufacturing

1. **Generate the profile.** `python cad/cycloid_profile.py`. The script asserts E·N < R and checks that the pin radius is below 0.8 × the minimum radius of curvature of the pin-centre epitrochoid (no undercut); it prints that radius. **Before sending the DXF:** change the eccentric bore from the Ø22 in the script to **Ø26 G6 (C25)** and keep **Ø22 G6 (C20)** (docs/09 §11 H6), and add the output-pin holes Ø9.80 / Ø7.60 +0.01/0. Have Ø3 wire start holes drilled at two output-pin centres before hardening; the shop uses them to align and pin the two-disc stack.
2. **Blanks:** 100Cr6 plate, ground both faces parallel within 0.005 mm, thickness 8.00 / 6.00 −0.005. Through-harden to 60–62 HRC and temper **before** EDM (route A, docs/09 §5.3). Witness coupon with each batch for hardness.
3. **Wire EDM (route A):** stack the two discs of a pair on the alignment pins and cut profile, bore and pin holes in one setup: rough cut + 3 skim cuts. Specify to the shop: profile ±0.005 mm, bore and pin-hole position ±0.005 mm, Ra ≤ 0.6 µm. Mark both discs on the same lobe ("lobe 1") before unstacking. - Route B (42CrMo4 Q&T + nitride): cut, then gas nitride; press a hardened 100Cr6 sleeve into the bore as the needle raceway; allow for 0.005–0.010 mm nitride growth in the profile offset. - Route C (jig/profile grinding from the DXF): best accuracy (±0.002 mm, Ra ≤ 0.2); use it if the EDM sets fail the run-in or efficiency targets.
4. **Recast removal:** lap the profile flanks and bore lightly with fine diamond paste on a hardwood or brass lap (2–5 µm), then ultrasonic clean. Do not round the lobe tips.
5. **Ring:** wire EDM the 31 semicircular grooves (Ø5.00 / Ø4.00 F7) on the pin circle, then nitride. Pins must roll freely in the grooves.

6. **Eccentric sleeves:** turn, harden, then grind the OD on a fixture that sets E; grind both sleeves of a set in the same setup so E matches within 0.003 mm. Keyway before hardening.
7. **Output pins:** ISO 8734 hardened dowels; tap the ends M4 (C20: M3) by EDM-drill or buy pins with tapped ends (ISO 8735).

E Assembly (bench pre-check of each set before it goes into an actuator)

1. Clean all parts; oil lightly.
2. On a plate: ring pins into the ring, both discs on a ground test mandrel with the sleeves, phased 180°.
3. Rotate the mandrel through 30 turns (one output turn): smooth, no tight spot, no knock.
4. Measure the free lost motion of the output pins by hand with a dial indicator on a lever: record.
5. Tag the set (ring, disc pair, sleeves) with one serial. Parts of a set stay together.

F Electrical

None. (The actuator's drive limits are set in P04 §F.)

G Testing and measurement

Check	Method	Accept
Disc profile	CMM (≥ 72 points per lobe) or optical comparator at 20× against the DXF	±0.005 mm everywhere
Over-pins dimension	two gauge pins Ø5 (Ø4) in diametrically opposite lobe valleys, micrometer; nominal from the CAD	within ±0.005 mm of nominal; the two discs of a pair within 0.003 mm
Bore and pin holes	bore gauge; CMM position	bore G6; pin-hole position ±0.005 mm; bore-to-profile concentricity ≤ 0.005 mm
Hardness	Rockwell on the witness coupon; file test on a non-functional face	60–62 HRC (route A)
Eccentricity of sleeves	V-block + dial indicator	E 0.900 / 0.800 ±0.003 mm, pair within 0.003 mm
Ring-pin circle	CMM	Ø84.000 / Ø68.000 ±0.005 mm, groove pitch error ≤ 0.005 mm
Bench pre-check (E above)	hand rotation, lost motion	smooth; lost motion recorded
In the actuator	P04 §G (run-in, overload, backlash ≤ 7 arcmin, efficiency ≥ 0.80)	as listed

P05 KM-130x50-LC motors and KA-130K / KA-130H actuators

Gate G1

The custom motor (magnetic design, laminations, winding, magnets, rotor) is built step by step per docs/13 §3, with the Mk II changes in docs/09 §6.1 (50 mm stack, N50UH, 3 turns × 2 paths, potted, liquid jacket). This phase assembles the actuator around it. Build the jackets in **P05.1** (below) before step 2.

Mk II values (`docs/generated/actuator_family.md` , docs/04 §5, docs/05 §6.1):

Class	Joints	Motor	Gear	Peak / continuous out	No-load speed	Reflected inertia	Mass	Drive
KA-130K	knee ×2	KM-130x50-LC	2-stage planetary **17:1**	**493 / 295 N·m** (263 N·m at the HC2's 75 A continuous)	12.5 rad/s	0.400 kg·m²	4.96 kg	HC2, 150 A pk / 75 A cont
KA-130H	hip pitch ×2	KM-130x50-LC	2-stage planetary **14:1**	**406 / 243 N·m** (217 N·m at 75 A)	15.2 rad/s	0.271 kg·m²	4.76 kg	HC2

Knee and hip share the motor, jacket, housing, stage 2 and drive; only stage 1 differs (docs/09 §5.1). The motor's thermal rating (18.85 N·m = 84 A) assumes the jacket reaches ≤ 0.25 K/W copper-to-coolant: that is the critical number of this phase. In service the HC2's 75 A continuous limit sets the usable continuous torque (docs/09 §11 H8).

A Visual reference

- Cutaway and exploded views (K1 finned version; Mk II replaces the fins with the jacket): `renders/18_KA130K_cutaway.png` , `renders/19_KA130K_exploded.png`
- Drawing sheets (K1 geometry; read with docs/09 §5.1 and §11 H1–H3): `cad/out/drawings/ka130k__front_housing.svg` , `ka130k__motor_housing_finned.svg` , `ka130k__rotor_hub.svg` , `ka130k__s2_ring_gear_z54_m1.25.svg` (stage 2 unchanged in Plan A), `ka130k__s2_carrier_output.svg`
- STEP: `cad/out/step/ka130k__*.step`
- Motor design: R10 §5.1 and docs/04 §5.1 (KM-130x50-LC); K1 sheet for comparison `docs/generated/km_motor_designs.md`
- Envelope: `docs/img/envelope_KA-130K.png` , `envelope_KA-130H.png`

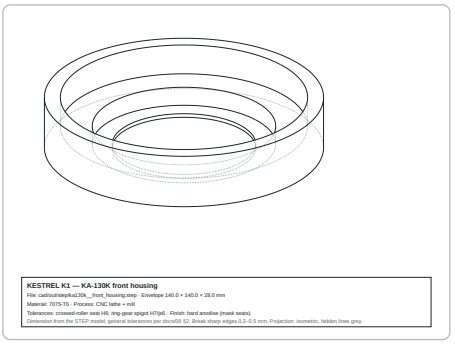
Axial stack (rear → output): rear cap · **HC2 drive bolted to the jacket's rear face** (the jacket is its cold plate; drip lip above it) · motor encoder AksIM-2 · housing sleeve (≈ Ø145) with the **liquid jacket** and the potted KM-130x50-LC stator · rotor (N50UH magnets, CF sleeve) on the hub carrying the stage-1 sun · stage 1 nested inside the Ø85.5 rotor bore (ring on the stage-1 spider) · front housing with the stage-2 ring z54 · stage 2 (4 × z18 planets) · output carrier · **THK RB9016** crossed roller · output flange (12× M5 PCD 105). No through-bore on the knee with the Plan A gears; Ø12 on the hip (docs/09 §11 H2).



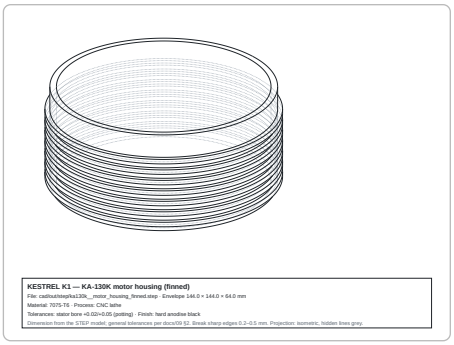
RENDER Cutaway and exploded views (K1 finned version; Mk II replaces the fins with the jacket)



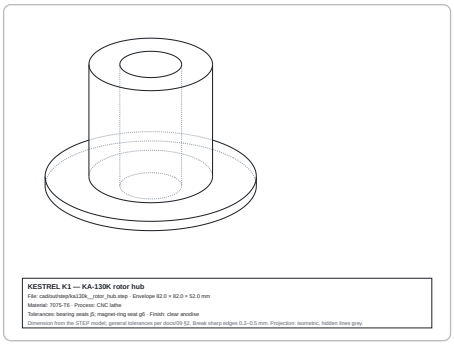
RENDER Cutaway and exploded views (K1 finned version; Mk II replaces the fins with the jacket)



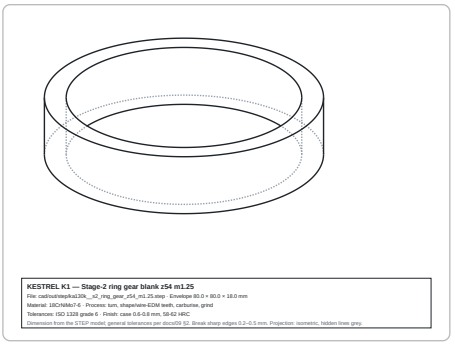
DRAWING ka130k / front housing



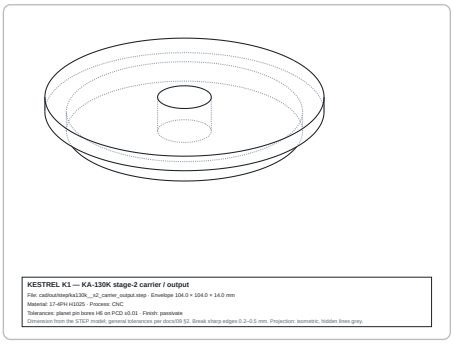
DRAWING ka130k / motor housing finned



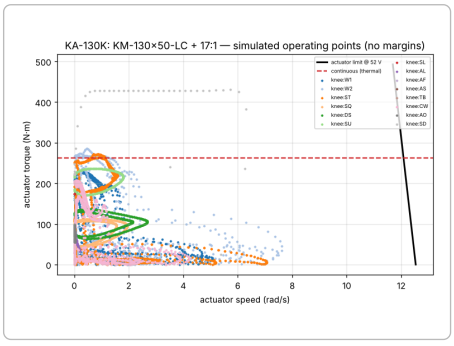
DRAWING ka130k / rotor hub



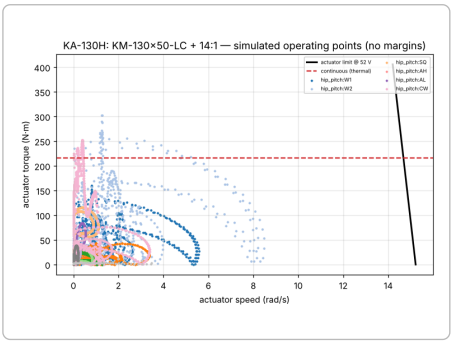
DRAWING ka130k / s2 ring gear z54 m1.25



DRAWING ka130k / s2 carrier output



PLOT Envelope



PLOT Envelope

B Parts list (per actuator; build 2 × K and 2 × H, plus one spare motor and jacket)

BOM	Item	Qty
B001, B002, B101, B102	KM-130x50-LC motor: 50 mm NO20 lamination stack, 42 N50UH arcs, winding set (10 × Ø0.67 mm per conductor, 3 turns × 2 paths), potting ≥ 1.5 W/m·K, rotor back iron + CF sleeve (docs/13 §3, docs/09 §6.1)	1 set
B103	Liquid jacket with two-start helix, 4 EPDM O-rings, leak-test ports, 2 coolant ports (P05.1)	1
B007 (K) / B105 (H)	Planetary set per docs/09 §5.1 Plan A. K: stage 1 m0.8 z16 / 4 × z18 / z52; H: stage 1 m0.8 z24 / 3 × z18 / z60; both stage 2 m1.25 z18 / 4 × z18 / z54, 18 mm face. Planet pins 100Cr6 (Ø5 stage 1, Ø10 stage 2); INA K5×8×8 and K10×14×13 needle cages	1
B013	Crossed roller **THK RB9016 (90 × 130 × 16)** (hold H1: the CAD seat is OD 120)	1
B016	Rotor bearings SKF 61808-2RZ (40 × 52 × 7) + wave spring	2 + 1
B017	AksIM-2 × 2 (motor and output)	2
B020	NTC TDK B57861S0103F045: 2 in the end turns + 1 at the jacket outlet	3
B021	Jacketed housing set: rear cap with HC2 bay and coolant ports, stage-1 spider, front housing, rotor hub, stage-1 carrier, stage-2 carrier/output, output flange (7075-T6 / 17-4PH per docs/09 §3)	1
B025	HC2 150 A GaN drive (annular, 6-layer) with leak pad, phase busbars ≈ 12 mm² ≤ 60 mm	1
B081	Output flange 12 × M5 × 14 12.9; base flange 12 × M5 12.9; stage-2 ring 12 × M4 12.9; crossed-roller clamp ring 12 × M4 12.9; 2 × Ø6 × 12 dowels per flange; HeliCoil Plus M5 × 10 and M4 × 8; housing O-rings EPDM/FKM 70; output radial lip seal (double lip, FKM) for the Ø90 spigot; ePTFE vent	set
—	Grease Klüber Isoflex Topas NB 52 ≈ 25 g; Loctite 243, 638, 222; silicone grease for EPDM O-rings (never mineral grease on EPDM)	—

C Tools

As P03, plus a 2 t arbor press, an oven to 150 °C (jacket shrink fit), the magnet-safe rotor insertion jig (printed guide + guide rod, docs/13 §3.7), a rotor-locking pin, a hi-pot tester, an LCR meter (also for L_s, needed for the active-short-circuit check), the coolant loop bench (B131, P01), the test pack (P06) for the 150 A tests.

D Manufacturing

- **Housings:** CNC 7075-T6 per docs/09 §2–§3. No fins: the housing sleeve closes the jacket's helix. Crossed-roller seat Ø130 H7 (hold H1). Coaxiality of rotor-bearing, ring and crossed-roller seats ≤ 0.01 mm.

- **Gears:** gear shop, 18CrNiMo7-6 carburised and ground, crowned with tip relief (docs/09 §5.1). Stage-2 parts are common to K and H: order 4 + 1 sets.
- **Motor:** docs/13 §3, plus the FEMM check in docs/11 §4.1 item 1 **before the Ø130 × 50 stack and the N50UH magnets are ordered**.
- **Jacket:** P05.1.

E Assembly

1. **Motor first-article tests before building the actuator** (docs/13 §3.10, with the KM-130x50-LC model values in docs/09 §6.1): R, L (record L_s), back-EMF, hi-pot, locked-rotor Kt.
2. **Stator in its jacket and sleeve:** P05.1 §E, including the pressure test. Do not continue with a jacket that has not passed P05.1 §G.
3. **NTCs:** two in the end turns (one at the hot-spot side near the cable exit), one in the jacket outlet boss.
4. **Rear rotor bearing** into the rear wall of the sleeve (Loctite 638 on the outer ring), wave spring.
5. **Rotor hub:** press the front bearing onto the hub (inner ring only); fit the stage-1 sun and the motor AksIM-2 ring.
6. **Insert the rotor with the jig.** The N50UH magnetic pull is several hundred newtons. Keep fingers clear.
7. **Stage-1 spider** (stage-1 ring + front rotor-bearing seat) into the rotor bore and onto the sleeve: dowels, 8.8 screws at the housing torque, Loctite 222. This captures the rotor.
8. **Stage 1:** planets with needle cages on the stage-1 carrier pins (4 on the knee, 3 on the hip); grease; mesh. Turn the rotor: free, no tight spots.
9. **Front housing with the stage-2 ring z54:** spigot H7/js6, 2 dowels, **12 × M4 12.9 at 4.5 N·m**, Loctite 243. This joint carries ≈ 75 % of the output torque (the ring reaction), so the K1 value of 2.5 N·m is not enough. Stage-2 sun on the stage-1 carrier; 4 planets on the output carrier.
10. **Crossed roller RB9016** onto the output carrier, then into the front housing; clamp ring 12 × M4 12.9 at 4.5 N·m in a star, two passes. Output runout ≤ 0.02 mm (dial indicator).
11. **Output AksIM-2 ring** on the carrier; readhead set with its shim.
12. **HC2:** thermal pad, then bolt the HC2 to the jacket's rear face; phase busbars to the stator leads; encoders; NTCs. Route every coolant hose **below and away from** the board; no fitting above it. Rear cap on (O-ring).
13. **Output flange:** radial lip seal, 2 × Ø6 dowels, **12 × M5 12.9 at 8.5 N·m**, Loctite 243, witness marks. This joint is also the overload slip fuse (935 N·m vs 535 N·m momentary, SF 1.75); do not over-torque it.
14. Cap both coolant ports with plugs (or dry-break halves) until P07.

F Electrical

- DC input 12 AWG through XT60H from the leg zone (≤ 0.5 m); HC2 12 V aux on a Gecko-MT 2-pin (2 × 20 AWG); J-LINK IN/OUT (docs/05 §6.2).
- Drive setup from the **measured** motor test (not the model): pole pairs 21, Kt, R, L.
- Current limits: start at 30 A; raise to 150 A only on the dyno, **with coolant flowing**, after the thermal test. Firmware DC-bus current limit 45 A per HC2.
- Hardware defaults alongside STO (docs/05 §8): **active short circuit** below a speed threshold after STO (passive damping ≈ $K_m^2 \cdot N^2$ = 478 N·m·s/rad at the knee actuator, 324 at the hip), enabled only after the phase short-circuit current λ/L_s is measured below 150 A; field weakening off until G1 completes.
- **Safety function:** the HC2's STO is uncertified, so on the dyno and through commissioning the external safety relay + DC contactor in the HC2 feed (P02 §F) is the stop that counts. Every e-stop test in §G is run with it.
- Coolant interlock: the drive limits current to 30 A unless the coolant-flow OK signal (loop bench flow sensor now, robot loop sensor later) is present. On loss of flow in service the RT layer derates to the air-only rating (0.64 × the jacketed continuous, docs/03 §9.6) and ends the session.

G Testing (gate G1; R10 §7 items 1, 3, 4 and 5 run here, item 2 in P05.1)

Test	Pass (KA-130K / KA-130H)
FEMM check of KM-130x50-LC (before the die, docs/11 §4.1)	Kt within ±10 % of 0.224 N·m/A; T_sat near 30 N·m; N50UH demagnetisation margin at 204 A and 120 °C
Gear run-in (R10 §7 item 4): 30 min each direction at 25 % then 50 % peak, 50 % speed; drain and inspect grease; ISO 6336 tooth-root check on paper	no metal flakes; stable temperature and noise
Kt at the output vs model (0.224 × N × 0.92 = 3.50 / 2.89 N·m/A rms)	within ±8 %; update `calc` if outside
150 A locked-rotor test (R10 §7 item 3): output locked through the transducer and reaction arm; 150 A for 2 s, then 99 A for 10 s; torque vs current from 10 to 150 A	≥ 493 / 406 N·m at 150 A; the curve bends no earlier than the model (T_sat ≈ 30 N·m at the motor); no flange slip; HC2 within its thermal rating
Active short circuit: measure L_s, compute λ/L_s; then trigger ASC at the speed threshold	phase current < 150 A peak; braking torque as modelled
Efficiency map, both directions	η recorded (model 0.92); feed into `ETA`
k_L measurement (R10 §7 item 5): voltage-limited torque–speed curve with the bus held at 52 V	fitted k_L ≥ 0.75; otherwise update `K_L` in `actuators.py` and re-run `run_all.py`
Torque–speed: points from the envelope plot with ×1.2 / ×1.15 margin at 52 V	all reachable
Thermal step (critical): 75 A continuous (263 / 217 N·m) at 2 rad/s, coolant 0.5 L/min at 25 °C inlet, to steady state	winding rise over coolant inlet ≤ 45 K (0.25 K/W × ≈ 170 W copper loss at 75 A); HC2 temperature within rating
Walking duty replay (W1/W2 knee or hip torque series ×1.0) for 30 min with coolant	winding ≤ 100 °C; no drift in Kt
Coolant-loss response: stop the flow at 75 A	drive derates on the interlock / thermal model before the 130 °C warning; log the time
Backlash / stiffness	backlash ≤ 10 arcmin; stiffness recorded
Endurance 2×10^5 walking cycles	backlash growth < 2 arcmin; gear inspection clean
Static overload: rotor locked with the pin, 590 N·m (1.1 × 535 N·m momentary) on the output through the arm, 1 s × 3	no flange slip (witness marks), no gear damage on teardown of the first article

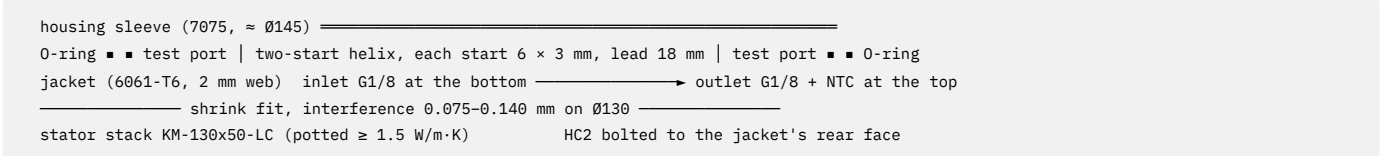
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P05.1 — Liquid-cooling jackets (KA-130K and KA-130H)

One jacket per KM-130x50-LC (4 + 1 spare). Making the jacket and the first-article thermal test are described in docs/13 §3.9, the motor acceptance tests in docs/13 §3.10; the specification is docs/09 §6.2 and docs/03 §9.3. This sub-phase is the build-and-test sequence. One deliberate difference: the shrink-fit interference here is **0.075–0.140 mm**, not docs/13's 0.06–0.08 mm, because aluminium on steel loses ≈ 0.15 mm of interference on Ø130 in a uniform 100 K hot soak. The loop that feeds the jackets is built in P08.1; on the bench they run from the loop bench (B131, P01).

A Visual reference

- Housing envelope (K1 finned sleeve, to be replaced by the jacket + sleeve): `cad/out/drawings/ka130k__motor_housing_finned.svg`
- Section:



B Parts list (per jacket)

BOM	Item	Qty
B103	Jacket, 6061-T6, bore Ø129.860 +0.025/0, two-start 6 × 3 mm helix on the OD, leak-test port between each O-ring pair	1
B021	Housing sleeve, 7075-T6 (part of the jacketed housing set)	1

BOM	Item	Qty
B103	EPDM 70 O-rings, 3.53 mm section, AS568 -2xx size for the ≈ Ø140 bore (docs/13 §3.9)	4 + 4 spare
B126	Port fittings: G1/8 (1/8 BSPP) ISO 1179 stud ends with bonded seals and barbs for the 6 mm ID hose, stepless ear clamps	2
B020	NTC for the outlet boss	1
—	Silicone grease for EPDM (e.g. Molykote 111); blanking plugs for the test ports	—

C Tools & equipment

Oven to 150 °C, heat-resistant gloves, press fixture with a positive stop and a guide sleeve, outside micrometer 125–150 mm, bore gauge, hand pump with 0–6 bar gauge and shut-off valve, bucket of water, loop bench (B131), a ≥ 100 A low-voltage DC supply **or** the HC2 holding d-axis current, 4-wire milliohm meter, thermocouples, data logger.

D Manufacturing

- **Jacket (6061-T6):**
- bore Ø129.860 +0.025/0, Ra 0.8, cylindricity 0.01 mm, entry chamfer 2 × 15°, locating shoulder at one end;
- two-start helix, each start 6.0 wide × 3.0 deep, pitch 18 mm, lands ≥ 3 mm; deburr every groove edge (a burr cuts the O-ring);
- two O-ring grooves at each end per ISO 3601-2 for 18–25 % squeeze, with a leak-test port drilled between each pair;
- plenum grooves at both ends join the two starts to the ports; G1/8 ports spot-faced Ra 3.2 for bonded seals; cut the helix on a CNC lathe as a 2-start thread with an 18 mm lead and a 6 mm grooving insert, in several passes;
- coolant surfaces Type II anodised or trivalent chromate; the stator bore stays bare; the rear face flat to 0.05 mm for the HC2.
- **Housing sleeve (7075-T6):** bore H7 over the jacket lands, 15° O-ring lead-in, Ra 0.8 in the seal zones.
- **Stator:** potted and varnished per docs/13 §3.5; OD measured at 4 angles × 3 heights, 130 h7.

E Assembly

1. Measure stator OD and jacket bore; record the actual interference. **It must be 0.075–0.140 mm**; outside that, do not assemble.
2. Heat the jacket to 150 °C (20 min soak, ≤ 30 min total; 6061-T6 loses temper with longer exposure). Stator at room temperature.
3. Press the stator into the jacket in **one continuous stroke** against the shoulder, phase leads aligned to the slot. Hold under light load for 5 minutes while it cools.
4. At room temperature, check the stator tooth-tip bore: not more than 0.02 mm smaller than before (the shrink fit compresses the stack).
5. Pot the end turns into the jacket with the thermally conductive epoxy (≥ 1.5 W/m·K) so end-turn heat also reaches the coolant (docs/13 §3.9 step 4). Grease the EPDM O-rings with silicone grease, fit them, and slide the jacket into the sleeve without twisting. Secure it axially (shoulder + retaining ring or screws per the CAD).
6. Fit the port fittings with bonded seals to the fitting maker's torque. **Inlet at the bottom, outlet at the top when the robot stands**, so air purges upward.
7. Glue the outlet NTC into its boss with thermal epoxy. Plug the leak-test ports.

F Electrical

- NTCs (2 end turns + 1 outlet) to the HC2 NTC inputs; the outlet temperature also goes to the RT layer for the coolant model.
- Leak pad on the HC2 edge facing the inlet fitting → HC2 input **and** the safety MCU leak line (docs/05 §8: a leak at an HC2 trips the stop chain).
- Phase leads leave through the sealed slot on the side away from the ports.

G Testing (per jacket; R10 §7 item 2)

Test	Method	Pass
O-ring pair check	air at 2 bar into each leak-test port, channel open	≤ 0.01 bar drop in 5 min (each seal pair proven on its own)
Proof	fill the channel with water, isolate, 3 bar for 10 min	no weep (dry paper towel at both test ports and every joint)
Leak (docs/13 §3.9)	dry, air at 2 bar under water for 10 min; then 2 bar pressure decay for 10 min at stable temperature	no bubbles; ≤ 5 mbar drop
Flow and pressure drop	loop bench, 0.2–0.8 L/min	record Δp at 0.5 L/min (expected ≈ 1.4–1.8 kPa, docs/03 §9.3)
Thermal resistance (first on a K1 KM-130x45 stator if one exists, then every KM-130x50-LC)	rotor locked or removed; inject copper loss of 100, 150 and 215 W (≥ 100 A DC through two phases, or the HC2 holding d-axis current); coolant 0.5 L/min at 25 °C; winding temperature from resistance, R_T = R_20 · (1 + 0.00393 · (T − 20)); R_th = (T_winding − T_coolant,mean) / P_cu (docs/13 §3.9)	**≤ 0.25 K/W** meets the R10 rating. 0.25–0.30 K/W: accept (docs/13 pass limit) and derate the continuous rating by √(0.25 / R_th) in `calc`. **> 0.30 K/W: reject** (check potting voids and the fit)
Cleanliness	flush with deionised water until clear, then drain and cap	no debris in the filter

P06 Two hot-swap 16S4P battery packs with in-pack BMS and FET switch

Gate G0 (packs, bay)

Hazard: each 1.15 kWh pack can deliver ≈ 1.2 kA into a short and can burn violently. Build packs in the battery area (P00), with insulated tools, never alone, and never leave charging unattended. Build **pack A completely and pass its gate before you weld a single cell of pack B.**

Mk II values (`docs/generated/mass_budget.md` , `power_budget.md` ; architecture in docs/05 §3–§4, bay in docs/03 §10):

Item	Per pack	Robot (A + B)
Configuration	16S4P, 64 × Molicel INR-21700-P50B	2 packs; robot-side OR-ing and tie switch (KES-PPM, P06.1)
Voltage	59.2 V nominal, 67.2 V full, 48 V cut-off (3.0 V/cell); minimum loaded bus 52 V	same
Capacity / energy	19.4–20 Ah, **1.15 kWh** (≥ 1.12 kWh with minimum-capacity cells)	2.24–2.30 kWh, 1,951 Wh usable
Mass / envelope	5.9 kg budget; ≈ 150 × 98 × 185 mm in `params.py` (see §D note)	11.8 kg
Runtime	—	**4.9 h normal mixed, 4.3 h continuous walking** (16S8P row of the power budget)
Current	100 A Class T fuse inside; robot simultaneous peak 3.08 kW ≈ 59 A = **14.8 A/cell when one pack carries it** (P50B 60 A continuous)	7.4 A/cell with both
Internal resistance	≈ 51 mΩ (16 × 12.8 mΩ / 4); one-pack sag at 59 A ≈ 3 V	

A Visual reference

- Layout and series path (K1 single-pack drawing; the Mk II arrangement is in §D): `diagrams/battery_pack.svg`
- Power architecture (K1 drawing; Mk II single-line diagram in docs/05 §4.1): `diagrams/power_architecture.svg`
- Torso placement (two packs side by side at y = ±52 mm, low in the torso): `renders/17_torso_ghost.png`

Pack block diagram (identical in A and B):

```
cells 16S4P — B+ — 100 A Class T fuse — back-to-back 100 V FETs (driven by the BQ76952) — connector P+
          └ B− — 0.5 mΩ shunt ———————————————————————————————————— connector P−
KES-BMS16: BQ76952 (16S AFE) + BQ77216 (secondary OV, separate taps) + CAN-FD MCU ── connector CAN-FD + wake
pilot loop through the connector (pilot pin ≥ 2 mm shorter) → FETs stay off unless seated AND the robot or charger
has completed the CAN handshake  ⇒  a removed pack has dead terminals
```

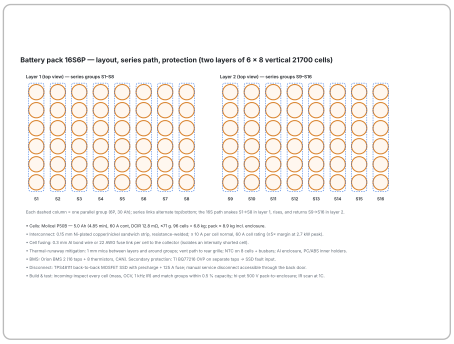


DIAGRAM Layout and series path (K1 single-pack drawing; the Mk II arrangement is in §D)

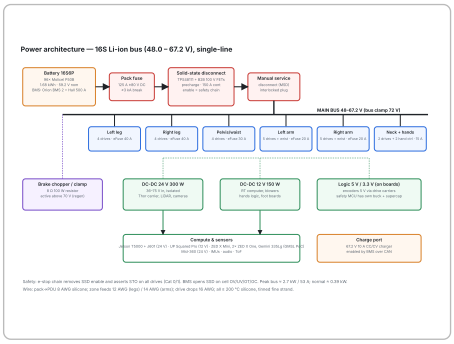


DIAGRAM power architecture



RENDER Torso placement (two packs side by side at y = ±52 mm, low in the torso)

B Parts list (per pack)

BOM	Item	Qty	Spec
B028	Molicel INR-21700-P50B	64 (+ matching spares from the 200-cell buy)	5.0 Ah (4.85 min), 60 A, ≤ 71 g [ds R3]; authorised distributor, lot-numbered invoices; one lot per pack
B029	**KES-BMS16** per-pack BMS: TI **BQ76952** 16S AFE, TI **BQ77216** secondary OV on separate taps, CAN-FD MCU, in-pack back-to-back 100 V N-FET switch, 0.5 mΩ shunt, 8 cell NTC + 1 FET NTC, pilot/wake	1	custom, TI TIDA-010208 pattern (docs/05 §3.2)
B031	Pack fuse **100 A UL Class T** + holder (Littelfuse JLLN-100 / Eaton Bussmann JJN-100 class)	1 + 1 spare	**verify ≥ 125 V DC rating and ≥ 20 kA DC interrupting on the datasheet**; prospective fault ≈ 1.2 kA
B032	Enclosure: 6061 machined shell (≤ 1.3 kg), 1 mm mica liners, two-piece PC or PC/ABS cell holders (never PLA), Ni-plated Cu busbars, carry handle, IP54/IP67 face gasket, vent path	1	clear anodise
B032	Parallel collectors 0.15 mm Ni-plated Cu + 0.15 mm pure Ni sandwich; series links ≥ 4 mm² Ni-plated Cu (e.g. 0.3 × 15 mm); per-cell fuse link (0.3 mm Al bond wire or 22 AWG fuse link)	1 set	
B113	In-pack thermal-event sensor (vent gas / pressure rise)	1	docs/06 §1
B111	Pack half of the hot-swap connector: blind-mate, keyed, ≥ 80 A per power contact, ≥ 100 V DC, power make-first / break-last, pilot make-last / break-first (≥ 2 mm stagger), CAN-FD + wake, IP2X unmated	1	part not yet chosen (Anderson SBS with auxiliary contacts or Stäubli CombiTac class): confirm stagger and ratings on its drawing
—	10 kΩ NTCs, 22 AWG silicone tap wires (fused 1 A at the BMS), 8 AWG silicone main leads, Kapton tape, fish paper, foam-free compression pads, 15 W silicone heater film (docs/03 §10.2)	1 set	
—	Insulating cover for the pack connector, fitted whenever the pack is out of the robot	1	

C Tools

Battery spot welder (kWeld kit with a ≥ 1500 A source, or a laser-weld service), insulated hand tools (P00), 4-wire IR meter (1 kHz) or battery tester, precision scale (0.01 g), DC electronic load ≥ 100 A, the dock charger (B037) or a programmable 16S charger, thermal camera, insulation tester (500 V), oscilloscope with a current probe, a pack test harness that closes the pilot loop and talks CAN to the BMS.

D Manufacturing

- **Cell arrangement (docs/03 §10.1):** cells lie fore-aft, in two layers of 4 × 8. Each 4-cell column is one parallel group: 8 groups per layer, 16 in series. Series path: S1 at the front layer top, down the front layer to S8, doubled link to the rear layer, up to S16. Cell polarity alternates group by group; series links alternate between the outer face and the mid-plane. BMS, fuse and handle in the rear end cap.
- **Envelope note:** 2 × 4 × 8 cells fill the 150 × 98 × 185 mm box of `params.py` with ≈ 2 mm to spare; the end cap needs ≈ 25 mm more length. Confirm the pack CAD with the CAD owner before machining (docs/09 §11 H12).
- **Enclosure:** 6061 CNC, ≤ 1.3 kg including the end cap; keyed T-rail shoe on the floor, PTFE/POM wear strips on the sides, over-centre cam-lever boss on the end cap (P06.1), vent path to the rear grille. Mica on every inner face.
- **Cell holders:** two-piece 4 × 8, bore = measured cell OD + 0.2 mm.
- **Busbars:** cut collectors and series links; pre-bend series links into a U across each junction.

E Assembly (per pack)

1. **Incoming inspection, 100 % of cells:** mass (reject > 71.0 g or < 68.5 g), OCV (all within ±20 mV), 1 kHz IR (reject > 10 % above the median), wrap and top insulator undamaged. Log by serial.
2. **Matching:** sort 64 cells into 16 groups of 4 with equal summed capacity (IR and OCV as proxies, or a 1C capacity test on a sample).
3. **Front layer (S1–S8):** cells into the holder with alternating group polarity per §D. Check polarity twice with a meter before any welding.

4. **Weld:** - parallel collectors first (pure nickel on the cell, copper over it), then series links; - 3–4 welds per cell contact; test-weld a sacrificial cell and peel-test it (the nugget should tear the nickel); - per-cell fuse link between each cell and its collector.
5. **Rear layer (S9–S16):** same. 1 mm mica sheet between the layers. Join S8 → S9 with a doubled series link.
6. **Cell taps:** solder each group's 22 AWG tap (fused 1 A at the BMS end) to its collector, in order B0…B16. **Verify the tap voltages ascend in ≈ 3.6 V steps before plugging into the BMS.** Separate taps for the BQ77216.
7. **Thermistors:** 8 on cells (centre, corners, near the series links) and the FET NTC, Kapton and thermal paste; heater film on the outer face of the cell block; thermal-event sensor in the end cap.
8. **Main path:** B+ → fuse → in-pack FETs → connector P+; B− → shunt → connector P−. 8 AWG, ≤ 150 mm, twisted.
9. **End cap:** BMS board, FET bank bolted to the aluminium (thermal pad), connector half with its gasket, pilot and CAN-FD to the connector signal pins. Close with foam-free compression pads holding the cell stack.
10. Fit the insulating connector cover. Label: pack serial, cell lot, build date, capacity (after §G).

F Electrical

- **BMS settings (per pack):**
- cell OV 4.20 V, UV 3.00 V, balance from 4.05 V at 10 mV delta;
- charge 0–45 °C, discharge −10–60 °C; below 0 °C no discharge above C/5 until the heater has warmed the pack;
- DCL 80 A continuous (the robot-side LM5069 limit, docs/05 §4.6), derated below 3.3 V/cell and above 50 °C; CCL 20 A (5 A/cell standard charge), ramping to 0 at 4.15 V; short-circuit detection set above the robot-side limits so the robot side clears first;
- CAN-FD to the KES-PPM and the safety MCU bus; reports V, I, T, SoC, SoH, CCL/DCL, fault flags; keeps its own serial and cycle log.
- **Pack switch enable = AND** (pilot loop closed, CAN handshake with the robot or charger complete, BQ76952 protections OK, BQ77216 OK). Pulling the pack breaks the pilot first, so a removed pack always has dead terminals.
- OR-ing, hand-off, tie and inrush control are on the robot side (KES-PPM, P06.1, docs/05 §4.2–§4.4). The pack does not need its own precharge.

G Testing (pack gate, every pack)

Test	Pass
Insulation, pack terminals → enclosure, 500 V DC (pack switch closed on the test harness)	> 100 MΩ
Dead terminals: pack unseated (no pilot, no CAN), 10 kΩ load across P+/P−	< 1 V
Pilot without handshake: pilot closed, CAN silent	switch stays open
Tap voltages / BMS readings	all 16 groups within 10 mV of a reference meter
Short-circuit on the pack switch into a 10 mΩ shunt (12 V supply first, then the pack)	BQ76952 SCD opens the FETs; FETs undamaged; fuse intact
Discharge 1C (20 A) to 3.2 V/cell	max cell < 45 °C on the thermal camera; groups within 30 mV
Peak discharge 80 A for 10 s, then 59 A for 2 min	FET temperature rise < 40 K; fuse and connector < 70 °C
Charge CC/CV to 4.20 V/cell under BMS control (dock charger at 10 A per pack, then the 20 A CCL)	balancing works; no cell > 4.21 V
Capacity to 3.0 V/cell	≥ 1.10 kWh
BMS trips (OV/UV by tap override, OT with a thermistor in hot water, OC on the load)	FETs open; CAN reports the fault
Handle pull test, 4 × pack weight (24 kg) for 1 min	no deformation

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P06.1 — Hot-swap battery bay and pack power path

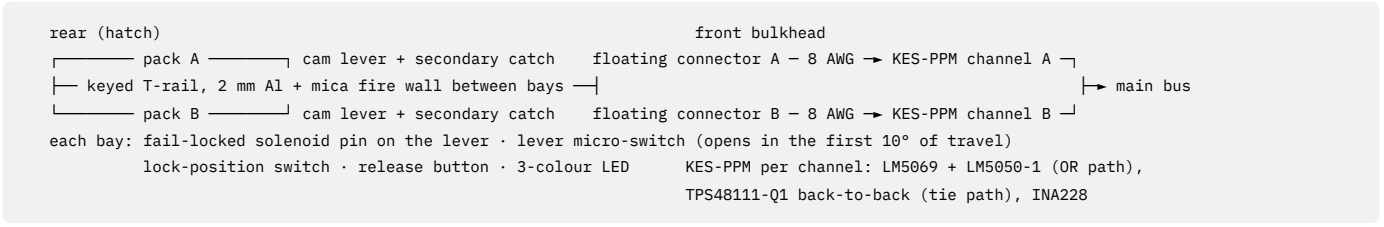
The bay holds packs A and B side by side in the lower torso (pack centres x = −30, y = ±52, z = +130 mm). Packs slide in fore-aft from the back through gasketed hatches, on a keyed T-rail, and dock on a floating connector at the front bulkhead (docs/03 §10). The robot-

side power path (KES-PPM, B030) is built and bench-tested here with a DC load and installed in the torso in P08.

Programme rule: until gate G6, a hot swap is done **only with the robot supported in the gantry harness (or on the leg stand) and every drive in STO**; logic bus, computers and coolant pumps stay powered. The standing swap of docs/03 §10.3 / docs/05 §4.5 is a G7-onward capability. Two people for every swap.

A Visual reference

- Torso placement: `renders/17_torso_ghost.png`
- Bay and power path (top view):



B Parts list

BOM	Item	Qty
B112	Bay mechanics per bay: 7075 channel pair tied into the spine and lower rib (B064), PTFE/POM guide strips, keyed T-rail; over-centre cam lever with hardened keeper pin (Southco draw-latch class) and a separate secondary catch; fail-locked solenoid lock pin; lever micro-switch (Omron D2SW sealed class) and lock-position switch; release button and 3-colour LED; keyed manual override behind the back cover	2
B111	Robot half of the hot-swap connector on a ± 1 mm / ± 1° floating mount with 2 guide pins that engage before the contacts	2
B030	**KES-PPM** pack power-path board: per channel TI **LM5069** hot-swap controller + TI **LM5050-1** ideal-diode controller (OR path), TI **TPS48111-Q1** back-to-back tie switch, TI **INA228** monitor, 100 V ≈ 2 mΩ FET banks (Nexperia PSMN2R0-100SSF class), local CAN-FD MCU, solenoid drivers, hardware unlock AND; plus the dock charge-inlet channel	1
B079	Amphenol SurLok Plus 5.7 mm pair per channel (PPM side), 8 AWG tinned silicone ≤ 0.4 m per pack, twisted	2 sets
—	Back-cover switch (NC) wired into the e-stop chain (docs/05 §8)	1

C Tools

As P06, plus a DC electronic load ≥ 60 A, an oscilloscope with two current probes, a regenerative DC source or a second supply with a series diode for the back-feed test, a stopwatch.

D Manufacturing

- Bay channels: pack-to-guide clearance 0.3–0.5 mm, 3 mm lead-in chamfers, the T-rail keyed so a pack enters one way only; either pack fits either bay. 2 mm aluminium + mica fire wall between the bays.
- Connector carrier: ± 1 mm radial / ± 1° float on rubber-bushed shoulder screws; guide pins longer than any contact.
- Cam-lever keeper positioned so the last 8 mm of lever travel draws the pack home onto the connector with ≈ 300 N preload; the lever micro-switch opens in the first 10° of opening travel.

E Assembly

1. Mount guides, T-rails, levers, keepers, solenoid locks and switches on the bay frame (in the torso during P08; on a bench frame for this test).
2. Mount the floating connector carriers; set the float with feeler gauges.
3. Wire each pilot loop: lever micro-switch (closed only when the lever is home) in series with the connector pilot pins.
4. 8 AWG pairs from each connector to its KES-PPM channel (SurLok Plus 5.7 mm), P-clipped every 80 mm.
5. CAN-FD from both connectors to the PPM MCU and the robot CAN-FD bus; 120 Ω terminations only at the bus ends.

F Electrical (docs/05 §4)

- **Modes:** *Diode* (default): both OR paths on, tie paths open; the higher pack carries the load; no regeneration into the packs (the chopper takes it). *Tied*: both tie switches closed only when both packs are healthy, |V_A − V_B| < 0.5 V at rest and SoC < 98 %; equalisation current ≤ 4.4 A. *Single pack*: the RT controller caps actuator power at 1.8 kW. *Swap*: actuator power capped at 1.5 kW (in this programme: drives in STO).
- **Insertion:** power pins mate (pack switch still open) → pilot closes, BMS wakes, CAN handshake → BMS closes its switch → PPM enables the LM5069 with its power limit; a higher pack takes over gradually, a lower pack waits reverse-biased → tie only later, if the tie conditions hold.
- **Removal interlock (hardware AND; firmware cannot override):** the solenoid of pack X retracts only if pack Y is present, V_Y ≥ 54 V, Y's LM5069 PGOOD is high, X's path current < 1 A with its tie open, and firmware says "swap permitted". If the lever is forced, the pilot breaks ≥ 2 mm (≥ 6 ms at 0.3 m/s) before the power pins, against < 0.1 ms to shed the channel: the power contacts always break at zero current.
- **Fusing coordination:** TPS48111 tie short-circuit 150 A (1.2 µs) / overcurrent 80 A 10 ms; LM5069 breaker ≈ 120 A, current limit 80 A with a fault timer for 2 s peaks; in-pack 100 A Class T fuse as backup.
- **Regeneration:** into tied packs (limited by the BMS CCL over CAN); otherwise the brake chopper (2 × 5 Ω 100 W in parallel, B036) clamps at 70 V on / 68 V off.
- **Power-off for service:** keyed main switch (PPM sheds every path) **and** pull both packs. There is no mechanical service disconnect in Mk II.

G Testing (bay gate, part of G0)

Test	Pass
Interlock refuses: request a release of A with B absent, B below 54 V, or A carrying > 1 A	solenoid stays locked in every case
Forced lever: open A's lever with the solenoid overridden while A carries 20 A (bench load)	A's path current < 1 A before the connector moves; no arc marks on the contacts
Pilot stagger: slow withdrawal, continuity tester on pilot and power pins	pilot breaks ≥ 2 mm before the power pins
Hot swap on a 15 A load (idle power), both packs tied: release A, pull, insert a fresh A 5 V above B	load uninterrupted; bus dip < 2 V; logic bus (24 V / 12 V) does not reset; A takes over through the LM5069 ramp; tie closes only after convergence
Insert a pack 5 V **below** the bus	its OR path stays reverse-biased; < 0.5 A until the bus falls to it
Equalisation: tie at 0.5 V and 1.0 V difference	≈ 4.4 A and ≈ 8.9 A (docs/05 §4.3 table) within ±20 %
One pack alone at 59 A for 2 s (at 20 % SoC)	sag ≈ 3 V; bus stays ≥ 52 V; no trip
Back-feed 10 A with both packs in diode mode	no current into the packs; chopper clamps the bus ≤ 72 V
Swap time, robot in the gantry with drives in STO, two people	≤ 3 min from request to both packs online
Repeat the hot swap 50 × (docs/05 §14)	no arc marks, no under-voltage events, connector resistance change < 10 %

P07 Legs: hip, thigh with knee four-bar, shank with ankle push-rods, foot, leg coolant lines

Gate G2

Build the **left leg completely and pass G2 before building the right leg.** Before machining any leg part, clear holds H1, H4, H5, H7 and H9 in docs/09 §11: the K1 thigh plate and knee hub cannot carry the Mk II torques as detailed.

A Visual reference

- Hip + knee joint assemblies (ghost): `renders/09_hip_knee_joints_ghost.png`
- Ankle + foot (ghost): `renders/16_foot_ankle_ghost.png`
- Skeleton rear: `renders/11_skeleton_rear.png`
- Knee four-bar geometry and G(θ): `diagrams/knee_linkage.svg`, `data/knee_fourbar.json`
- Ankle push-rods: `diagrams/ankle_linkage.svg`
- Drawings: `cad/out/drawings/l_hip__yaw_roll_yoke.svg`, `l_hip__roll_pitch_bracket.svg`, `detail__thigh_plate.svg`, `detail__knee_hub.svg`, `l_shank__shank_upper_fitting.svg`, `detail__ankle_cross.svg`, `detail__foot_plate.svg`, `l_thigh__knee_pushrod.svg`
- Leg limb interface, coolant loops through the hip: docs/03 §11 and §13.1

Leg chain (left), with the joint frame origins in the zero pose:

Joint	Actuator	Location
hip yaw	KA-90P (stays on the pelvis; the leg detaches at its output, docs/03 §13.1)	pelvis bowl, axis vertical, 75 mm above the hip centre (0, 105, 990 mm)
hip roll	KA-110P	axis x through the hip centre, actuator body at x = −118 mm (gluteal)
hip pitch	KA-130H (jacketed, ≈ Ø145)	axis y through the hip centre (0, 105, 915)
knee actuator	KA-130K (jacketed)	axis y at z = 755 (160 mm below the hip); ≈ 15 mm running clearance to the hip actuator
knee joint	—	axis y at z = 495
ankle actuators	2 × KA-90P	at z = 400 (A, lateral rod) and z = 300 (B, medial rod)
ankle cross	—	z = 75; sole at z = 0



RENDER Hip + knee joint assemblies (ghost)



RENDER Ankle + foot (ghost)



RENDER Skeleton rear

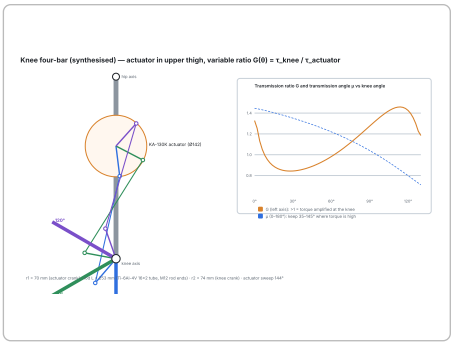


DIAGRAM Knee four-bar geometry and G(θ)

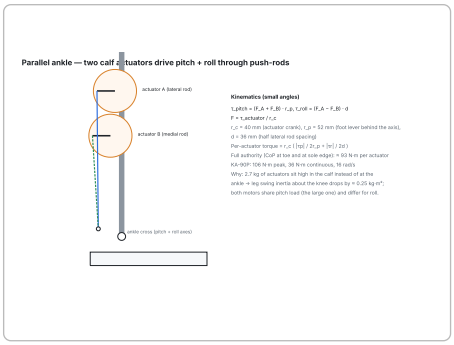
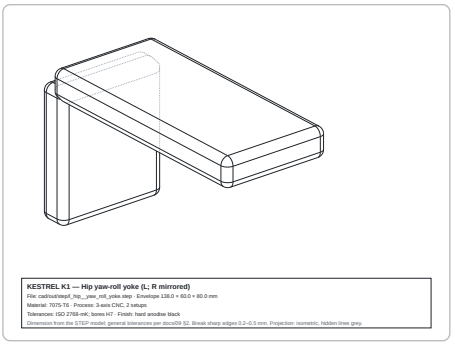
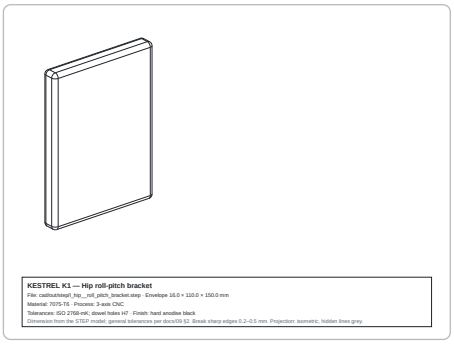


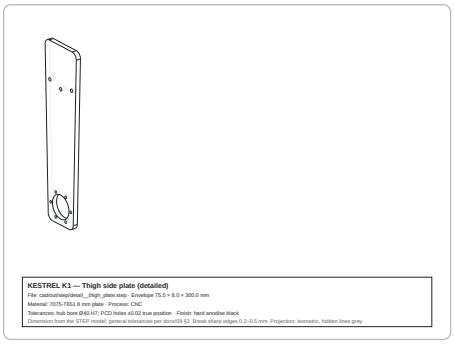
DIAGRAM Ankle push-rods



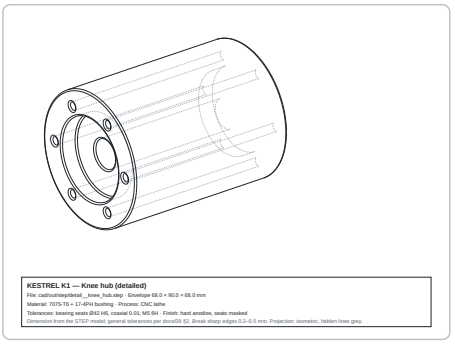
DRAWING l hip / yaw roll yoke



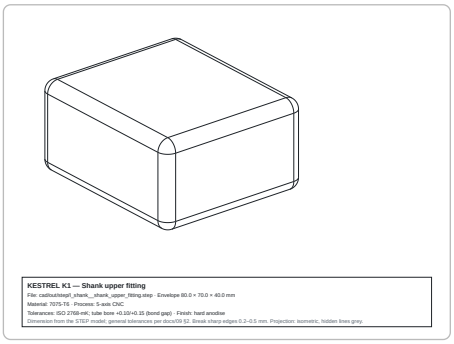
DRAWING l hip / roll pitch bracket



DRAWING detail / thigh plate



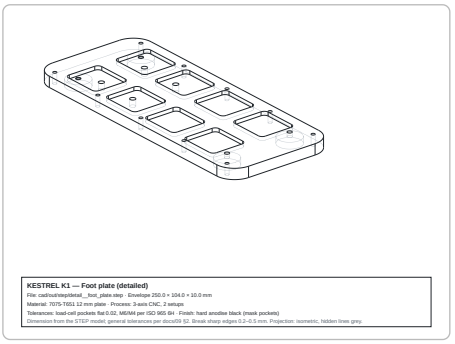
DRAWING detail / knee hub



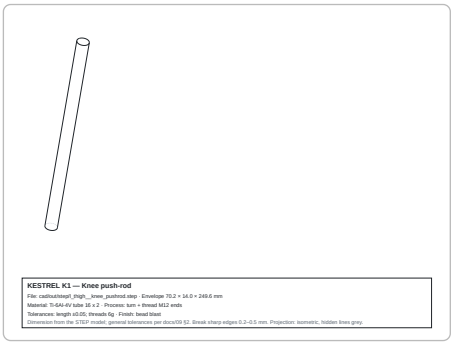
DRAWING l shank / shank upper fitting



DRAWING detail / ankle cross



DRAWING detail / foot plate



DRAWING l thigh / knee pushrod

B Parts list (one leg)

BOM	Item	Qty
(P03/P05)	KA-90P ×2 (ankle A, B), KA-110P ×1, KA-130H ×1, KA-130K ×1, tested; hip-yaw KA-90P is part of the pelvis (P08)	5
B065	Yaw–roll yoke (carries the leg-side V-flange), roll–pitch bracket	1 + 1
— (docs/03 §13.1)	Leg limb interface: pelvis-side and leg-side V-flanges (Ø70 H7/g6 spigot, 1 round + 1 diamond dowel, face O-ring), 3-segment 17-4PH V-band with a single M8 12.9 T-bolt and spring safety catch, torque-limiting T-handle (10 N·m), blind-mate service plate (power, EtherCAT in/out, STO, leg ID, 2 dry-break couplings)	1 set
B066	Thigh plates (medial + lateral), CF tube Ø40 × 2 × 150, end fittings; **hip adapter ring with the full 12-hole PCD 105 pattern (hold H4)**	1 set
B067	Knee push-rod: Ti-6Al-4V, Ø16 × 2 mm section with **solid threaded ends** (see §D), L = 252.9 mm between ball centres; rod ends SKF **SIKAC 12 M** (RH) and **SILKAC 12 M** (LH), PTFE-lined; 2 jam nuts M12 × 1.25	1

BOM	Item	Qty
B068	Actuator crank (r1 = 70 mm), knee crank (r2 = 73.7 mm at 229.4° in the shank frame) **made integral with the knee hub (hold H5)**, knee bearings per the revised hub drawing	1 set
B069	Shank upper fitting, CF tube Ø40 × 2 × 230, lower fitting, 2 ankle push-rods (Ti 10 × 1.5 with solid M8 ends) with SKF SIKAC 8 M / SILKAC 8 M rod ends, 2 actuator cranks r_c = 40 mm	1 set
B070	Ankle cross (Ti-6Al-4V), 4 × INA HK1212 drawn-cup needle bearings, **2 × Ø12 h6 pins in 100Cr6, 60–64 HRC, ground** (not Ti or 17-4PH: hold H13), foot plate, foot levers (r_p = 52 mm, ±36 mm lateral), replaceable NBR sole	1 set
B051	4 × load cells + foot board	1 set
B077	Ankle TPU gaiter	1
B125, B126	Coolant: 1 dry-break pair on the service plate (supply + return), 6 mm ID EPDM/Norprene hose ≈ 2.6 m, G1/4 fittings, clamps, rolling-loop C-trough for the hip-pitch crossing, U-loop clip at hip roll	1 set
B127, B128	Flow sensor for this leg branch (in the torso, P08.1); NTC probes at the knee-jacket inlet and the hip-jacket outlet	1 + 2
—	Polyurethane 90A bumpers for every joint end stop; knee stops sized for ≈ 31 J at the actuator (docs/03 §6)	1 set
B081	Fasteners 12.9: M5 (KA-130 flanges, 12 + 12; KA-110P flanges, 12 + 12), M4 (KA-90P, 8 + 8), M6 (foot clevis, 4), M4 (sole carrier, 12), M3 (shell bosses); Ø6 / Ø4 dowels; HeliCoils; 3M DP490 structural epoxy	set

C Tools

Torque wrenches (1–25 N·m), torque-limiting T-handle for the V-band, dial indicators, height gauge, V-blocks, bonding fixture for the CF tubes (printed alignment jig), digital protractor (0.05°), leg stand (P01), loop bench with dry-break halves (P01), hand pump with gauge.

D Manufacturing

- **CF tube bonding (thigh and shank):** 1. Grit-blast the 7075 fitting bores (Ra 3–5 µm) and abrade the tube ends (P120). 2. Clean with IPA. 3. DP490 with a 0.10–0.15 mm bond gap (glass-bead shims in the adhesive), 25 mm engagement. 4. Hold in the printed alignment jig so the fitting bores are coaxial within 0.05 mm and the clocking is correct. 5. Cure 24 h, then post-cure at 60 °C for 2 h. 6. Proof-test one coupon per batch (docs/09 §9).
- **Knee push-rod:** a Ø16 × 2 tube has a 12 mm bore, so it cannot be tapped M12 (tap drill 10.2 mm), as K1 specified. Machine it from Ø16 Ti-6Al-4V bar instead: 1. bore Ø12 through the middle, leaving 25 mm solid at each end; 2. turn M12 × 1.25 male threads, 24 mm long, RH at one end and LH at the other; 3. stress-relieve (595 °C, 2 h, inert), dye-penetrant inspect; 4. set the length to 252.9 ± 0.1 mm between rod-end ball centres; lock with the jam nuts at 25 N·m, Loctite 243. Ankle push-rods the same way in Ø10 bar with M8 × 1.25 ends (jam nuts 15 N·m).
- **Ankle pins:** 100Cr6, harden 60–64 HRC, grind Ø12 h6, Ra ≤ 0.2; retaining screw groove before hardening.
- **Foot sole:** cast NBR/PU tread 10 mm in a printed mould, bonded to a 1 mm aluminium carrier screwed to the foot plate (12 × M4), so it can be replaced.

E Assembly (left leg)

1. **Pelvis-side V-flange** onto the hip-yaw KA-90P output (P08), or onto the stand carriage plate in the same pattern: 2 dowels, 8 × M4 12.9 at 4.5 N·m, Loctite 243.
2. **Leg-side V-flange** onto the yaw–roll yoke (dowels, 12.9 screws at the structural torque).
3. **Hip roll actuator** (KA-110P) into the yoke at x = −118 mm, output pointing +x. **Base flange 12 × M5 12.9 on PCD 115 at 8.5 N·m**, Loctite 243 (K1's 10 × M4 base would slip at the Mk II torque).
4. **Roll–pitch bracket** onto the KA-110P output: 2 × Ø6 dowels, **12 × M5 12.9 on PCD 100 at 8.5 N·m**, Loctite 243. Check the roll axis passes through the hip centre: dial indicator on a test bar through the KA-130H seat, rotate the roll, runout ≤ 0.1 mm.
5. **Hip pitch actuator** (KA-130H) into the bracket, coaxial with the hip centre, output facing medially. Base 12 × M5 12.9 at 8.5 N·m. Fit the jacket hoses before the thigh goes on (access is lost afterwards).
6. **Thigh** (sequence depends on the detailed CAD, hold H9): hip adapter ring onto the KA-130H output (2 × Ø6 dowels, 12 × M5 12.9 at 8.5 N·m), thigh plates to the ring, CF tube

assembly between the plates, then the **KA-130K** at z = 755, output lateral, base 12 × M5 at 8.5 N·m. Torque every screw before the next part blocks its access, and witness-mark it.

7. **Knee hub** (crank integral) between the lower plate ends, bearings per the revised hub drawing; set the preload with the shim stack so the hub turns with 0.2–0.4 N·m drag and no play.
8. **Shank:** shank upper fitting onto the knee hub (pattern per the revised hub drawing, hold H5). Mount both ankle actuators on the fitting brackets (A at z = 400 output lateral; B at z = 300 output medial).
9. **Knee four-bar, zero setting:**
10. Set the knee to 0° (straight) with the Ø4 zero pin through the knee hub and plate.
11. Command the KA-130K to its zero (output encoder), so the actuator crank sits at **φ0 = 277.1°** in the thigh frame.
12. Fit the actuator crank (r1 = 70 mm) on the KA-130K output flange (12 × M5 12.9 at 8.5 N·m, dowels).
13. Install the push-rod and adjust its length (LH/RH threads) until both zero pins slide in freely. Torque the jam nuts.
14. **Check the knee ratio** at 0°, 40°, 80°, 120° from the actuator and knee encoders: the incremental ratio must match G(θ) in `data/knee_fourbar.json` within ±3 % (1.33 at 0°, 0.87 at 40°, 1.17 at 80°, 1.43 at 120°).
15. **Ankle:**
    1. Ankle cross into the shank lower fitting (pitch pin) and the foot clevis (roll pin, 4 × M6 12.9 at 14 N·m into the foot plate), HK1212 bearings pressed into the cross (Ø18 N6), pins greased, retaining screws with Loctite 243.
    2. Fit the push-rods from the actuator cranks (r_c = 40 mm) to the foot levers (52 mm behind the axis, ±36 mm lateral).
    3. Zero: foot flat on a surface plate, both ankle actuators at encoder zero, rod lengths adjusted until the foot is level within 0.2° in pitch and roll.
16. **Hard stops:** fit the polyurethane bumpers at each joint end of range and check they contact 1–2° before the linkage limits. For the knee, put the stops on the **actuator crank** side so the actuator's own torque into a stop does not load the push-rod (hard stop at 132° knee).
17. **Coolant routing (docs/03 §11):** supply from the service plate → U-loop behind the hip-roll actuator → rolling loop in the C-trough concentric with the hip-pitch axis (radius 85 mm) → **knee jacket first** → back across the hip pitch → **hip-pitch jacket** → return to the service plate. Bend radius ≥ 30 mm, clips every 80 mm, grommet at every edge, hoses never above an HC2. Fit the knee-inlet and hip-outlet NTCs.
18. **Foot board** with 4 load cells in the foot plate pockets; sole carrier on top; ankle gaiter.

F Electrical

- **Leg zone feed:** 10 AWG silicone pair from the PDU leg eFuse (60 A fast / 25 A I²t, docs/05 §5), SurLok Plus 3.6 mm at the service plate. docs/05 routes it through the hip hollow shaft; with the Mk II gears there is no such bore (hold H2), so run it in the hip service loop alongside, and separated from, the coolant hoses.
- Drops: HC2 (hip pitch, knee) 12 AWG with XT60H, ≤ 0.5 m; Everest (hip roll, ankle A/B) 16 AWG with XT30. HC2 12 V aux 2 × 20 AWG (Gecko-MT 2-pin).
- EtherCAT order: hip roll → hip pitch → knee → ankle A → ankle B (J-LINK OUT → IN); the hip yaw sits upstream on the pelvis.
- Foot board (load cells + IMU) → SPI bridge on the ankle-A carrier (default) or its own EtherCAT slave.
- Cables cross each joint in a service loop with 30 mm slack each side, clipped every 80 mm.
- **Knee torque limit (drive parameter table, interpolated in θ):** τ_act ≤ min(493 N·m, 535 / G(θ), F_rod,max · r2 · |sin μ(θ)| / G(θ)) with F_rod,max = 12 kN (rod end C0 ≈ 30 kN unverified / 2.5). docs/04's 535 / G(θ) rule alone still allows 21 kN in the rod at 130° (hold H7).

Knee angle	0°	20°	40°	60°	80°	100°	120°	130°
G(θ) / transmission angle μ	1.33 / 152°	0.86 / 142°	0.87 / 128°	0.98 / 112°	1.17 / 91°	1.38 / 66°	1.43 / 36°	1.18 / 20°
Actuator limit (N·m)	**312**	493	493	493	457	388	**364**	**256**
Joint torque available (N·m)	415	424	429	483	535	535	520	302

The calc owner must confirm the deep-squat (DS, ≈ 125°) and 50 kg lift (LM) point clouds sit under this table before G4.

G Testing (gate G2: leg on the stand)

Test	Pass
Torque audit: every flange screw re-checked at its set torque after the first powered session	no rotation; witness marks intact
Joint zeroing: five leg encoders (plus hip yaw on the stand) vs zero pins	±0.05°
Range of motion, slow (5 °/s), each joint, with hoses and cables fitted	full software range, no interference, no hose kink or rub, bumpers contact as designed
Kinematics: foot pose vs forward kinematics at 20 poses (height gauge / tracker)	≤ 1.5 mm
Knee ratio check	G(θ) within ±3 %
Knee torque limit: command 493 N·m at 0°, 120° and 130° against a locked foot	measured actuator torque follows the table ±5 %
Leg coolant: loop bench on the service-plate couplings, 0.5 L/min, 1.5 bar leak test with the leg cycled through its range 100 ×	no leak; flow ≥ 0.45 L/min at every pose; coupling re-mate dry
Hose-loop fatigue (start here, continue on the stand; docs/03 §11)	10^5 cycles before G4, 10^6 before G8, no chafe or leak
Gravity compensation, leg hanging	holds with < 10 % torque error vs the model
Loaded squat on the stand: counterweight sets 56 kg (half the 112.7 kg design mass), to 90° knee, 20 cycles	tracking error < 2°, temperatures logged, no noise
Mk II payload on the stand: 71 kg equivalent (half of 112.7 kg + 30 kg), 20 squats; 81 kg equivalent (half of 112.7 kg + 50 kg) single slow lift	within the torque table; KA-130 windings within the dyno model + 10 %; no slip marks
Single-leg stance loading: 113 kg equivalent for 60 s	KA-130K winding rise within 10 % of the dyno model
Ankle: CoP sweep heel → toe at full load	ankle actuators ≤ 129 N·m; load-cell CoP within ±5 mm of the target
Stair-step replay (ST scenario, ×0.5 speed, then ×1.0)	no limit violations; temperatures and coolant ΔT logged
Leg-zone current log during the ST replay	peak < 60 A, I^2t below the eFuse setting
Limb interface (docs/03 §13.1): 1.5 × 546 N·m bending, 260 N·m torque on the fixture	no 0.02 mm feeler entry; no slip
E-stop at each test	joints go limp (ASC damping on the KA-130), the stand catches the load

P08 Pelvis, waist, torso frame, battery bay, power distribution and compute

Gate G2/G3

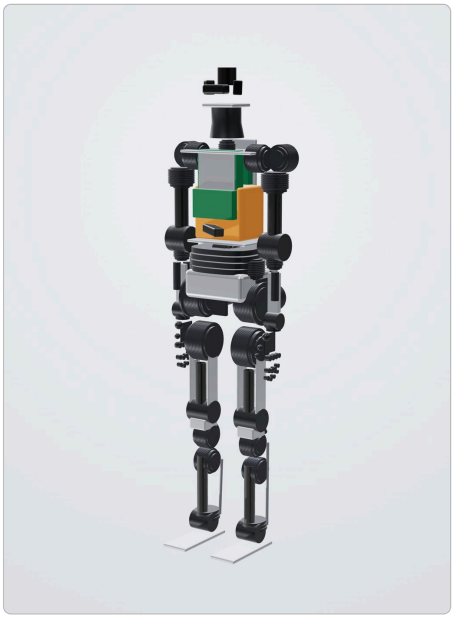
The liquid-cooling loop is built and tested in **P08.1** (below) once the torso and both legs are together.

A Visual reference

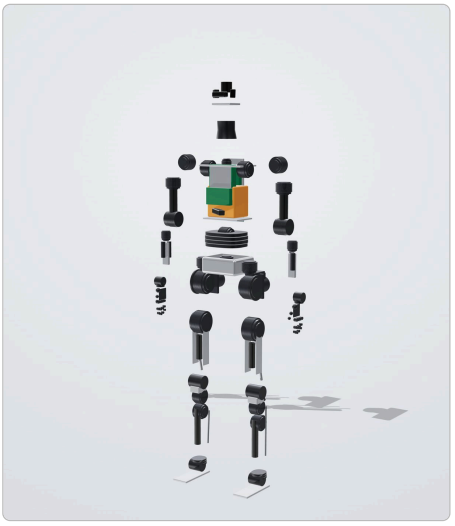
- Torso internals (K1 layout; Mk II adds the two-pack bay, the radiator in the upper back and the Thor cold plate): `renders/17_torso_ghost.png`
- Skeleton: `renders/10_skeleton_front.png` , `renders/13_exploded_skeleton.png`
- Drawings: `cad/out/drawings/pelvis__pelvis_bowl.svg` , `torso__spine_plate.svg`
- Power and comms (K1 drawings; Mk II single-line diagram in docs/05 §4.1): `diagrams/power_architecture.svg` , `diagrams/comms_topology.svg`
- Battery bay: docs/03 §10; loop layout: docs/03 §9.2; routing through the waist: docs/03 §11



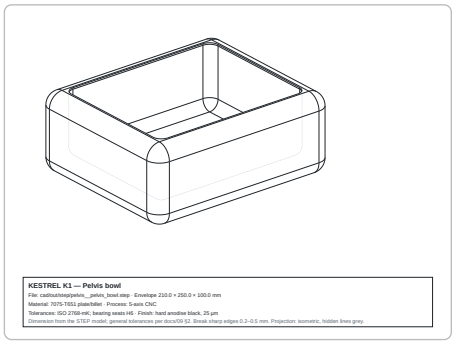
RENDER Torso internals (K1 layout; Mk II adds the two-pack bay, the radiator in the upper back and the Thor cold plate)



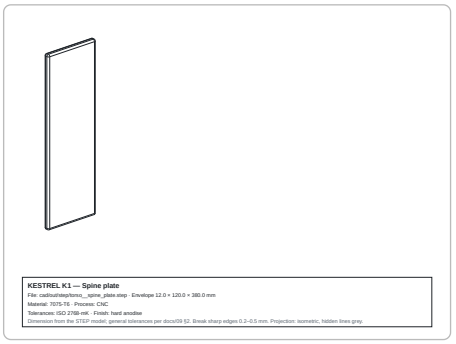
RENDER Skeleton



RENDER Skeleton



DRAWING pelvis / pelvis bowl



DRAWING torso / spine plate

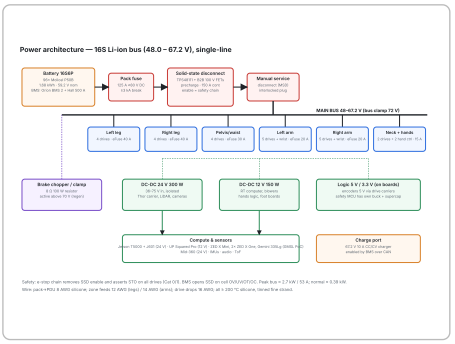


DIAGRAM power architecture

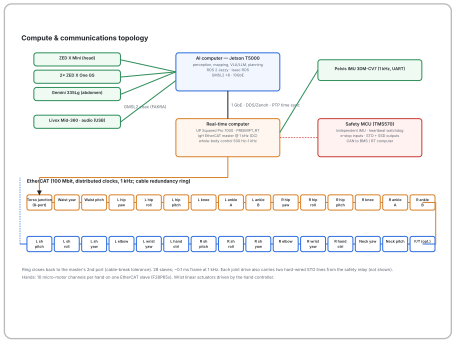


DIAGRAM comms topology

B Parts list

BOM	Item	Qty
B063	Pelvis bowl (7075, 5-axis) with an M12 lifting-eye insert	1
(P03)	KA-90P ×3 (hip yaw ×2 with pelvis-side V-flanges, waist yaw)	3
(P03)	KA-110P (waist pitch)	1
B064	Spine plate, upper/lower ribs, CF closeout panels, battery-bay channels and fire wall, M12 lifting-eye insert on the top rib	1 set
B112, B111, B030	Battery bay mechanics, robot-side connectors, KES-PPM (built and tested in P06.1)	1 set
(P04)	KH-25 shoulder-pitch actuators	2
(P04)	KH-14 neck-yaw actuator	1
B035, B036	PDU KES-PDU2 (logic / actuator bus split, 7 zone eFuses, brake chopper, telemetry), 2 × 5 Ω 100 W brake resistors	1 + 2
B033, B034	DC-DC 24 V 300 W and 12 V 150 W, 36–75 V input (input range must cover 67.2 V + the 70 V clamp: **verify**)	1 each
B038–B040, B124	Jetson T5000 + J601 carrier + NVMe; liquid cold plate for the Thor module	1
B041	UP Squared Pro 7000 (RT computer)	1
B042, B050	Safety MCU board + IIM-46230	1
B044	Safety relay with delayed-contact expansion + wired pendant + 2 on-robot buttons	1
B049	MicroStrain 3DM-CV7 on a 4-point silicone isolator (pelvis)	1
B047	Gemini 335Lg (abdomen, 35° down)	1
B080	2 electronics blowers (RT computer, PDU, DC-DCs) + MJF ducts + filters	1 set
—	Keyed main switch; back-cover switch; commissioning DC contactors for the two leg zones (Gigavac GX11 class, P02 §F)	1 + 1 + 2
(P06)	Battery packs A and B	2
B077	Waist TPU bellows	1
B081	Fasteners 12.9: 8 × M4 per KA-90P flange, **12 × M5 per KA-110P flange**, M4/M5 rib screws; dowels; HeliCoils	set

C Tools

As P07, plus thermal paste and pads, crimp tools (XT60H, XT30, SurLok, Gecko-MT, JST GH), a label printer, the gantry with both M12 lifting eyes fitted.

D Manufacturing

- **Pelvis bowl:** 5-axis 7075 with all seats (hip yaw ×2, waist yaw) bored in one setup for relative position ≤ 0.02 mm; hard anodise with the seats masked; key-locked M12 insert for the lifting eye.
- **Spine and ribs:** CNC 7075 plates; battery-bay channels with PTFE/POM strips and the keyed T-rail (P06.1 §D); 2 mm aluminium + mica fire wall between the bays; CF closeouts cut from 2 mm CF sheet and bonded/screwed.
- **Waist-pitch interface:** the spine plate carries the **12 × M5 PCD 100** pattern and 2 × Ø6 dowel holes for the KA-110P output (K1 drawings show 10 × M4 on PCD 85).

E Assembly

1. **Pelvis:** fit both hip-yaw KA-90P (outputs down, pelvis-side V-flanges already on) and the waist-yaw KA-90P (output up) into the bowl seats: 8 × M4 12.9 at 4.5 N·m each, Loctite 243. Mount the IMU isolator at (−60, 0, +50 mm) from the hip-centre midpoint, x-axis forward ±0.5°.
2. **Waist:** yoke onto the waist-yaw output (2 dowels, 8 × M4 12.9 at 4.5 N·m). KA-110P (waist pitch) into the yoke, axis y at z = 1085 mm: **base 12 × M5 12.9 on PCD 115 at 8.5 N·m**.
3. **Torso frame:** spine plate onto the waist-pitch output (2 × Ø6 dowels, **12 × M5 12.9 on PCD 100 at 8.5 N·m**, Loctite 243, witness marks), then the ribs and bay channels. KH-25 shoulder-pitch actuators on the upper rib at y = ±135 mm, outputs outward, axes at z = 1420 mm (8 × M4 at 4.5 N·m). Neck-yaw KH-14 on top.
4. **Battery bay:** install the P06.1 mechanics, floating connectors and KES-PPM; pilot loops and CAN-FD as tested in P06.1.

5. **Electronics bay:** - Jetson + J601 on the front of the upper ribcage, the module's thermal transfer plate on the liquid cold plate (thermal paste); the cold-plate hoses stay capped until P08.1; - RT computer and safety MCU on the rear of the spine plate (conduction pad), electronics blower stirring the sealed core; - PDU, PPM and DC-DCs beside the battery bay, **above and upstream of every coolant joint**, over nothing wet.
6. **Hang the torso** from both M12 lifting eyes in the gantry. **Mount each leg** at its V-band limb interface (docs/03 §13.1): leg on its trolley, offer it up axially so the guide pins engage the service plate, close the V-band, M8 12.9 T-bolt at 10 N·m with the torque-limiting T-handle, check the spring safety catch.
7. Leave the battery bay empty until P08.1 and §G are complete.

F Electrical (docs/05 §5–§9)

- **Actuator-bus zones (PDU eFuses):** leg L / leg R 60 A fast, 25 A I²t, 10 AWG pair (SurLok Plus 3.6 mm), in the hip service loop (hold H2); pelvis/waist 20 A, 14 AWG; arm L / arm R 20 A, 14 AWG through the shoulder (52 V to the forearm buck); neck 10 A, 16 AWG. Commissioning contactors in both leg-zone feeds (P02 §F).
- **Logic bus (not cut by the e-stop):** 24 V 300 W → J601 (XT30, 12 AWG), Mid-360, GMSL camera power through the J601; 12 V 150 W → RT computer, coolant pumps, radiator fans, blowers, foot boards, HC2 aux, LEDs; safety MCU on its own 12 V → 3.3 V supply with a 200 ms supercap hold-up.
- **Pack path:** each pack → 8 AWG → KES-PPM channel (SurLok Plus 5.7 mm) → PDU busbar. Keyed main switch to the PPM (sheds every path).
- **EtherCAT:** RT computer port 1 → torso junction → waist → left leg → right leg → arms → neck → back to port 2. 28 slaves per docs/05 §7 (30 if the KW-L2 boards are their own slaves, docs/09 §11 H11).
- **E-stop chain:** FORT receiver + pendant socket + 2 on-robot buttons + back-cover switch → safety relay (back of the spine). Instantaneous contacts → STO_A/STO_B to all 26 drives and the forearm-buck disables; delayed contacts (0.5–1.5 s) → actuator-zone eFuse enables. Safety MCU inputs: tilt, winding > 150 °C, bus OV/OC, RT heartbeat, comms loss, **leak at any HC2, coolant > 75 °C**.
- **CAN-FD power bus:** BMS A, BMS B (through the pack connectors), PPM MCU, PDU MCU, safety MCU, RT computer; 120 Ω at the ends only.
- **Grounding:** BUS– bonded to the chassis only at the PDU star point (10 mm² strap); mask the anodise at the bonding pads. Packs carry no chassis bond of their own.

G Testing (gate G2/G3 preparation)

Test	Pass
Insulation, bus → chassis with the star bond lifted, 500 V	> 10 MΩ (then refit the bond)
Power-up sequence with one pack: pack → PPM inrush → logic bus → DC-DCs → computers → zone eFuse soft starts	rails within ±3 %; no zone powers before its enable; leg zone soft-start ≈ 70 ms
Second pack inserted, both tied (P06.1 sequence on the robot)	tie closes only after convergence
eFuse trip test per zone with an electronic load	trips at the set current ±10 %
Stop sequence: e-stop while the logic bus is loaded	STO at t = 0 (< 10 ms), zone eFuses open at 0.5–1.5 s, commissioning contactors open, logic bus and pumps stay up, no computer reboots
EtherCAT: all slaves present, ring redundancy (pull one cable mid-run)	OP state maintained; 1 kHz DC, jitter logged < 20 µs
Waist yaw/pitch range with the hoses fitted (after P08.1)	full range, no hose strain, bellows clear
Safety MCU trip paths (tilt the torso > 35° in the gantry, kill the heartbeat, wet a leak pad)	Cat 1 sequence then STO; leak trips the chain

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P08.1 — Liquid-cooling loop (pumps, radiator, compensator, lines, quick-disconnects)

Layout, components and arithmetic: docs/03 §9; electrical: docs/05 §13; specification summary: docs/09 §6.2. The loop is **sealed and fully filled with a spring-diaphragm compensator, no reservoir open to air**, because the robot tilts, lies down and falls. Loop volume ≈ 0.47 L; torso part ≈ 1.5 kg.

A Visual reference

- Loop (docs/03 §9.2):

```
radiator 240 mm + 2 × 120 mm IP68 fans (upper back, wet duct)
  ↓ outlet (coolest)
compensator 40 mL @ 0.3 bar + fill/bleed valve + pressure sensor
  ↓
Thor cold plate
  ↓ split
pump L — check valve — supply L — waist clock-spring — hip-yaw loop — QD (limb interface) — hip-roll U-loop
       — rolling loop across hip pitch — KNEE jacket — back across hip pitch — HIP-PITCH jacket — return L ─┐
pump R — check valve — supply R — (right leg, mirrored) ──────────────────────────────────── return R ──────┤
  ↑ return manifold ←──────────────────────────────────────────────────────────────────────────────────────┘
```

B Parts list

BOM	Item	Qty
B120 / docs/03 §9.4	Radiator 240 × 120 mm, ≤ 30 mm core, ≈ 20 W/K, aluminium core (Alphacool NexXxoS ST30 240 class or aluminium equivalent; Koolance HX-720 is heavier)	1
B121	120 mm 12 V PWM fans with tach, **IP68** (San Ace waterproof class)	2
B119 / docs/03 §9.4	12 V brushless pumps, ≥ 0.5 L/min at ≥ 30 kPa, tach, glycol-compatible (TCS Micropumps M510 or Laing DDC class)	2 + 1 spare
—	Check valves ≤ 3 kPa cracking, EPDM	2
— (docs/09 custom)	Compensator 40 mL: EPDM membrane in a PA12 housing, spring-loaded, fill/bleed valve	1
—	Supply and return manifolds (6061)	2
B124	Thor liquid cold plate (6061)	1
B126	Hose 6 mm ID × 10 mm OD EPDM or Norprene ≈ 6 m, G1/8 / G1/4 fittings with bonded seals, stepless ear clamps, grommets	1 set
B125	Dry-break non-spill coupling pairs on the two leg service plates (supply + return each)	4 halves × 2
B127, B128, B130	2 flow sensors (one per branch), NTC probes (supply/return per branch), loop pressure sensor 0–2.5 bar	2 + 4 + 1
B129	Leak pads (HC2 ×4, pumps, manifold, cold plate) + 1 m leak-sensing cable on the torso floor under every fitting	1 set
—	Coolant: 40 % propylene glycol with an aluminium-safe (mixed-metal) inhibitor, premixed with deionised water (DOWFROST or an aluminium-engine OAT coolant at 40 %)	1 L

C Tools & equipment

Vacuum pump with a gauge reading ≤ 50 mbar and a fill rig (valve, sight tube, coolant container), hand pressure pump with a 0–4 bar gauge, refractometer (glycol concentration), drip trays and spill kit, smoke pen, the loop bench (P01) for pre-testing pumps and radiator, a padded table to lay the robot on.

D Manufacturing

- **Compensator:** PA12-MJF housing, EPDM membrane, compression spring set for 0.3 bar at half stroke, fill/bleed valve in the top; proof-test the housing at 3 bar.
- **Wet duct and radiator shroud:** MJF PA12 in the upper back, fans pulling air through the radiator to the rear grille; the duct drains outward.
- **Hose guides:** waist clock-spring former (4 hoses as a flat ribbon, 1.5 turns round the waist-yaw housing, inside the bellows); hip-yaw partial clock-spring in the pelvis shell; hip-pitch C-trough concentric with the axis at radius 85 mm (PA12-MJF); U-loop clips at waist pitch and hip roll. Bend radius ≥ 30 mm everywhere.
- **Manifolds:** 6061, anodised, ports for the pumps, check valves, sensors and drains.

E Assembly

1. Pre-test both pumps and the radiator on the loop bench: flow–pressure curve of each pump at the branch's ≈ 23 kPa; radiator UA with a known heat input.

2. Mount radiator, fans and shroud in the upper-back wet duct; compensator at the radiator outlet; pumps with check valves and the manifolds below the electronics, over the leak cable.
3. Connect the Thor cold plate (already on the module, P08 §E step 5).
4. Route the four hoses through the waist on the clock-spring former, then the waist-pitch U-loop, into the pelvis, and to the pelvis-side couplings of each limb-interface service plate. Clip every 80 mm; grommet every edge.
5. Mate both legs (P08 §E step 6): the service plates close the couplings.
6. **Dry leak test:** air at 1.5 bar with the pumps isolated, 15 min at stable temperature.
7. **Vacuum fill (docs/03 §9.8):** pull the loop to ≤ 50 mbar through the compensator valve, hold 10 min (rise ≤ 5 mbar), then let the premixed coolant in under vacuum so the jacket helices fill without trapped air. Pressurise to 0.3 bar.
8. Run both pumps 10 min with the robot laid on each side and on its back (padded table, gantry slack), then upright; re-check pressure and top up through the compensator.
9. Check the glycol concentration with the refractometer (40 ± 3 %). Label the fill date.

F Electrical (docs/05 §13)

- Pumps on the 12 V logic bus, each on its own PWM high-side switch with current sense and tach into the PDU MCU; fans PWM + tach. The logic bus stays up after a stop, so the pumps keep cooling the jackets.
- Flow sensors (I²C), NTCs, pressure sensor, pump tach → PDU MCU; leak pads and the leak cable → PDU MCU **and** the safety MCU (a leak at an HC2 trips the stop chain; coolant > 75 °C trips it too).
- Software responses (docs/03 §9.6): pump failure → that leg's KA-130 units to the air-only rating (0.64 ×); pressure < 0.15 bar → both legs to the air-only rating and stop at the next safe posture; supply > 65 °C → Thor to its 40 W mode, speed limit, then stop.
- Sensor and leak wiring runs above the coolant lines, never below.

G Testing (part of gate G3)

Test	Pass
Dry leak test, 1.5 bar air, pumps isolated, 15 min	≤ 0.01 bar drop
Vacuum hold before filling, 10 min at ≤ 50 mbar	rise ≤ 5 mbar
Flow per branch at nominal pump speed	0.5 ± 0.05 L/min each; branches within 10 % of each other
Joint-cycling leak test: waist yaw, waist pitch, hip yaw, hip roll and hip pitch through full range, 1,000 cycles, loop at 0.3 bar	pressure steady (± 0.02 bar after temperature correction); no damp on any leak pad; no hose rub
Radiator capacity: known heat into the loop (jackets by DC injection, P05.1 method) at 25 °C ambient	UA ≥ 18 W/K (docs/05 §13)
Thor at 100 W synthetic load for 30 min, 25 °C ambient	coolant supply ≤ 45 °C; module below its throttle temperature
Pump-failure simulation: stop pump L	flow alarm within 2 s; left-leg derate applied; check valve stops back-flow through the stopped branch
Leak detection: damp cloth (deionised water) on each pad and on the leak cable	each input reports; HC2 pads trip the chain
Coupling re-mate: unmate and re-mate a leg 10 ×	≤ 0.5 mL lost per cycle; compensator absorbs it; 2 bar decay test of the couplings after 100 mates (docs/05 §14)

P09 Arms: shoulders, upper arms, elbows, forearms, wrists

Gate G2

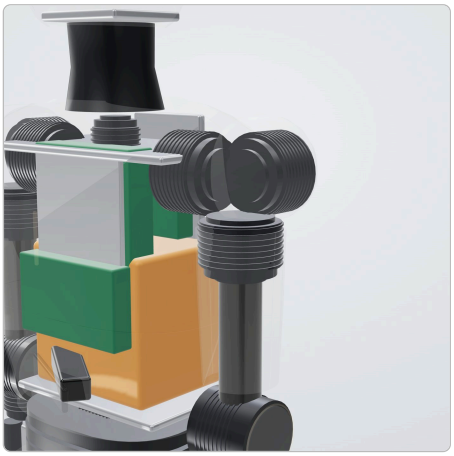
The wrist pitch/roll actuators are built in **P09.1 KW-L2** (below) before step 4.

A Visual reference

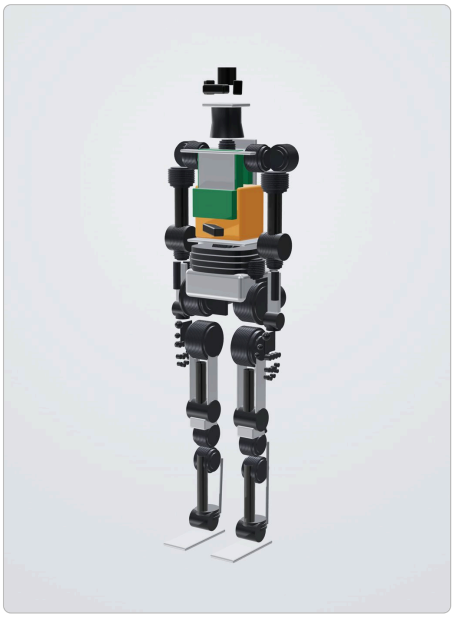
- Shoulder (ghost): `renders/15_shoulder_ghost.png`
- Skeleton: `renders/10_skeleton_front.png`
- KH-25 cutaway (K1 strain-wave version): `renders/20_KH25_cutaway.png`
- Arm limb interface: docs/03 §13.2

Arm chain (left), zero pose (arm hanging, palm facing the thigh):

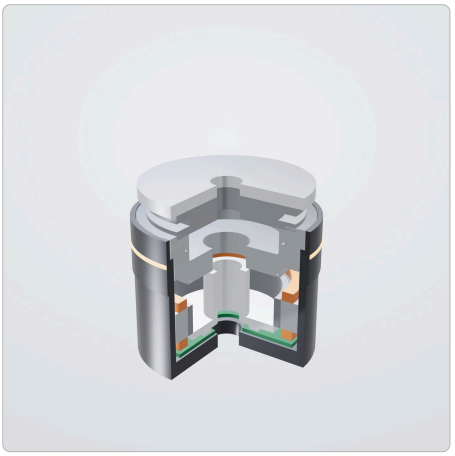
Joint	Actuator	Location
shoulder pitch	KH-25 (KH-C25 cycloid, limit 160 N·m)	in the chest, axis y at (0, 135, 1420)
shoulder roll	KH-25	axis x, body behind the shoulder centre at x = −45
shoulder yaw	KH-20 (KH-C20 cycloid, limit 105 N·m)	axis along the arm at z = 1340
elbow	KH-20	axis y at z = 1120
wrist yaw	KH-14 (limit 23 N·m)	axis along the forearm at z = 1072
wrist pitch/roll	2 × KW-L2 linear actuators	z = 985; wrist cross at z = 860



RENDER Shoulder (ghost)



RENDER Skeleton



RENDER KH-25 cutaway (K1 strain-wave version)

B Parts list (one arm)

BOM	Item	Qty
(P04)	KH-25 (roll), KH-20 ×2 (yaw, elbow), KH-14 (wrist yaw), tested with their drive limits stored	4
— (docs/03 §13.2)	Arm limb interface at the shoulder-pitch output: Ø80 V-band with M6 T-bolt and safety catch, spigot + 2 dowels, service plate (power, EtherCAT, STO, arm ID; palm-camera coax)	1 set
B071	Shoulder bracket, roll-output bracket, upper-arm CF tube Ø50 × 2 + fittings, elbow-output bracket, forearm frame, Ti-6Al-4V wrist cross with 4 × INA HK0609 drawn cups on 100Cr6 Ø6 h6 pins	1 set
(P09.1)	KW-L2 wrist actuators with M5 rod ends	2
B107, B114	KES-WRIST driver board and the forearm 24 V buck (LM5146-Q1)	1 + 1
B108	Wrist-cross absolute angle sensors (MT6835 + diametric magnet), pitch and roll	2
B054	VL53L5CX ToF module (forearm)	1
B117	Palm-camera link: torsion-rated 50 Ω micro-coax through elbow, wrist yaw and wrist cross, FAKRA-Z break at the forearm	1

BOM	Item	Qty
—	Clock-spring cable cassettes for shoulder pitch (240° range) and shoulder roll (175°)	2

C Tools

As P07, plus a 0–50 kg spring balance or load cell, a 10 kg and a 15 kg test mass with a handle, a padded pendulum.

D Manufacturing

- Upper-arm CF tube bonding as in P07 §D.
- Wrist cross (Ti-6Al-4V): two perpendicular Ø10 N6 bores for the HK0609 drawn cups, axes intersecting within 0.01 mm. Lever arms 22 mm (pitch) and 16 mm (roll) to the KW-L2 rod ends. Pins 100Cr6 60–64 HRC, ground Ø6 h6 (needle raceways; never Ti).

E Assembly

1. **Arm-side V-flange and shoulder bracket** onto the KH-25 shoulder-pitch output (8 × M4 12.9 at 4.5 N·m, dowels); mount the KH-25 roll actuator on the bracket, axis x through the shoulder centre (test bar as in P07 step 4).
2. Fit the **clock-spring cassettes** at shoulder pitch and roll. Wind each to the middle of its travel at the zero pose.
3. **Roll-output bracket** → KH-20 shoulder-yaw actuator (axis along the arm) → upper-arm tube assembly → KH-20 elbow actuator at 300 mm (shoulder-to-elbow centre distance ±0.2 mm). KH-20 output flanges **8 × M4 12.9 at 4.5 N·m** (Mk II; K1's 8 × M3 slipped below the elbow momentary torque).
4. **Forearm:** elbow-output bracket → KH-14 wrist-yaw actuator (6 × M3 12.9 at 1.7 N·m) → forearm frame carrying the two KW-L2 actuators, the KES-WRIST board and the 24 V buck. The hand controller sits in the palm (P10).
5. **Wrist:** wrist cross on its pins (Loctite 243 on the retaining screws); KW-L2 rod ends to the cross levers (A at +x/+y, B at −x/−y). Zero: wrist straight, both screws at mid-stroke, rods adjusted until the hand axis is coaxial with the forearm within 0.3°. Fit the two wrist-cross angle sensors.
6. Bumpers at the elbow (0° and −145°) and the wrist limits (pitch ±70°, roll ±35°).
7. Route the palm-camera micro-coax with the arm harness: service loops at elbow, wrist yaw and wrist cross; FAKRA-Z break at the forearm.

F Electrical (docs/05 §5, §6.4, §9)

- Arm zone 14 AWG from the PDU (eFuse 20 A) through the shoulder service plate and the clock-springs; 52 V runs to the forearm.
- Forearm 24 V buck (LM5146-Q1, 24 V 10 A, hiccup limit 12 A) → KES-WRIST (KW-L2) and the hand's local 12 V buck, 16 AWG, XT30, ≤ 0.2 m. In STO the buck enable and its output load switch both open (2-channel), so the KW-L2s and the hand stop (Cat 0); the back-drivable screws let a held object slip, which is accepted.
- EtherCAT daisy chain: shoulder pitch → roll → yaw → elbow → wrist yaw → hand controller (which drives the KES-WRIST, docs/05 §7). Torque-flexure bridges on the KH-25 and KH-20 carriers.

G Testing (gate G2: arm on the bench, shoulder clamped)

Test	Pass
Range of motion (slow), all 7 joints	full range, no cable or coax snag (watch the clock-springs)
FK accuracy at the hand flange, 20 poses	≤ 2 mm
Drive limits in place: command 2 × over EtherCAT at shoulder, elbow, wrist yaw	output torque clamps at 160 / 105 / 23 N·m ± 5 %
Payload AH: 10 kg at full forward reach for 60 s	shoulder ≤ 160 N·m (requirement 90 N·m hold); winding rise within the dyno model; no gear-limit warnings
One-hand AO: 15 kg in the hand, elbow at 90°, for 60 s	elbow holds (≈ 59 N·m) without exceeding its thermal model; wrist pitch holds 14 N·m
Fast reach (AF scenario replay)	tracking error < 3°; speeds reach 6.2 rad/s shoulder / 7 rad/s elbow

Test	Pass
Collision detection: padded impact on the forearm at 0.5 m/s	stops within 50 ms; impact force logged
Wrist torque against a fixture: 19 N·m pitch (2 s), 29 N·m pitch (0.2 s), 10 N·m roll	holds; screws not back-driven; KW-L2 currents within P09.1 limits
Wrist speed: pitch 6 rad/s at no load	reached
Palm-camera link: 10^3 full-range elbow and wrist cycles while streaming	no link errors (docs/05 §9 torsion/flex test)

P09.1 — KW-L2 wrist linear actuators (4 units)

Build per docs/13 §4.4 with one Mk II change: **1 mm screw lead** (docs/09 §11 H10). Requirement per actuator (docs/04 §9): **754 N** at the repeated combined corner (19 N·m pitch + 10 N·m roll over the 22 / 16 mm levers), **972 N** momentary, **318 N** held for the 15 kg one-hand case, **132 mm/s** at 216 N, stroke ±30 mm. With the ECX FLAT 32 L (98.5 mN·m continuous, 540 mN·m stall, 10,600 rpm no-load at 24 V) and a 1 mm lead ($2\pi \times 0.88 / 0.001 = 5.53$ N per mN·m): 545 N continuous, 754 N at 136 mN·m, 972 N at 176 mN·m, ≈ 164 mm/s at 216 N. With the 2 mm lead of R10 and docs/13 the continuous force is only 272 N, below the 318 N hold.

A Visual reference

- Section:

```
ECX FLAT 32 L — rigid clamp hub — screw journal in a preloaded angular-contact pair (bearing block)
— ball screw Ø8 × 1 — flange nut in a push tube with an anti-rotation flat (POM-lined slot) — PTFE wiper
— Ti push tube — M5 rod end — wrist lever (22 mm pitch / 16 mm roll)          stroke ±30 mm + 5 mm to each stop
```

B Parts list (per actuator)

BOM	Item	Qty
B024	maxon ECX FLAT 32 L high torque, 24 V, Hall sensors (ECXA32LZF50E6ILACO1Y330A)	1 (+ 1 spare per robot)
B106	Ground ball screw **KSS Ø8 × 1** with flange nut (Ø6 × 1 only if its Ca ≥ 1.5 kN and C0a ≥ 3 kN; confirm in the KSS catalogue)	1
B106	Back-to-back pair of miniature angular-contact bearings for the screw journal, preloaded (e.g. SKF 706 CD/P4A class, 6 × 17 × 6; verify the axial rating ≥ 1 kN static)	1 pair
B106	7075 housing and bearing block, rigid clamp hub, Ti push tube, POM anti-rotation insert, PTFE rod wiper, ePTFE vent, O-rings	1 set
B106	M5 PTFE-lined rod end, female (SKF SIKAC 5 M class; verify C0 ≥ 3 kN) + jam nut	1
—	Elastomer end stops	2

C Tools & equipment

Torque screwdrivers 0.2–3 N·m, dial indicator (screw run-out), force gauge or load cell ≥ 1.5 kN in a test frame with a linear encoder, 24 V bench supply ≥ 10 A, the KES-WRIST board, a thermocouple on the motor housing.

D Manufacturing

- Housing and bearing block: 7075-T6, hard anodise; bearing seats per the bearing maker (housing H5/J5, journal h5 for a preloaded pair); motor-flange-to-screw axis coaxial ≤ 0.02 mm.
- Push tube: Ti-6Al-4V with the anti-rotation flat; rod-end thread M5 at the end.
- Screw journal: turned by the screw maker to your drawing (journal for the bearing pair and the clamp hub).

E Assembly

1. Press and preload the angular-contact pair on the screw journal (locknut to the bearing maker's preload torque).
2. Fit the bearing block to the motor flange; fit the rigid clamp hub to the motor shaft and the screw journal.
3. Check screw run-out at the nut position: ≤ 0.02 mm.
4. Fit the nut into the push tube; set the anti-rotation flat in its POM-lined slot; fit the wiper and the housing O-rings.
5. Fit the rod end at mid-stroke; lock the jam nut (Loctite 243).
6. Wire to the KES-WRIST board (3 phases, Halls); home against the soft stops, never the hard stops.

F Electrical

- KES-WRIST (2 × DRV8353, 100 V FETs sized ≥ 15 A), 24 V from the forearm buck.
- Current limits from the force targets ($k_t \approx 21.6$ mN·m/A derived in docs/05 §6.4; use the datasheet value): continuous ≈ 4.2 A (nominal), 2 s ≈ 6.3 A (754 N), 0.2 s ≈ 8.1 A (972 N), with an I^2t model from maxon's winding time constant.
- Position from the Halls and the wrist-cross absolute sensors (B108); force estimated from current.

G Testing (per actuator; R10 §7 item 7, docs/13 §4.4 tests at the 1 mm lead)

Test	Pass
Confirm the maxon datasheet values (nominal and stall torque, k_t, mass) before the test	recorded; if stall < 300 mN·m, stop and revisit the design
Force–speed curve at 24 V on the test frame	≥ 754 N for 2 s; ≥ 972 N for 0.2 s; ≥ 132 mm/s at 216 N
Continuous: 318 N for 60 min (AO hold)	motor housing ≤ 80 °C, winding (by resistance) ≤ 100 °C
Continuous: 545 N for 10 min	winding within the maxon limit
Backlash	< 0.02 mm
Endurance: 10^5 cycles at ±25 mm and 300 N	no backlash growth > 0.01 mm; wiper and screw clean
Back-drive with the motor unpowered (STO)	back-drives smoothly (expected; a held object can slip)

P10 Hands Mk II (13 actuated DOF each, palm camera)

Gate G2

Build **one finger first** (index: 3 actuators, then a 4-DOF thumb) and run the single-finger endurance and force tests before building the rest. Architecture: docs/03 §8 (mechanical), docs/05 §6 (electrical); K1 hand CAD `cad/hand.py` is the 10-DOA version and is the starting point for the Mk II detailing.

Mk II targets (R7 #4, `mass_budget.md`): **13 DOA**, 4-DOF thumb, finger abduction, **≥ 15 N fingertip**, **≥ 200 N power grasp**, **≥ 20 kg hook/handle hold**, tactile threshold ≤ 5 g, palm camera, **1.15 kg** per hand.

A Visual reference

- Hand detail render (K1): `renders/08_hand_detail.png`
- Actuation map (K1 10-DOA diagram; Mk II allocation in the table below): `diagrams/hand_architecture.svg`
- CAD: `cad/hand.py` (palm, phalanges, joints, internals: `hand_internals()`), STEP `cad/out/step/l_hand__palm_shell.step`
- Reference design: ILDA hand (Kim et al., Nat. Commun. 2021) [R5]

Actuator allocation (docs/03 §8):

Finger	DOA	Actuators (gearhead)
Thumb	4	CMC opposition rotation (4:1), CMC flexion (high-force), MCP flexion (16:1), IP flexion (16:1, short tendon)
Index	3	MCP abduction (4:1), MCP flexion (high-force), PIP + DIP coupled (16:1)
Middle	2	MCP flexion (high-force), PIP + DIP coupled (16:1)
Ring	2	MCP flexion (high-force), PIP + DIP coupled (16:1, short tendon to DIP)
Little	1	coupled flexion (high-force)
Spread	1	ring + little abduction, coupled (4:1)

High-force flexors (5 per hand): maxon DCX 12 L + GPX 12 (B118). Standard units (8 per hand): maxon DCX 10 L + GPX 10 (B058). All drive KSS SR0401K Ø4 × 1 ball screws on miniature guides into crossed four-bar links; DIP joints follow their PIP through a second four-bar or a short Dyneema tendon.



RENDER Hand detail render (K1)

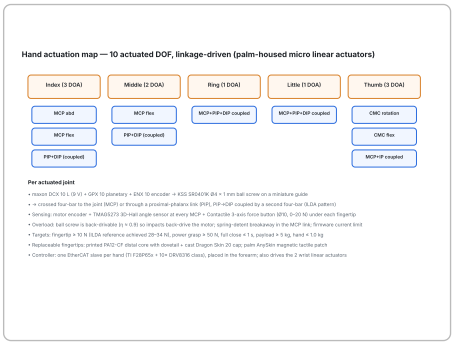


DIAGRAM Actuation map (K1 10-DOA diagram; Mk II allocation in the table below)

B Parts list (one hand)

BOM	Item	Qty
B118	High-force micro linear actuator: DCX 12 L 9 V + GPX 12 + ENX + KSS SR0401K Ø4 × 1 + miniature guide	5
B058	Standard micro linear actuator: DCX 10 L 9 V + GPX 10 (16:1 flexors, 4:1 abduction/rotation) + ENX 10 + KSS SR0401K + MGN5-class guide	8
B059	TMAG5273 3D Hall angle sensors + Ø3 × 1 mm diametric N52 magnets (MCP, PIP, thumb joints)	16
B052	Contactile 3D force buttons Ø10, 0–20 N	5
B053	Palm magnetic tactile skin (5 × MLX90393 + magnetised Dragon Skin 10 + iron microparticles, AnySkin pattern)	1

BOM	Item	Qty
B116	Palm camera TechNexion VLS-GM2-AR0234-C-S191-IR (global shutter, GMSL2) behind a sealed PC window, 20° toward the fingertips	1
B060	Hand controller Mk II (F28P65x, 13 H-bridges, EtherCAT, local 12 V buck from the 24 V forearm rail, KW-L2 interface), in the sealed palm	1
B061	Hand mechanical set: 7075 palm frame, 4-DOF thumb base on a 2-bearing pivot, abduction pivots, Ti-6Al-4V Ø2 pins, 17-4PH bushings, MJF PA12-CF links, Dyneema SK99 tendon (0.5 mm) with crimp terminals, cast Dragon Skin 20 fingertip caps, palm shell with camera window	1 set
B081	M1.6/M2 screws, E-clips, 0.2 mm PTFE washers, Loctite 222	set

C Tools

Stereo microscope, fine tweezers, M1.6 drivers, torque screwdriver 0.05–0.5 N·m, reamers Ø2 H7, small press for pins, silicone casting kit (vacuum chamber), printed moulds, force gauge 0–100 N, a 50 mm test cylinder on a load cell (grasp force), a 20 kg handle hook test mass, the finger endurance rig (printed frame + load spring + cycle counter), precision weights 1–10 g (tactile threshold).

D Manufacturing

- **Palm frame (7075):** CNC; holds 13 actuators in two staggered rows, guide rails, MCP pivot bores (Ø2 H7), the thumb-base pivot bearings, the abduction pivots and the controller/camera bay. Hard anodise.
- **Phalanx and four-bar links:** MJF PA12-CF, reamed to Ø2 H7 after printing, bushings pressed in. ILDA crossed four-bars for MCP and PIP; DIP ≈ 0.8 × PIP through a second four-bar (index, middle) or a short tendon (ring, thumb IP).
- **Tendons:** 0.5 mm Dyneema SK99 over Ø4 PTFE pulleys, pre-stretched at 50 N for 1 h, terminated with crimp sleeves; set length with a printed jig.
- **Fingertip caps:** Dragon Skin 20 (Shore 20A) cast over a PA12-CF distal core with a dovetail; the Contactile button sits under 1.5 mm of silicone.
- **Palm skin:** Dragon Skin 10 with iron microparticles, cast 2 mm, magnetised (AnySkin method [R5]).
- **Palm window:** 2 mm PC, AR coated, bonded and O-ring sealed in front of the camera (the palm is a sealed zone, docs/03 §12).

E Assembly

1. **Actuator sub-assemblies:** motor + gearhead + ball-screw coupling (set screw on a flat, Loctite 222), nut carriage on the guide, rod end. Check smooth travel; no-load current over the full stroke < 60 mA (DCX 10 L) / < 80 mA (DCX 12 L).
2. **Fingers:** phalanges with pins + PTFE washers + E-clips; four-bar links; magnets in each MCP and PIP pin end and the TMAG5273 flex PCB in the facing link; tendons for ring DIP and thumb IP.
3. **Thumb base:** opposition-rotation actuator (4:1) turning the thumb base on its 2-bearing pivot; CMC flexion (high-force) and MCP flexion actuators; IP by tendon.
4. **Abduction:** index abduction actuator; spread actuator driving ring and little abduction through a coupled link.
5. **Mount the actuators** in the palm frame and connect each nut carriage to its crank or link.
6. **Overload breakaway:** the spring detent in each MCP crank must release at 1.5 × the stall force (set with the force gauge) and re-seat by hand.
7. **Palm camera and controller:** camera behind the sealed window, controller board in the palm bay; micro-coax from the camera to the forearm FAKRA-Z break (P09).
8. Fingertip caps (dovetail slide-on, one drop of silicone adhesive), palm skin (4 × M2), palm shell.
9. **Wiring:** motor leads and encoder flex to the in-palm controller; only power (24 V), EtherCAT and the camera coax cross the wrist, with service loops through the wrist cross.

F Electrical

- Hand controller fed from the forearm 24 V rail (XT30, 16 AWG) through its local 12 V buck; EtherCAT IN/OUT; it also commands the forearm KES-WRIST board (docs/05 §7). The 24 V rail is disabled in STO (Cat 0 for the hand).
- Each micro-motor on an H-bridge with a current limit from the maxon data sheet for the 9 V winding on the 12 V supply: continuous current for holds, short-time 2 × for grasps, I^2t in firmware; stall protection.

- I²C buses: 2 for the TMAG5273 sets (address-select via mux), 1 for the Contactile buttons, 1 for the palm skin.
- Palm camera: GMSL2 power-over-coax from the J601 (ports 4–5, docs/05 §7).

G Testing (gate G2: hand)

Test	Pass
Each joint range and the finger angle sensors vs a protractor	±1°
Fingertip force, straight and bent (force gauge), every finger	**≥ 15 N**
Power grasp on the 50 mm instrumented cylinder (short-time)	**≥ 200 N** total; ≥ 150 N held for 60 s
Hook/handle hold: 20 kg on the four fingers for 60 s	no slip; motor temperatures < 60 °C; nut force ≤ 560 N (KSS Ø4 × 1 Ca) by current
Thumb opposition: thumb tip meets each fingertip	all four reached; pinch ≥ 15 N index–thumb
Close time, open → full fist	≤ 1.0 s
Back-drive: push a fingertip with 20 N, motors off	finger yields without damage
Breakaway: overload the MCP	detent releases, re-seats
Endurance (first finger, then first thumb): 10^4 close/open cycles at 50 % load	force loss < 10 %; pin wear < 0.05 mm; tendon creep within the adjuster range
Tactile: Contactile and palm skin, 5 g weight and 0.1–20 N ramp	5 g detected; monotonic, calibrated
Palm camera: stream at rate through the wrist with the arm cycling	no drops; image clear through the window
Mass of the finished hand (without F/T sensor)	≤ 1.15 kg (budget); record the actual

P11 Neck, head, cameras, LiDAR, calibration

Gate G3

A Visual reference

- Head detail: `renders/07_head_detail.png`
- Sensor placement: docs/06
- CAD: head shell `cad/out/step/head__head_shell.step`; visor and side windows in `cad/build_robot.py`



RENDER Head detail

B Parts list

BOM	Item	Qty
(P04)	KH-14 neck pitch, drive limit 23 N·m stored (neck yaw installed in P08)	1
B072	Neck yoke, head frame (6061 plate + PA12-CF bracketry)	1 set
B045	ZED X Mini (behind the visor, 10° down)	1
B046	ZED X One GS (temples, yaw ±55°)	2
B048	Livox Mid-360 (crown)	1
B055	ReSpeaker XVF3800 + speaker	1
B054	VL53L5CX (head rear)	1
B075	Visor (PC, smoked) + side windows + LiDAR window	1 set
—	GMSL2 FAKRA-Z coax, 1.2 m × 4; Ethernet (Mid-360); USB (mic)	—

C Tools

ChArUco calibration board (A1, rigid), AprilTag ground board, a laptop with Kalibr and the ZED SDK, a level, a lint-free optics kit.

D Manufacturing

- Head frame: a 6061 plate carrying the camera bracket (stiff: cameras must not move relative to each other, so first mode > 150 Hz), the LiDAR seat and the neck-pitch flange.
- Visor: thermoform 3 mm PC over a printed buck; cut the aperture windows; AR + hard coat on the camera apertures (outsourced optical coating, or buy coated PC windows and bond them in).

E Assembly

1. Neck yoke on the neck-yaw output (6 × M3 12.9 at 1.7 N·m, dowels); neck-pitch KH-14 in the yoke (axis y at z = 1555).
2. Head frame on the neck-pitch output (6 × M3 12.9 at 1.7 N·m, Loctite 243, dowels).
3. Cameras on the rigid bracket: ZED X Mini centred 66 mm forward of the neck axis at z = 1655; the two ZED X One GS at the temples, yawed ±55°. Torque the camera screws per

the manufacturer and add thread-locker.

4. Mid-360 on its seat, its top ≈ 2 mm below the crown window. Ethernet + 24 V. Mic array at the lower rear behind its grille.
5. Route the 4 coax cables and the Ethernet through the neck with a twist-capable service loop (neck yaw ±80°, pitch −35/+50°).
6. Fit the head shell (two halves) and the visor (gasketed). Clean the windows.

F Electrical

- GMSL2 coax to the J601 deserialiser ports 0–3 (ports 4–5 are the palm cameras, P10). Power-over-coax from the J601, which sits on the 24 V logic bus, so the cameras stay up through an e-stop.
- Mid-360: 24 V logic bus (9–27 V input [ds R4]), Ethernet to the Thor 10GbE switch port, PTP enabled.
- Hardware frame sync: all GMSL cameras on one J601 frame-sync line.

G Testing and calibration (part of gate G3)

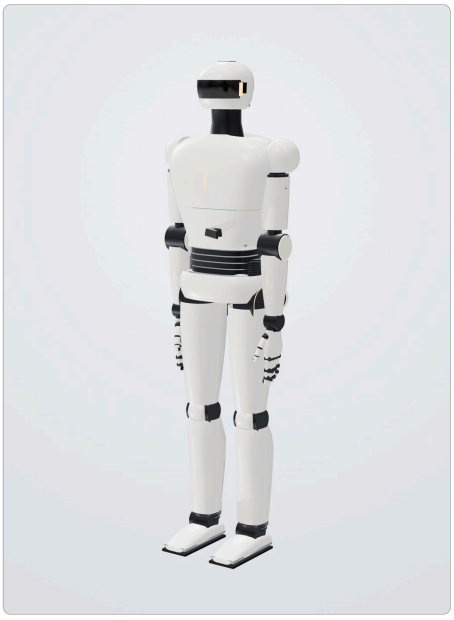
Test	Pass
All streams at rate (ZED X Mini 30–60 fps, One GS 30 fps, Gemini 30 fps, 2 palm cameras, Mid-360 10 Hz) with PTP timestamps	no drops over 10 min; time offset < 1 ms
Stereo + intrinsics calibration (Kalibr / ZED tool)	reprojection RMS < 0.3 px
Camera–IMU extrinsics + time offset (Kalibr)	converged; time offset stable within 0.5 ms
LiDAR–head extrinsics	plane residual < 1 cm
Head–body kinematic calibration (AprilTag board, 30 poses)	≤ 2 mm / 0.2° at 1 m
Neck range with cables, 1,000 cycles	no intermittent camera link errors

P12 Harness, shells, final leak check and final assembly

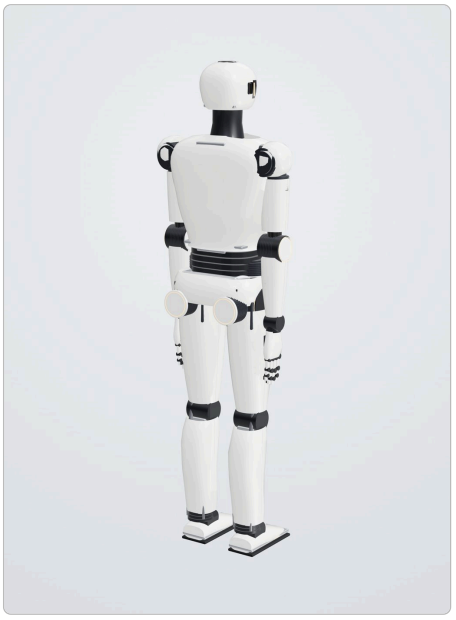
Gate G3

A Visual reference

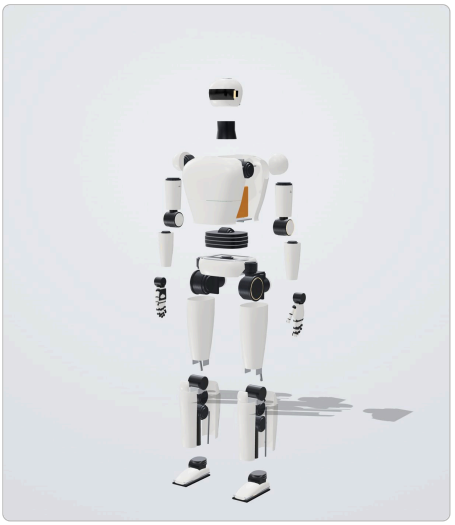
- Fully assembled: `renders/05_threequarter_front.png`, `renders/06_threequarter_rear.png`
- Exploded (shell panel splits): `renders/12_exploded.png`
- Ghost (shells over the structure): `renders/14_ghost_threequarter.png`
- Shell STEP files: `cad/out/step/*front*.step`, `*rear*.step`, `torso__chest_plate.step`, `torso__abdomen_plate.step`, `torso__back_door.step`, `pelvis__pelvis_*.step`, `l_shoulder__deltoid_cap.step`, `head__head_shell.step`, `l_foot__foot_shell.step`
- Sealing zones (IP65 body, IP67 legs and hands): docs/03 §12; routing through joints: docs/03 §11



RENDER Fully assembled



RENDER Fully assembled



RENDER Exploded (shell panel splits)



RENDER Ghost (shells over the structure)

B Parts list

BOM	Item	Qty
B073	MJF PA12 shells (≈ 42 panels), including the two gasketed battery hatches in the back door and the radiator grille	1 set

BOM	Item	Qty
B074	Primer + 2K satin warm-white + matte graphite + clearcoat (shop)	1
B076	N52 disc magnets Ø10 × 3 + steel pads, captive M3 quarter-turn studs, M3 heat-set inserts, 3 mm silicone isolators	1 set
B077	Waist bellows, ankle gaiters	3
B078/B079	Harness wire (8/10/12/14/16 AWG silicone), igus chainflex CF Robot, connectors (SurLok Plus, XT60H, XT30, Gecko-MT, JST GH, IP67 sensor connectors), sleeving, heat-shrink, P-clips, labels	1 set
B080	Electronics-blower filters	2
—	EPDM D-profile hatch gaskets, PTFE drain plugs for the limbs	1 set

C Tools

Heat-set insert iron, harness nail-board (printed 1:1 from the CAD routing), crimp tools, insulation tester, label printer, paint shop (outsourced) or a spray booth, pressure-test kit and refractometer (P08.1).

D Manufacturing

- **Shells:** order the STEP files from an MJF bureau (KAD3D / Formero AU [R6]). 1. **Before printing, add bosses:** Ø6 × 6 mm with a Ø4.2 hole for M3 heat-set inserts, every 100–150 mm along the edges, matching the frame standoffs; add the battery hatches (2 quarter-turn fasteners each, EPDM gasket) and the radiator inlet/outlet grilles of the wet duct. 2. Bead blast, or vapour smooth for a premium finish. 3. Fit-check every panel on the robot **before** painting, with the hoses and service loops in place. 4. Paint: adhesion promoter → 2K primer → satin warm white (25–30 GU) on the shells, matte graphite on the joint caps, soft clearcoat. 5. Mask the magnet and insert locations.
- **Light guides:** machined clear PC with diffuser film; amber LED strips (2.2 mm) behind.

E Assembly

1. **Harness:** - build each zone harness on the nail-board; label both ends (zone-joint-function); - test continuity 100 % and insulation at 500 V (> 100 MΩ); - install in the service loops (P07 §F, P09 §F); P-clip every 80 mm; 30 mm slack each side of every joint; - sensor and leak wiring above the coolant lines, never below; nothing electrical under a coolant fitting.
2. **Cooling recheck:** with both packs out, inspect every hose loop for rub marks after the G2/G3 runs; check the loop pressure (0.3 bar cold), glycol concentration and that the compensator is mid-stroke. Run the electronics blowers and the radiator fans and check the airflow direction with a smoke pen.
3. **Bellows and gaiters:** waist bellows clamped to the pelvis and torso rims over the hose clock-spring; ankle gaiters over the push-rods.
4. **Shells:** - press heat-set inserts into the bosses; - fit silicone isolators on the frame standoffs; - mount the inner panels with screws, the service panels with quarter-turn studs + magnets; - check the 1.0 ± 0.3 mm panel gaps with feeler gauges, and that no shell touches a moving part or a hose through the full range of motion (run the range slowly with the shells on).
5. **Final torque audit:** re-torque every actuator flange and apply paint-pen witness marks: KA-130 12 × M5 8.5 N·m; **KA-110P 12 × M5 8.5 N·m**; KA-90P, KH-25, **KH-20** 8 × M4 4.5 N·m; KH-14 6 × M3 1.7 N·m; V-band T-bolts 10 N·m (legs) with the safety catches closed.
6. **Weigh:** full robot and each module. **If the mass exceeds 112.7 kg (the design mass; the estimate is 102.5 kg), update `params.py` and re-run `calc/run_all.py` before commissioning.** Expect the loop (+0.3 kg) and the pack enclosures (docs/09 H12) to push it up.

F Electrical

- Final insulation test: bus → chassis with the star bond lifted (> 10 MΩ at 500 V), then refit the bond. Both packs out, keyed main switch off.
- LEDs on the 12 V logic bus through the safety MCU (status protocol: docs/02 §7); pack-bay LEDs from the PPM.

- Verify that every connector is IP2X (finger test) and latched, and every unmated connector capped.

G Inspection before commissioning

Check	Pass
Range of motion with shells on, all joints	no contact, gaps maintained, no hose or coax strain
Mass and CoM (two-scale method in the zero pose)	mass ≤ 112.7 kg; CoM within ±20 mm of the `calc` model (re-run with the measured mass)
Thermal soak: 30 min hanging in the gantry, all systems on, pumps running	all temperatures within limits; coolant supply ≤ 40 °C at 25 °C ambient
Coolant: pressure after the soak and a full range-of-motion run	0.3 ± 0.05 bar cold; no damp on any leak pad
Pack swap with the shells on (robot in the gantry, drives in STO)	≤ 3 min; hatches seal
Water: light spray on the shells (IPX4-style hand spray, 5 min) with power **off** and both packs out	no water inside the torso core, the actuators or the pack bays after opening (the IP65/IP67 tests of docs/03 §12 come later)
Documentation: build log complete, serials recorded, calibration files committed, design holds signed off	—

P13 Software bring-up and commissioning hand-off

Gate G3–G10

This phase takes the assembled robot from "all joints move" to the walking, manipulation and payload gates. Detailed test procedures and pass criteria are in docs/11. The architecture is in docs/07.

A Visual reference

- Software layers: `diagrams/software_stack.svg`
- Gates: `diagrams/test_gates.svg`
- Simulation model: `sim/kestrel_k1.xml` (MuJoCo, generated with the sized torque limits)

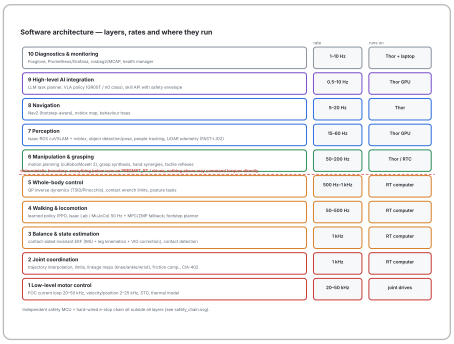


DIAGRAM Software layers



DIAGRAM Gates

B Software components

Component	Install
RT computer	Ubuntu 24.04 + PREEMPT_RT, IgH EtherLab 1.6 (igc), ros2_control (Jazzy), Pinocchio, ProxQP/OSQP, ONNX Runtime
AI computer	JetPack 7, ROS 2 Jazzy, Isaac ROS (cuVSLAM, nvblox), ZED SDK, Livox SDK2, Foxglove bridge
Safety MCU	monitor firmware (heartbeat, tilt, temperatures, e-stop, leak, coolant temperature, relay outputs)
HC2 drives	custom firmware on F28P65x (TI C2000 SDK + EtherCAT slave stack), CiA-402
PPM / PDU MCUs	hot-swap state machine, zone eFuse control, coolant sensing and pump control (docs/05 §4, §13)
Hand controllers	finger loops, tactile reflexes, KW-L2 commutation through the KES-WRIST board
Development PC	MuJoCo simulation, training package (see the training add-on), Foxglove

C Tools

Gantry + harness (P01), wireless e-stop, cage, foam floor, a second person, a laptop with Foxglove, payload boxes for G10 (a rigid box with handles on two sides and a quick-release strap, filled in 5 kg steps with sandbags).

D Configuration files to produce (commit them)

1. `config/joints.yaml` : joint order, encoder offsets, directions, soft limits, torque limits: - gear-limited classes at their repeated ratings (KH-25 160, KH-20 105, KH-14 23 N·m), duplicated by the drives' stored current limits (P04 §F); - the **knee angle-dependent limit table** (P07 §F) and the HC2 continuous limit (75 A ≈ 263 / 217 N·m); - linkage tables (`data/knee_fourbar.json` , ankle/wrist geometry with the 1 mm KW-L2 lead).
2. `config/ethercat.xml` : slave order (28 slaves; 30 if the KW-L2 boards become slaves, docs/09 H11), PDO maps (CST mode: target torque, position, velocity; actual position × 2, torque/current, temperatures, status), DC settings.
3. `config/safety.yaml` : tilt limit 35°; winding warn 130 °C / trip 150 °C; **coolant supply warn 55 °C, derate 65 °C, trip 75 °C; loop pressure < 0.15 bar → air-only rating**; leak → stop; heartbeat 2 ms; comms timeout 50 ms.
4. `config/power.yaml` : single-pack actuator power cap 1.8 kW; swap cap 1.5 kW; DIODE/TIED rules (|ΔV| < 0.5 V, SoC < 98 %); BMS CCL → regen cap.
5. `config/estimator.yaml` : IMU noise (CV7: 1.5 °/h bias instability, 30 µg/√Hz [ds R4]), kinematic noise, contact thresholds; payload mass and CoM parameters for G10.

E Bring-up sequence (each step gated in docs/11)

1. **G3 hanging:** robot suspended, feet clear, coolant loop running. Joint-by-joint at 10 % torque limits, then gravity compensation, then whole-body slow motions. Check every sensor stream, the estimator (base attitude vs the gantry inclinometer), the coolant sensors, and a pack swap with the drives in STO.
2. **G4 standing:**
3. lower until the feet carry 20 % of the weight, then 50 %, then 100 % with the harness slack-taut;
4. balance controller (WBC + IEKF);
5. push tests ≤ 20 N·s;
6. practise Cat 1 and Cat 0 stops caught by the harness.
7. **G5 stepping:** the ZMP walker (`calc/kestrel/gait.py` patterns through the WBC) for in-place steps, then 0.1–0.3 m/s, with the harness taking ≥ 10 % load.
8. **Learned policy:** train in simulation (training add-on), evaluate in MuJoCo against the ZMP walker on the regression suite, deploy with the harness at 50 % action scale first.
9. **G6–G9:** free walking in the cage, stairs, endurance and manipulation per docs/11.
10. **G10 payload:** 30 kg sustained and 50 kg momentary, in steps, harness attached (docs/11 §7.1).

F Electrical / safety in this phase

- Every session starts with the e-stop functional check (press → drives disabled → reset), including the commissioning contactors on the HC2 feeds; their uncertified STO is never the only stop (docs/09 H8).
- Torque limits are raised in steps (25 % → 50 % → 100 %) only after the previous level has run for 1 h without faults. **No unrestricted high-torque operation near people at any stage**; nobody in the cage while drives are enabled.
- The standing (no-STO) pack swap is enabled only after G6, with the robot in the swap posture and the 1.5 kW cap.

G Exit criteria

Gate G10 signed (docs/11): G9 passed (1 h untethered walking at ≥ 0.8 m/s in the cage, 0.18 m stairs, pick-and-place), 30 kg carried 20 m at 0.6 m/s and a 50 kg momentary lift with all margins logged, thermal and coolant margins logged, and a teardown inspection with no unexpected wear.

App. A Testing and commissioning gates

Nothing on KESTREL K1 Mk II is proven until it passes the tests below. Movement is introduced progressively: restrained → low speed → supported → free → loaded, with independent shutdown at every step. Numbers come from `docs/generated/` (design mass 112.7 kg, Mk II actuator family, payload scenarios CW, SQ, LM, AH, AO) and R10 §7.

1. Gate checklist (sign and date each)

Gate	Entry condition	Exit criteria (all required)
G0 Bench electrical	P02 complete	EtherCAT 1 kHz with distributed clocks, 1 h, 0 lost frames; STO < 10 ms from every e-stop device; HC2 external-contactor stop verified; drive-level current limits proven independent of the RT software; safety-MCU trips verified; **pack gate (P06 §G) for both packs and bay gate (P06.1 §G)** passed
G1 Actuator	each actuator built	P03/P04/P05 §G tables passed, **including the R10 §7 validation tests of §4.1**; P05.1 jacket tests and P04.1 cycloid measurements passed; P09.1 KW-L2 tests passed; measured Kt, η, R_th, k_L, thermal time constant and backlash fed back into `calc/` and the requirements re-checked
G2 Limb	limb assembled on its stand	P07/P09/P10 §G tables passed, including the knee torque-limit table, the leg coolant test and the stand payload cases
G3 Hanging	robot assembled, suspended, P08.1 loop filled	every joint, sensor, coolant sensor and estimator verified; Cat 0/1/2 stops tested (logic bus stays up); thermal soak OK; pack swap in STO ≤ 3 min; P12 §G inspection passed
G4 Standing	G3	stand 10 min unassisted (harness slack-taut); recovery from 20 N·s pushes in 4 directions; sit/stand × 20; single-pack standing for 10 min
G5 Supported stepping	G4	100 steps in place, 50 m at 0.3 m/s with the harness carrying ≥ 10 % load; no falls caught by the harness in the last 20 m
G6 Free walking (cage)	G5	0.3 → 0.8 → 1.2 m/s; turns; slopes 5° → 15°; 1 h cumulative; harness slack throughout; coolant supply and knee winding logged
G7 Stairs & terrain	G6	0.10 → 0.18 m steps up/down × 50 each; gravel, mats, cables on the floor; standing pack swap (no STO) × 10 with the harness slack-attached
G8 Endurance	G7	2 h mixed duty with thermal and coolant margins logged; 10 h cumulative walking; hose-loop fatigue 10^6 cycles complete; teardown: gears, cycloids, bearings, rod ends, bonds, jacket O-rings, quick-disconnects
G9 Manipulation & AI	G8	5 kg pick/place × 100; tool use; skill API rejects out-of-envelope requests; people exclusion zones enforced
G10 Payload (Mk II)	G9	§7.1 table passed in order: 10 kg per hand at reach, 15 kg one-handed, 30 kg squat, 30 kg carried 20 m at 0.6 m/s, 50 kg momentary lift; post-test flange audit and teardown of one knee and one shoulder

2. Test environment and safe operating limits

- **Cage:** 4 × 3 m mesh panels, foam floor tiles, operator outside with the wireless e-stop on a lanyard. The gantry harness is attached from G3 to G5 and slack-attached in G6–G7 and **for every G10 payload test**.
- **Two-person rule** from G3 and for every pack swap. Nobody inside the cage while drives are enabled, and **no unrestricted high-torque operation around people** at any stage of this programme.
- **Limits until G6:** torque limits 50 % of the class peak (the gear-limited KH classes at most their repeated ratings, the knee additionally by the angle table of P07 §F), speed 50 %, workspace fences in software (base within the cage, hands below 1.8 m).
- **HC2 stop function:** the HC2 is custom and its STO is uncertified. Through commissioning (until G6 and a recorded STO validation, §5) each leg-zone feed has a DC contactor opened by the safety relay's instantaneous contacts; that external chain is the safety function for the knees and hip pitches.
- **E-stop behaviour (docs/05 §8):** STO at t = 0, actuator-zone eFuses open at 0.5–1.5 s, logic bus and coolant pumps stay up. The e-stop is not an isolation: for any hands-on work, keyed main switch off and both packs out.
- **Hot swap:** until G6 only with the robot supported and all drives in STO; standing swaps from G7.
- **Coolant:** no powered tests with a known leak; any damp leak pad stops the session; spills on the cage floor are cleaned before walking.
- **Abort criteria:** any unexpected noise or smell, temperature above warn, coolant supply > 55 °C or pressure < 0.15 bar, estimator divergence, EtherCAT errors > 0 per minute, a pack

fault, or a witness mark that has moved. Stop and investigate before continuing.

3. Structural validation

What	How	Pass
Hand calculations	`calc/kestrel/structure.py` (generated)	all SF ≥ 1.5 (structural case) / ≥ 2.0 (fatigue): **met at the requirement loads**, including the KA-110P 12 × M5 flange (SF 2.35)
Design holds	docs/09 §11 H1, H4, H5, H7, H9, H12, H13	each signed off by the owner **before metal is cut**
Knee push-rod at actuator capability	rod force vs knee angle with the P07 §F torque table; rod-end and buckling check over 0–130°	rod force ≤ 12 kN everywhere; rod-end C0 confirmed from the purchased part
FEA	pelvis bowl, yokes, hip adapter ring, knee hub, thigh plates, shank fittings, foot plate, ankle cross, limb-interface V-flanges under N/M/X loads (CalculiX/FreeCAD FEM or Fusion); thigh and knee also at the actuator's full capability (≤ 535 N·m at the joint)	von Mises < σy/1.5 (X); < endurance/2 (N); modes > 40 Hz (legs), > 25 Hz (arms) with the heavier Mk II actuators
Proof load	each leg on the stand: 1.5 × the structural-case vertical load (1.5 × 2.76 kN ≈ **4.1 kN**) for 30 s, joints locked	no permanent set > 0.05 mm (dial indicators), no bond creep
Limb interfaces	V-band fixture: 1.5 × 546 N·m bending, 260 N·m torque (docs/03 §13.1)	no 0.02 mm feeler entry; no slip
Bonded joints	coupons: 1.5 × axial + torsion	no failure
Fatigue	leg stand, 10^6 cycles of the walking load profile (≈ 140 h at 2 Hz) on one leg before G8	no cracks (dye penetrant on Ti and 7075 hot-spots)
Drop	the structural case (0.15 m) on the stand with a **113 kg** sled	no yield; actuators may slip their flanges (inspect and re-torque)
Payload proof (before G10)	leg stand at the LM case: 81 kg equivalent per leg, slow, 5 cycles	no slip marks, no permanent set

4. Actuator validation (P03–P05, P04.1, P05.1, P09.1 §G)

Item	Method	Feeds back into
Joint torque validation	dyno torque vs current across the speed range	`MOTORS` Kt, `ETA`
Motor thermal testing	thermal step to steady state; walking-duty replay from `data/scenario_series.npz`; for the KA-130, the jacket R_th and the HC2's 75 A limit (§4.1)	derates; **the jacket thermal resistance is now the critical result** (knee continuous margin with HC2: 263 × 0.84 / 142 ≈ 1.56)
Gearbox endurance	10^5 cycles (2×10^5 for KA-130) of the scenario torque profile; backlash and efficiency before and after	gear spec, maintenance interval
Efficiency map	speed × torque grid, motoring and regenerating	power budget (`calc/kestrel/power.py`)
Overload	KA family: rotor locked, 1.1 × momentary on the output; KH cycloids: §4.1 item 6	flange slip setting, gear rating
Drive-level limits	command 2 × the stored limit over EtherCAT	KH classes clamp at 160 / 105 / 23 N·m ± 5 %

4.1 Mk II validation before building more than one (R10 §7)

#	Test	Where	Method	Pass / feeds back
1	**FEMM check of KM-130x50-LC** (also KM-115x32-N50 and KM-85x26-N50-4t before their dies)	desk, before ordering laminations and N50UH magnets	2-D magnetostatic model (docs/13 §3.1): Kt, saturation curve, demagnetisation at peak current and temperature	Kt within ±10 % of 0.224 N·m/A; T_sat near 30 N·m; N50UH demagnetisation margin at 204 A and 120 °C. Update `MOTORS`
2	**Jacket thermal-resistance dyno test**	P05.1 §G, first on a K1 KM-130x45 stator if available	DC copper loss 100 / 150 / 215 W, 0.5 L/min at 25 °C, winding temperature by resistance	R_th ≤ 0.25 K/W meets the rating; ≤ 0.30 K/W accept and derate by √(0.25 / R_th); > 0.30 reject. Feed F_cool into `actuators.py`
3	**150 A drive locked-rotor test**	P05 §G on the dyno, test pack as the source, external contactor in circuit	150 A for 2 s and 99 A for 10 s into the locked output; torque vs current 10–150 A	≥ 493 / 406 N·m at 150 A; saturation no earlier than modelled; no flange slip; HC2 within rating. Update `T_pk`

#	Test	Where	Method	Pass / feeds back
4	**Gear run-in, efficiency, ISO 6336**	P03 / P05 §G	run-in at 25 % then 50 % peak both directions; efficiency map; backlash; ISO 6336 bending and pitting on paper	clean grease, η recorded (replace 0.92), backlash ≤ 10 arcmin, ISO 6336 SF ≥ 1.4 bending / 1.0 pitting at 1.1 × peak
5	**k_L measurement**	P05 §G (also P03 for the RI motors)	voltage-limited torque–speed curve with the bus held at 52 V	fitted k_L ≥ 0.75; otherwise update `K_L` and re-run `run_all.py`
6	**Cycloid overload** (C25 and C20)	P04 §G	rotor locked, 250 / 160 N·m on the output through the 1 m arm, 1 s × 3 each direction; backlash, stiffness, efficiency vs speed	lost-motion change < 1 arcmin, no pin or lobe marks, backlash ≤ 7 arcmin, η ≥ 0.80 at rated. Confirms the CYC rating targets in `actuators.py`
7	**KW-L2 force–speed**	P09.1 §G	confirm the maxon datasheet values; force–speed at 24 V on a test frame with the **1 mm lead** screw	≥ 754 N for 2 s, ≥ 972 N for 0.2 s, ≥ 132 mm/s at 216 N, 318 N for 60 min with winding ≤ 100 °C

5. Electrical validation

Item	Method	Pass
Insulation	500 V bus → chassis (bond lifted); 1000 V motor windings → stack; repeat bus → chassis after the first coolant fill and after the IPX4 spray	> 10 MΩ / > 100 MΩ
Packs	P06 §G for each pack (capacity, balance, 1C thermal, 80 A peak, BMS trips, dead terminals)	as listed
Hot-swap power path	P06.1 §G; then docs/05 §14: pull and insert each pack 50 × while walking on the gantry (from G7)	no bus dip below 52 V, no drive under-voltage, no arc marks; interlock refuses unsafe releases
Single-pack peak	3.1 kW from one pack at 20 % SoC on the gantry	sag consistent with 51 mΩ; bus ≥ 52 V; RT power cap engaged
Power quality	scope the bus during a W2 replay and an SD drop on the stand in diode mode	bus stays 48–72 V; chopper engages above 70 V
eFuses	electronic-load trip per zone	±10 %
HC2 STO validation	scope STO_A/STO_B → gate-supply removal and PWM disable → phase current, read-back of both channels, 100 trials; fault-inject one channel	phase current zero < 10 ms; read-back detects every injected fault. Only after this may the commissioning contactors be removed (and not before G6)
E-stop response measurement	scope from the button contact to phase current zero, for each device; wireless out-of-range timeout; zone eFuse opening at 0.5–1.5 s; logic bus stays up	≤ 10 ms wired; wireless heartbeat timeout measured and logged (not published by the vendor [R2])
EtherCAT timing	log DC sync error and cycle jitter over 1 h with all slaves (28, or 30 per docs/09 H11)	jitter < 20 µs, sync error < 1 µs
EMC pre-scan	near-field probe + LISN on the bus during walking; ESD ±8 kV air on exposed metal	no resets, no EtherCAT errors
Coolant electrical	leak pads and cable wetted with deionised water; pump stall; flow sensor unplugged	each produces its alarm; HC2 leak trips the chain; pump stall → derate

6. Sensor calibration and validation

See docs/06 §4 for procedures. Validation: walk a 20 m closed loop and compare the estimator against a motion-capture or total-station reference. Pass: drift < 1 % of distance without vision; < 0.2 % with VIO + LiDAR. Palm cameras: intrinsics and hand-eye calibration against the wrist F/T flange (≤ 2 mm at 0.3 m).

7. Control-system testing (progressive)

Step	Environment	What is tested
Software-in-the-loop	MuJoCo with `sim/kestrel_k1.xml` + the RT stack through ros2_control	stand, walk 0.8/1.2 m/s, stairs, pushes, payload scenarios: regression suite before every hardware session
Single joint (G1)	dyno	torque, impedance, limits, faults, coolant interlock
Single limb (G2)	leg/arm stand	tracking, gravity comp., load cases, linkage maps, knee torque table
Balance (G4)	standing, harness	WBC, IEKF, push recovery, controlled stops

Step	Environment	What is tested
Supported walking (G5)	harness ≥ 10 % load	ZMP walker first, then the learned policy at 50 % action scale
Progressive locomotion (G6–G7)	cage	speed, turns, slopes, stairs, terrain
Full-system endurance (G8)	cage	2 h duty, thermal and coolant margins, inspection
Payload (G10)	cage, harness slack-attached	§7.1

7.1 Mk II payload tests (gate G10)

Run in this order; each row first at 50 % of the mass, then 100 %. The payload is a rigid box with handles on two sides and a quick-release strap, filled with sandbags in 5 kg steps; **never a person holding or handing over the load while drives are enabled**; the box is placed on a stand at the start height by the operator with the robot in STO. Catch table under the box path for the lift tests.

#	Test (scenario)	Procedure	Pass
P1	**10 kg per hand at full reach** (AH, AL)	lift 10 kg per hand from hanging to forward horizontal in 0.8 s, hold 60 s	shoulder ≤ 160 N·m (requirement 108 repeated / 90 hold); winding within the thermal model; no gear-limit or flexure alarms
P2	**15 kg one-handed** (AO)	elbow at 90°, 15 kg in one hand, 60 s, then walk 5 m at 0.3 m/s	elbow ≤ 105 N·m limit (requirement 79 repeated, 59 hold); KW-L2 holds wrist pitch (14 N·m) within its I^2t; balance maintained
P3	**30 kg squat** (SQ)	squat and rise holding 30 kg at the chest, × 10	knee within the P07 §F table; waist pitch ≤ 288 N·m repeated; tracking < 3°
P4	**30 kg sustained carry** (CW)	walk 0.6 m/s for 20 m carrying 30 kg at the chest, × 5; then 10 min of repeated pick–carry–place	knee and hip-pitch windings within the model + 10 %; coolant supply ≤ 55 °C; no single-pack event; CoP inside the sole margin
P5	**50 kg momentary lift** (LM)	squat-lift a 50 kg box from thigh height, × 3, harness slack-taut, catch table under the box	waist pitch ≤ 379 N·m and knee ≤ 535 N·m (momentary requirements, logged from current); no flange slip (witness marks); no structural noise
P6	Post-test inspection	flange torque audit; teardown of one knee and one shoulder actuator	no slip, no gear or cycloid marks, no bearing damage

Not in the sized scenarios — do not attempt in this programme: stairs with a 30 kg load (R7 #1 target), one-handed loads above 15 kg, and any payload test without the harness. Add the scenario to `calc/kestrel/scenarios.py` and re-run `run_all.py` first.

Sim-to-real checks: - compare measured joint torques (current × Kt) against the inverse-dynamics model for identical motions, including the payload cases. Expected agreement ±15 %; investigate anything larger. - re-identify inertial parameters if the mass of any module deviates > 5 % from `params.py`.

8. Failure modes (FMEA summary)

Failure mode	Effect	Detection	Mitigation
Knee/hip motor overheats in sustained walking or carrying	derate → slow, then Cat 1 stop	winding NTC + jacket-outlet NTC + thermal model	liquid jacket; HC2 75 A continuous limit; system-level thermal derate (docs/05 §13)
Coolant leak	loss of cooling; conductive coolant on electronics	loop pressure < 0.15 bar, leak pads and cable, branch ΔT	dry-break couplings, drip lips, drives above and upstream of joints, conformal coat; leak at an HC2 trips the chain; air-only rating 0.64 × (docs/03 §9.6)
Pump failure	one leg loses cooling	tach = 0, branch ΔT	check valves isolate the branch; that leg to the air-only rating; finish the task at reduced duty and sit
Coolant over-temperature	reduced margins	supply NTC	Thor to 40 W mode at 65 °C, speed limit, trip at 75 °C
Cycloid lobe or pin damage on impact	arm position loss, noise	dual-encoder mismatch, flexure torque vs current	drive-level current limits at the repeated rating; collision detection on the flexure
KH-14 strain-wave ratcheting	wrist/neck position loss	dual-encoder mismatch	drive limit 23 N·m; momentary rating 2 × repeated
Push-rod or rod-end failure	knee/ankle loss of drive → fall	encoder mismatch (actuator vs joint)	angle-dependent knee torque limit (rod ≤ 12 kN); buckling SF; inspection interval; harness during testing
Flange slip	joint offset	dual-encoder mismatch; witness marks	slip = overload fuse on the KA family; re-zero + re-torque procedure

Failure mode	Effect	Detection	Mitigation
HC2 STO fault (uncertified)	drive could keep torque after a stop	STO read-back mismatch, FAULT_N	commissioning contactor in each leg-zone feed; safety MCU refuses reset on inconsistent STO
EtherCAT cable break	loss of slaves	master WKC error	ring redundancy; Cat 1 stop if the ring is broken in two places
Pack cross-current on insertion	cell stress, fuse	INA228, BMS current	DIODE default; tie only at $\|\Delta V\| < 0.5$ V (≤ 4.4 A)
Pack pulled under load	arc at 67 V, connector damage	pilot loop, lever switch	fail-locked solenoid with hardware AND; pilot breaks ≥ 2 mm first
Single remaining pack overloaded	brown-out, fall	PPM, BMS DCL	single-pack power cap 1.8 kW; swap cap 1.5 kW; interlock refuses a release if the other pack < 54 V
Cell internal short / thermal runaway	fire	BMS (dT/dt, voltage), in-pack gas sensor, smoke	per-cell fuse links, mica barriers, fire wall between bays, in-pack FET switch, supervised charging, battery-area storage
RT computer crash	loss of control	safety-MCU heartbeat	Cat 1 → STO; harness
Estimator divergence	fall	covariance monitor, IMU vs kinematics residual	Cat 1 stop
Foot slip	fall	load-cell CoP jump, foot IMU	conservative friction cone ($\mu = 0.5$ assumed vs 1.0 sole); speed limits
Payload slip or shift	fall, dropped load	wrist F/T, CoM estimate jump	quick-release strap; harness for every G10 test; catch table
Hand overload	finger damage	current	breakaway detents, back-drivable screws, current limits
KW-L2 overheating in long one-hand holds	wrist droop	I^2t model, current	1 mm lead (545 N continuous vs 318 N needed); hold-time limit in the skill API
AI requests unsafe motion	collision	skill-API envelope check	requests are advisory; envelope enforced in the RT layer

9. Uncertainties and assumptions register

#	Assumption	Value used	How verified
1	Mass growth	+10 % (design 112.7 kg on a 102.5 kg estimate)	weigh modules (P12); re-run `run_all.py`
2	KM-130x50-LC Kt, saturation	model (R10 §5.1)	FEMM, then dyno (§4.1 items 1, 3)
3	Jacket thermal resistance	0.25 K/W (F_cool 2.4)	P05.1 thermal test (§4.1 item 2)
4	HC2 continuous current	75 A → 263 / 217 N·m usable continuous (below the 295 / 243 N·m motor rating)	HC2 thermal test on the jacket
5	Planetary efficiency	0.92	dyno
6	Cycloid efficiency and ratings	0.85; C25 160 / 250 N·m, C20 105 / 160 N·m (targets)	§4.1 item 6
7	Strain-wave efficiency (KH-14)	0.75	dyno
8	Voltage-limited speed factor	k_L = 0.75 (calibrated on a Ø115 TQ motor)	§4.1 item 5 at 42 poles
9	KW-L2 motor data	maxon catalogue values (98.5 / 540 mN·m, 10,600 rpm) with a 1 mm lead	§4.1 item 7
10	Radiator capacity	UA ≈ 20 W/K	loop bench (B131) and P08.1 §G
11	Rod-end static rating	≈ 30 kN (unverified)	purchased part datasheet
12	Gait model	flat-footed ZMP with dynamics filter (conservative knee); learned gait differs	measured torques vs model
13	Minimum loaded bus	52 V (3.25 V/cell); 3 V one-pack sag at 59 A	single-pack peak test
14	Prices	mostly estimate/quote	quotes before ordering
15	Unverified components	DC-DC bricks, Class T fuse DC rating, pack connector stagger, STO signal levels, HC2 (custom), wireless e-stop response time	datasheets/quotes before design freeze; tests in §5

10. Reporting

Each gate produces a short report in `logs/gates/Gx.md` : test list, data plots, deviations, decisions, the updated calc outputs, and the design holds closed. Keep failed attempts too; they are the most valuable data.



DIAGRAM Test gates

App. B Manufacturing package

This is the shop-floor specification for **KESTREL K1 Mk II**. The engineering numbers come from the generated files (`docs/generated/actuator_family.md` , `structure_checks.md` , `mass_budget.md` , `power_budget.md`) and from research note R10 (Mk II actuators). Where this page and an older document disagree, this page and the generated files win.
Read §11 (design holds) before you order any metal.

1. What is in the package and its status

Asset	Location	Status
Full robot assembly (named parts)	`viewer/kestrel_k1.glb` , generated by `cad/build_robot.py`	**envelope CAD**: sizes, positions and clearances for packaging; not every hole is modelled
Manufacturable part models (STEP + STL)	`cad/out/step/` , `cad/out/stl/`	shells, frames, tubes, rods, brackets: printable/machinable **as envelopes**; add the holes from §4 and clear the holds in §11 before ordering
Detailed load-path parts	`cad/out/step/detail__*.step` (ankle cross, foot plate, thigh plate, knee hub)	detailed: bores, bolt patterns, pockets. **Thigh plate and knee hub are on hold (§11 H4, H5)**
KA-130K actuator parts (full revolve)	`cad/out/step/ka130k__*.step`	**K1 geometry** (finned housing, z20/z50 stage 1, ID90/OD120 bearing). Mk II changes in §5, §6, §11 H1–H3
KH-25 actuator parts	`cad/out/step/kh25__*.step`	**K1 strain-wave geometry**, superseded by the KH-C25 cycloid (§5.3); use only the flexure and encoder parts as references
Cycloid discs	`cad/out/step/cycloid_KH-C25_disc.step` , `cycloid_KH-C20_disc.step` ; DXF from `python cad/cycloid_profile.py`	profile correct for 30:1; **disc bore must change (§5.3, H6)**
Drawing sheets (projected view + title block)	`cad/out/drawings/*.svg` (19 sheets)	visual reference; dimensions come from the STEP and the tables here
Gear, cycloid, lamination, winding and jacket specifications	§5–§7 below, R10 §5, docs/13	specification level

Status in one line: a complete preliminary design package. Each envelope part needs a detailing pass (holes, threads, fillets, datum features) to the standard of the four detailed parts, then FEA (docs/11 §3).

2. General tolerances, fits and standards

Item	Standard / value
General machining tolerances	ISO 2768-mK (medium linear, K geometric) unless noted
Limits and fits	ISO 286-1/-2. Every bearing, dowel, gear and pin seat gets an explicit fit on the drawing (table below)
Geometric tolerances on actuator seats	coaxiality of all bearing/gear seats in one housing ≤ 0.01 mm; seat face perpendicularity ≤ 0.005 mm; bore one housing in one setup
Threads	ISO metric coarse, 6H/6g. Every tapped hole in aluminium that carries a 12.9 screw or is reassembled gets a stainless free-running wire insert (Böllhoff HeliCoil Plus, length 2 × d)
Surface finish	shaft bearing seats Ra 0.4; housing bearing seats Ra 0.8; needle raceways (pins, cams, disc bores) Ra ≤ 0.2; gear bores and dowel holes Ra 0.8; O-ring bores Ra 0.8, groove flanks Ra 1.6; general Ra 3.2; anodised faces Ra 1.6 before anodise
Edge break	0.2–0.5 mm on all edges unless noted; O-ring lead-in chamfers 15–20°
Fasteners	ISO 4762 socket head, **property class 12.9, black oxide + oil or zinc flake** (never electro-zinc-plated 12.9: hydrogen embrittlement). Structural joints only with 12.9
Fastener preload	VDI 2230, 70 % of proof, μ_thread ≈ 0.12 with Loctite 243 wet: M3 ≈ 3.6 kN, M4 ≈ 6.9 kN, M5 ≈ 9.9 kN, M6 ≈ 14 kN, M8 ≈ 25 kN

Tightening torques (calibrated wrench, star pattern, two passes 50 % → 100 %):

Joint type	M3	M4	M5	M6	M8	Retention
12.9 structural (flanges, brackets, cranks), into steel or HeliCoil	1.7 N·m	**4.5 N·m**	**8.5 N·m**	14 N·m	34 N·m	Loctite 243, paint-pen witness mark
8.8 / A4-70 housing closures into aluminium	1.2 N·m	2.5 N·m	5.0 N·m	—	—	Loctite 222 (removable)
Rod-end jam nuts	—	—	—	—	—	M8 15 N·m, M12 25 N·m, Loctite 243
Lifting eye bolts DIN 580 M12	—	—	—	—	—	hand-tight to full seat, never loaded sideways

ISO 286 fits used in this package (deviations in mm):

Seat	Part	Shaft	Housing / bore
Rotor bearings 61808 (40 × 52 × 7)	KA-130, KH-C25 rotor hub	Ø40 k5 (+0.002/+0.013)	fixed side Ø52 K6 (+0.004/−0.015); floating side (wave spring) Ø52 H6 (0/+0.019)
Rotor bearings 61806 (30 × 42 × 7)	KA-90P, KH-C20	Ø30 k5 (+0.002/+0.011)	Ø42 K6 / H6 as above
Crossed roller THK RB9016 (90 × 130 × 16)	KA-130 output	Ø90 h5 (−0.015/0) for C1 preload, g5 (−0.027/−0.012) for normal clearance	Ø130 H7 (0/+0.040)
Crossed roller THK RB6013 (60 × 90 × 13)	KA-110P, KA-90P output (B014)	Ø60 h5 (−0.013/0)	Ø90 H7 (0/+0.035)
Crossed roller THK RB5013 (50 × 80 × 13)	KH-C25 output (B015)	Ø50 h5 (−0.011/0)	Ø80 H7 (0/+0.030)
Crossed roller THK RB4010 (40 × 65 × 10)	KH-C20 output (B015)	Ø40 h5 (−0.011/0)	Ø65 H7 (0/+0.030)
Crossed-roller fits	all	confirm against the THK fit table for the clearance class you buy (docs/03 §5 uses H6 / k5 with C1 preload); the values above are the usual inner-ring-rotating choice	
Planet pins with needle cages (INA K-series)	all planetaries	pin h5, 100Cr6 60–64 HRC, Ra ≤ 0.2	planet bore G6, ≥ 58 HRC (carburised planets)
Cycloid eccentric needle raceways	KH-C25 / KH-C20	cam Ø20 / Ø16 h5, ≥ 58 HRC	disc bore Ø26 / Ø22 G6, ≥ 58 HRC
Drawn-cup needle bearings INA HK1212	ankle cross	pin Ø12 h6 (−0.011/0), 100Cr6 60 HRC, Ra ≤ 0.2	Ø18 N6 (−0.009/−0.020)
Dowels ISO 8734 (hardened, m6) Ø4 / Ø6	all flanges	dowel m6 (Ø6: +0.004/+0.012)	press side H7 (Ø6: 0/+0.012) + Loctite 638; slip side G7 (Ø6: +0.004/+0.016)
Gear ring spigots	planetaries, cycloid ring	spigot js6	bore H7
Stator into liquid jacket (shrink fit)	KM-130x50-LC	stator stack OD 130 h7 (−0.040/0)	jacket bore Ø129.860 +0.025/0 → interference 0.075–0.140 mm (docs/13 §3.9 gives 0.06–0.08 mm, which goes loose in a uniform 100 K hot soak: Al on steel loses ≈ 0.15 mm on Ø130)
Stator into air-cooled housing (bonded)	RI85-PH, RI115-PH, KM-115, KM-85, KM-50	stack OD per maker	bore +0.02/+0.05 over the stack for a thermally conductive epoxy film
Bonded CF tube in 7075 fitting	thigh, shank, upper arm	radial bond gap 0.10–0.15 mm, 25 mm engagement, 3M Scotch-Weld DP490 or Loctite EA 9466, grit-blast + primer	

3. Materials, processes, heat treatment and coatings by part family

Family	Material / temper	Process	Heat treatment	Finish / coating	Notes
Actuator housings, flanges, carriers (aluminium)	7075-T651 bar/plate	CNC lathe + 3-axis mill	none (T651)	Type III hard anodise black 25–40 µm (MIL-A-8625 Type III Cl. 2); **mask** bearing seats, stator bore, bonding and earth pads	anodise grows 50 % inward: machine critical bores after anodise or mask them
Liquid jacket and housing sleeve	6061-T6 bar	CNC lathe: helical 6 × 3 mm groove on the jacket OD, O-ring grooves, ports	none; shrink-fit heating ≤ 150 °C for ≤ 30 min	coolant groove: Type II anodise or trivalent chromate (MIL-DTL-5541 Type II); **stator bore bare**	6061 for its corrosion behaviour in glycol; see §6.2
Pelvis bowl, yokes, brackets, spine, ribs, thigh/foot plates	7075-T651 plate/billet	3- or 5-axis CNC (or waterjet + CNC finish)	none	hard anodise black; flatness 0.1 mm on plates	stress relief not needed (T651)
CF tubes	roll-wrapped CFRP Ø40 × 2 (thigh/shank), Ø50 × 2 (upper arm)	diamond saw, ends sanded P120	—	clear coat	Enhanced Composites (AU) or Rock West
Ti parts (push-rods, wrist/ankle crosses)	Ti-6Al-4V annealed bar/tube (AMS 4928 / AMS 4911)	CNC, sharp tools, flood coolant, low speed	stress relieve push-rods after rough machining (595 °C, 2 h, argon/vacuum)	bead blast	dye-penetrant every part; **never use Ti as a bearing raceway**
Bearing pins (ankle, planet, rod-end clevis)	100Cr6 (AISI 52100)	turn, harden, grind	through-harden 60–64 HRC	ground h5/h6, Ra ≤ 0.2	replaces the K1 17-4PH/Ti pins, which are too soft for needle rollers
Planetary gears (sun, planets)	18CrNiMo7-6	hob or shape	carburise 0.6–0.8 mm case, 58–62 HRC surface, core 30–40 HRC	ground to ISO 1328-1 grade 6, crowning + tip relief	gear job shop to §5 tables
Planetary ring gears	42CrMo4 Q&T 28–32 HRC (or 18CrNiMo7-6 carburised + ground)	wire EDM or gear shaping	gas nitride 0.3 mm, ≥ 600 HV after cutting	—	ISO 1328-1 grade 6–7
Planet carriers, cycloid output ring	17-4PH H1025 or 42CrMo4 Q&T	CNC	as stated	black oxide (steel)	
Cycloid discs	**100Cr6, hardened 60–62 HRC before cutting** (preferred) or 42CrMo4 Q&T + nitride with a pressed 100Cr6 raceway sleeve	wire EDM in the hardened state: rough + 3 skim cuts; or CNC jig/profile grinding	as stated	lap the profile to remove the EDM recast layer	§5.3
Cycloid ring	42CrMo4 Q&T 28–32 HRC	wire EDM pin grooves	gas nitride	—	ring pins roll in the grooves
Cycloid eccentric sleeves	100Cr6 60–62 HRC	turn, harden, grind OD and bore in one setup	as stated	ground h5, Ra ≤ 0.2	E matched within 0.003 mm per pair; 180° phase by a common key
Rotor back iron	1018 / C15 steel	turn	none	phosphate	magnets bonded, CF/Kevlar 0.3 mm sleeve, balance G2.5
Laminations	NO20-1200H 0.20 mm (Tata/Surahammars)	laser or wire-EDM cut, bonded stack	stress-relief anneal if laser-cut (per steel maker)	C-5 insulation coat	do not fall back to 0.35 mm for the KM-130x50 (R10 §3.2.5)
Shells (≈ 42)	PA12 HP MJF	MJF print (KAD3D, Formero AU [R6])	—	bead blast or vapour smooth → 2K primer, satin top coat	wall 2.0–2.4 mm; M3 heat-set inserts in bosses
Visor	PC 3 mm	thermoform over a printed buck, or machine	—	smoke tint + AR hard coat on apertures	

Family	Material / temper	Process	Heat treatment	Finish / coating	Notes
Bellows/gaiters	TPU 95A	FDM print or urethane cast	—	—	
PCBs	FR-4 4-layer 2 oz (power), 6-layer (hand, wrist, GaN drive)	JLCPCB / PCBWay assembly	—	acrylic conformal coat (IPC-CC-830); boards near coolant: silicone conformal coat	IPC-A-610 class 2
Battery enclosures (×2)	6061-T6	CNC	—	clear anodise; 1 mm mica liners	§8
Fingertip caps	Smooth-On Dragon Skin 20	cast in printed two-part moulds	—	—	dovetail onto a PA12-CF core

4. Interface standards (use on every envelope part when detailing)

Interface	Standard	Slip torque (μ 0.15) vs worst demand
KA-130K/H output flange	PCD 105, **12× M5 12.9** (HeliCoil), 2× Ø6 H7 dowels at 180°, spigot Ø90 h6	935 N·m vs 535 N·m momentary (SF 1.75)
KA-130K/H base flange	PCD 125, 12× M5 12.9, 2× Ø6 dowels	1,114 N·m
KA-110P output flange (Mk II)	**PCD 100, 12× M5 12.9** (HeliCoil), 2× Ø6 H7 dowels at 180°, spigot Ø80 h6	**891 N·m vs 379 N·m momentary (SF 2.35)**; K1's 10× M4 on PCD 85 slipped at 440 N·m (SF 1.16)
KA-110P base flange (Mk II)	**PCD 115, 12× M5 12.9**, 2× Ø6 dowels	1,025 N·m. The base carries the same reaction torque as the output, so it gets the same bolt class
KA-90P output and base flanges	PCD 75, 8× M4 12.9, 2× Ø4 H7 dowels, spigot Ø60 h6	310 N·m vs 149 N·m (SF 2.08)
KH-C25 output flange	PCD 70, 8× M4 12.9, 2× Ø4 dowels	290 N·m vs 136 N·m
KH-C20 output flange (Mk II)	**PCD 58, 8× M4 12.9** (was 8× M3), 2× Ø4 dowels	240 N·m vs 115 N·m elbow momentary; K1's 8× M3 gave 125 N·m (SF 1.09)
KH-14 output flange	PCD 42, 6× M3 12.9, 2× Ø3 dowels	68 N·m vs 12 N·m
Planetary stage-2 ring to housing (carries ≈ $(i_2 - 1)/i_2$ of the output torque)	KA-130: 12× M4 12.9 + 2 dowels; KA-110P: 12× M4 12.9 + 2 dowels; KA-90P: 8× M3 12.9 + 2 dowels, all on the ring flange PCD	KA-130 ≈ 650 N·m vs 375 N·m ring reaction; KA-110P ≈ 690 vs 350; KA-90P ≈ 190 vs 106
Cycloid ring to housing	8× M4 12.9 + 2 dowels (C25), 8× M3 12.9 + 2 dowels (C20)	carries the full output torque
Hollow cable bores (maximum the gear trains allow, §5)	Plan A: KA-130H Ø12, KA-130K / KA-110P / KA-90P none. Plan B: KA-130 Ø14, KA-110P Ø10, KA-90P Ø8. KH-C25 Ø12, KH-C20 Ø10, KH-14 Ø12	the Ø20 / Ø15 bores of docs/03 §11 do not fit inside any of the sun gears (§11 H2)
Coolant ports (KA-130 jacket)	2× G1/8 (ISO 228-1) with ISO 1179 stud ends and bonded seals, inlet at the bottom when standing	
Shell bosses	M3 heat-set insert, Ø6 boss, 3 mm silicone isolator	
Joint zero pins	Ø4 mm H7 hole through link + actuator housing at the zero position	

Flanges are friction joints. The dowels locate; the bolt preload carries the torque. On the KA family the slip torque is the designed overload fuse. On the KH-C25 and KH-C20 the slip torque is *above* the cycloid's momentary rating, so the arm is protected by the drive-level current limits in §5.3, not by slip.

5. Gear trains

5.1 Planetary sets (KA family): Mk II tooth counts

All sets are simple two-stage planetaries, 20° pressure angle, zero total profile shift (suns below z17 get x ≈ +0.3, planets the matching negative shift). **Plan A is the baseline tooth plan of docs/04 §5.2 and §7**; its KA-130 stage 2 is the K1 geometry checked in `structure_checks.md` (m1.25, z18, 18 mm face, Lewis SF 2.16) and drawn in `ka130k__s2_ring_gear_z54_m1.25.svg`. Every set satisfies the coaxial condition, the equal-spacing assembly condition (Zs + Zr)/N = integer and ≥ 0.6 mm planet-tip clearance, and leaves ≥ 2 m of rim between the planet root and its needle bore. The K1 table did not: its stage-1 sun z20 / ring z50 with 3 planets gives (20 + 50)/3 = 23.3 (planets cannot be equally spaced), and its z15 planets left 0.5 mm of steel round a Ø6 pin plus needles.

Class (ratio)	Stage 1 (module, sun / planets / ring, face)	Stage 2 (module, sun / planets / ring, face)	Ratio	Max through-bore
KA-130K (17:1)	m0.8, z16 / **4** × z18 / z52, b 14 (inside the Ø85.5 rotor bore)	m1.25, z18 / 4 × z18 / z54, b 18	4.25 × 4.0 = **17.0**	≈ Ø6 (stage-1 sun)
KA-130H (14:1)	m0.8, z24 / 3 × z18 / z60, b 12 (inside the rotor bore)	same parts as the knee	3.5 × 4.0 = **14.0**	Ø12
KA-110P (29:1)	m0.6, z15 / 3 × z30 / z75, b 10 (ring PCD 45 inside the Ø74 rotor bore)	m1.0, z24 / **4** × z34 / z92, b 16	6.0 × 4.83 = **29.0**	none (z15 sun)
KA-90P (28:1)	m0.6, z15 / 3 × z27 / z69, b 8	m0.8, z18 / 3 × z27 / z72, b 12	5.6 × 5.0 = **28.0**	none (z15 sun)

Through-bores (hold H2). docs/03 §11 still routes leg power and EtherCAT through Ø20 / Ø15 actuator bores. No sun gear above fits round those bores. If signal cables must pass through the axis, use **Plan B**; it keeps the ratios and opens KA-130 Ø14, KA-110P Ø10, KA-90P Ø8 (EtherCAT only; 10 AWG power and coolant still cross in external service loops). Decide before ordering gears.

Plan B (bore-compatible)	Stage 1	Stage 2	Ratio
KA-130K	m0.8, z25 / 3 × z26 / z77, b 14	m1.25, z24 / 4 × z26 / z76, b 18	4.08 × 4.17 = 17.00
KA-130H	m0.8, z25 / 3 × z17 / z59, b 12	same as KA-130K	3.36 × 4.17 = 14.00
KA-110P	m0.7, z22 / 3 × z38 / z98, b 10	m1.25, z18 / 3 × z30 / z78, b 16	5.45 × 5.33 = 29.09
KA-90P	m0.6, z23 / 3 × z34 / z91, b 8	m1.0, z17 / 3 × z31 / z79, b 10	4.96 × 5.65 = 27.99

Stage-2 root-stress screening (Lewis) at the delivered peak: Plan A KA-130 541 MPa (generated, SF 2.16 vs 1,170 MPa), KA-110P ≈ 423 MPa, KA-90P ≈ 518 MPa (docs/04 §7); Plan B KA-130 z24 ≈ 340 MPa at 1.1 × peak. **Run ISO 6336 (bending and pitting) in detail design**; R10 §5.1's m1.5 × 14 mm KA-130 stage 2 (the BOM line B007 quotes it) is the fallback if ISO 6336 asks for it. If the as-built ratio differs from the sized value, put it into `calc/kestrel/actuators.py`.

Item	Specification (Plan A)
Planet pins	100Cr6 60–64 HRC, ground h5: Ø5 (KA-130 and KA-90P stage 1), Ø6 (KA-110P stage 1), Ø10 (KA-130 stage 2), Ø12 (KA-110P stage 2), Ø8 (KA-90P stage 2)
Planet bearings	INA needle roller and cage assemblies on those pins: K5×8×8, K6×9×8, K10×14×13, K12×16×13, K8×12×10 (verify C/C0 against the planet load at 1.1 × peak); planet bore G6
Carriers	17-4PH H1025, two cheeks pinned + screwed, pin holes jig-bored in one setup (position ≤ 0.01 mm)
Ring gears	42CrMo4 Q&T, wire EDM, nitrided; spigot H7/js6, 2 dowels, 12.9 screws (§4)
Lubrication	Klüber Isoflex Topas NB 52 (or equivalent synthetic NLGI 2), 30–40 % of the free volume; O-ring sealed housings, V-ring at the output
Backlash target	≤ 10 arcmin at the output, measured with the two encoders
Inspection	supplier gear report (profile, lead, runout); incoming over-pin measurement on every gear; double-flank roll test if available

5.2 Strain-wave (KH-14 only)

KH-14 keeps the CSG-14-50 class component set (Leaderdrive LHSG-14-50, B012) with Harmonic Grease SK-1A (or 4B No.2 per the vendor), housing tolerances per the vendor's installation drawing (circular spline/wave generator concentricity ≤ 0.01 mm). Repeated-peak rating 23 N·m: the drive limit in §5.3 is mandatory.

5.3 Machined two-disc cycloid reducers (KH-C25, KH-C20)

Item	C25 (KH-25: shoulder pitch, roll)	C20 (KH-20: shoulder yaw, elbow)
Ratio	30:1 (31 ring pins, 30-lobe discs)	30:1
Ring-pin circle / pins	Ø84.000 ±0.005; 31 × Ø5 needle rollers ISO 3096 grade G2 × 19.8 mm (or ISO 8734 Ø5 m6 dowels, 60 HRC)	Ø68.000 ±0.005; 31 × Ø4 × 15.8
Eccentricity E	0.900 ±0.003 mm	0.800 ±0.003 mm
Discs	2 × 8.0 mm, mounted 180° apart, stack-cut as a matched pair	2 × 6.0 mm (7.0 mm if docs/13 §4.2's recommendation is adopted; same profile)

Item	C25 (KH-25: shoulder pitch, roll)	C20 (KH-20: shoulder yaw, elbow)
Profile	`cad/cycloid_profile.py`, 0.02 mm equidistant clearance	same
Output pins	8 × **solid Ø8** ISO 8734 m6, ≥ 58 HRC, on PCD 52, supported at both ends (output disc and output ring); disc holes Ø9.80 +0.01/0	6 × solid Ø6 on PCD 40; holes Ø7.60 +0.01/0
Eccentric bearings	needle roller and cage assembly per disc, Fw 20 / Ew 26, width ≤ 8 mm (INA K20×26×xx class; C0 ≥ 8 kN), running directly on the hardened cam and disc bore	Fw 16 / Ew 22, width ≤ 6 mm (INA K16×22×xx class)
Cam (built up)	two separate eccentric sleeves, 100Cr6 60–62 HRC, OD Ø20 h5, E 0.900, keyed 180° apart on a hollow input shaft (bore Ø12), so each disc and cage slides on in turn	sleeves Ø16 h5, E 0.800, shaft bore Ø10
Input/output support	input shaft journals Ø17: 61803 (17 × 26 × 5) in the output disc, 61808 rotor bearings at the motor end; output on THK RB5013 (50 × 80 × 13)	journals Ø15: 61802 (15 × 24 × 5) + 61806 rotor bearings; output on THK RB4010 (40 × 65 × 10)
Grease	Nabtesco VIGOGREASE RE0 (RV-reducer grease) or Molykote BR2 Plus	same
Rating targets (R10, to be proven on the dyno)	rated 80 / repeated **160** / momentary **250** N·m	52 / **105** / **160** N·m
Motor capability (generated)	KM-115x32-N50 → **362 N·m peak**	KM-85x26-N50-4t → **158 N·m peak**
Drive peak-current limit (set in the drive's non-volatile parameters, not only in the RT software)	**21 A rms** (160 N·m at η = 1), then trim on the dyno to 152–160 N·m	**18.8 A rms** (105 N·m at η = 1)

KH-14 on the same principle: 23 N·m / (50 × 0.106) = **4.3 A rms**.

Why the bearing change: in a cycloid the eccentric bearing carries roughly T_out/(N·E) of tangential load. At 160 N·m that is ≈ 3 kN per disc plus the radial component. A 6802 ball bearing (the size noted in `cycloid_profile.py`) is far too small, so the discs run on needle cages and their bore must be **Ø26 (C25) / Ø22 (C20)**, not the Ø22 modelled (hold H6). Output pins are solid and doubly supported because a Ø5 pin inside an Ø8 sleeve would see ≈ 1.7 GPa bending at 250 N·m.

Manufacturing routes for the discs (choose one; all need the measurement plan below):

Route	Accuracy	Notes
A. Harden 100Cr6 blanks to 60–62 HRC, then wire EDM: rough + 3 skims, bore and output holes in the same setup, 2 discs stacked	profile ±0.005 mm, Ra 0.4–0.6	preferred for a home builder; lap the profile and bore lightly to remove the 2–5 µm recast layer
B. Wire EDM 42CrMo4 Q&T, then gas nitride	±0.010 mm after nitriding growth	press a hardened 100Cr6 sleeve into the bore as the needle raceway
C. Jig/profile grinding from the DXF (Moore-type CNC jig grinder or a gear shop with a cycloid program)	±0.002 mm, Ra ≤ 0.2	best result, specialist and costly

6. Motors and cooling

6.1 Motor manufacturing specification (full procedure in docs/13 §3)

Item	KM-130x50-LC (KA-130K and KA-130H, 4 + 1 spare)	KM-115x32-N50 (KH-C25)	KM-85x26-N50-4t (KH-C20)	KM-50x20 (KH-14)
Laminations	0.20 mm NO20-1200H, Ø130 die (K1 die), 36 slots	Ø115 die, 36 slots	Ø85 die, 36 slots	Ø50, 12 slots
Stack	**50 mm (250 laminations)**, OD 130 h7	32 mm	26 mm	20 mm
Poles / magnets	42 × **N50UH** arcs, 3.0 mm thick, 50 mm long	40 × N50UH, 2.8 mm	42 × N50UH, 2.5 mm	14 × N42SH, 2.0 mm
Winding	double-layer tooth coils, **3 turns × 2 parallel paths**, ≈ 3.65 mm² per conductor (≈ 10 × Ø0.67 mm strands), grade 2 Class H 200 °C	3 turns	4 turns	18 turns
Kt / R_LL / Km (model)	0.224 N·m/A rms / 20.3 mΩ / 1.286	0.252 / 70.2 mΩ / 0.777	0.186 / 146 mΩ / 0.396	0.106 / 865 mΩ / 0.093
Continuous / peak motor torque	18.85 N·m (jacketed) / 31.5 N·m at 150 A	5.49 / 14.2 N·m	2.01 / 6.18 N·m	0.28 / 0.90 N·m

Item	KM-130x50-LC (KA-130K and KA-130H, 4 + 1 spare)	KM-115x32-N50 (KH-C25)	KM-85x26-N50-4t (KH-C20)	KM-50x20 (KH-14)
Impregnation and potting	vacuum dip in Class H varnish; slots and end turns potted with thermally conductive epoxy ≥ 1.5 W/m·K, Class H	same	same	varnish
Rotor	1018 back iron, magnets bonded with high-temperature epoxy, 0.3 mm CF/Kevlar sleeve, balanced G2.5	same	same	same
Acceptance tests	R (4-wire), L, back-EMF, hi-pot 1,000 V AC 1 min, locked-rotor Kt, cogging (docs/13 §3.10), plus the FEMM check of docs/11 §4.1 before cutting the die	same	same	same

The bought motors are CubeMars **RI115-PH** (KA-110P, 3 + 1 spare) and **RI85-PH** (KA-90P, 7 + 1 spare). The TQ ILM85x26 / ILM85x13 and the KM-130x45 / KM-130x40 of K1 are no longer used.

6.2 Liquid jacket and loop (KA-130K, KA-130H; layout in docs/03 §9, electrical in docs/05 §13)

Item	Specification
Jacket (B103)	6061-T6 sleeve, 2 mm web, stator shrink-fitted (fit in §2: heat the jacket to 150 °C, stator at room temperature, push home in one stroke against the shoulder). Housing OD at the jacket ≈ 145 mm
Channel	**two-start helix**, each start 6 mm wide × 3 mm deep, pitch 18 mm, on the jacket OD, closed by the 7075 housing bore (two starts quarter the pressure drop: ≈ 1.4–1.8 kPa at 0.5 L/min, docs/03 §9.3)
Seals	**two EPDM 70 O-rings at each end with a leak-test port between them**; radial static seal, grooves per ISO 3601-2 for 18–25 % squeeze; EPDM because it is the glycol-compatible elastomer; assemble with silicone grease (mineral grease swells EPDM)
Ports	G1/8 (1/8 BSPP) ISO 1179 stud ends with bonded seals and barbs for the 6 mm ID hose; **inlet at the bottom, outlet at the top when standing** (air purges upward), docs/13 §3.9
Drive mounting	the HC2 board bolts to the jacket's rear face, which is its cold plate (docs/05 §6.1); drive above the O-rings when standing, behind a drip lip
Thermal path	copper → potting → laminations → shrink fit → jacket → coolant; target ≤ 0.25 K/W copper-to-coolant-inlet, reject > 0.30 K/W (docs/11 §4.1)
Loop	sealed and fully filled, **no reservoir open to air** (a humanoid tilts and falls; the BOM's 60 mL reservoir B123 is replaced by the compensator): radiator → spring-diaphragm compensator (40 mL, EPDM membrane in a PA12 housing, 0.3 bar) with fill/bleed valve and pressure sensor → Thor cold plate → 2 pumps in parallel, one per leg branch, each with a check valve → per leg **knee jacket first, hip-pitch jacket second** → return manifold → radiator (docs/03 §9.2)
Flow	0.5 L/min per leg branch, ≈ 1.0 L/min through radiator and cold plate; branch pressure drop ≈ 23 kPa
Pumps / radiator / fans	**component list of docs/03 §9.4**: 2 brushless 12 V pumps (TCS Micropumps M510 or Xylem/Laing DDC class; ≥ 0.5 L/min at ≥ 30 kPa, tach, ≤ 0.15 kg, ≤ 10 W each); one 240 × 120 mm radiator, ≤ 30 mm core, ≈ 20 W/K, aluminium core (or a mixed-metal-safe radiator with the inhibited coolant); 2 × 120 mm IP68 PWM fans (San Ace waterproof class) in the wet duct. The BOM's Koolance PMP-420 / HX-720 (B119/B120) work but weigh ≈ 1.5 kg more (docs/05 §13). Radiator UA is an assumption: measure it on the loop bench (B131)
Coolant	**40 % propylene glycol with an aluminium-safe (mixed-metal) inhibitor**, freezing ≈ −21 °C (docs/03 §9.4). All wetted metal aluminium; **no copper or brass** in the loop. The BOM's insulative PC coolant (B122) is acceptable only if its freeze point and aluminium compatibility are confirmed
Hose	6 mm ID × 10 mm OD EPDM or Saint-Gobain Norprene (B126), rated ≥ 3 bar at 80 °C, minimum bend radius 30 mm; ≤ 3 kPa EPDM check valve per pump; external service loops at waist yaw, waist pitch, hip yaw, hip roll and hip pitch (docs/03 §11)
Quick-disconnects (B125)	non-spill dry-break pair per leg crossing (CPC LQ / Koolance QD3 class, 1/4 in), in the leg limb-interface service plate (docs/03 §13.1); verify ≥ 3 bar at 70 °C and re-test after 100 mate cycles
Sensing	2 flow sensors (Sensirion SLF3S-4000B, B127, calibrate on the coolant), NTCs at supply/return of each branch and the jacket outlets (B128), loop pressure 0–2.5 bar (TE M3200 class, B130), pump tach, 8 leak pads + torso leak cable (B129)
Mass	≈ 1.5 kg for the torso part of the loop with the docs/03 parts (radiator 0.60, 2 IP68 fans 0.30, 2 pumps 0.30, compensator/valves/manifold 0.10, coolant ≈ 0.15, blower 0.08), ≈ 0.3 kg over the 1.25 kg in `params.py`; loop volume ≈ 0.47 L (docs/03 §9.5, §9.7)
Test pressures	jacket proof 3 bar water 10 min, then 2 bar air under water 10 min and 2 bar decay ≤ 5 mbar in 10 min (docs/13 §3.9); loop vacuum-fill ≤ 50 mbar hold 10 min (≤ 5 mbar rise), then 1.5 bar leak test with the pumps isolated; operating 0.3 bar

7. Electronics manufacturing

- **Boards** (docs/05): Everest drive carrier ×22 (B027); **HC2 150 A GaN drive** ×4 + 1 (B025, custom; bring-up on the EPC9186HC2 board B109); **PDU KES-PDU2** (B035: logic/actuator bus split, 7 zone eFuses, brake chopper); **pack power path KES-PPM** (B030: LM5069 + LM5050-1 OR path and TPS48111-Q1 tie switch per pack, INA228, solenoid-lock driver, hardware unlock AND); **pack BMS KES-BMS16** ×2 (B029: BQ76952 + BQ77216 + in-pack

FET switch); safety MCU (B042); hand controller Mk II ×2 (B060); **KW-L2 driver KES-WRIST** ×2 (B107: 2 × DRV8353; driven by the hand controller per docs/05 §7, or its own F28P65x slave per the BOM, see H11); **forearm 24 V buck** (LM5146-Q1, B114 or on B107); foot board ×2. 4–6 layer FR-4, 2 oz copper on power layers, ENIG, JLCPCB/PCBWay assembly. Every board gets a test pad set and a scripted bring-up test (docs/10 P02).

- **Coolant-adjacent boards** (HC2, PDU, PPM): conformal coat (silicone or dielectric-tolerant acrylic), interdigitated **leak pads** (B129) on the board edge facing the nearest coolant fitting, and no fitting directly above any board.
- **Harness:** built on a printed nail-board from the CAD routing; every wire labelled; 100 % continuity and insulation-resistance test (500 V, > 100 MΩ) before installation.

8. Battery pack manufacture (two hot-swap packs)

Two identical **16S4P** packs, each 64 × Molicel INR-21700-P50B (B028): 59.2 V nominal, 67.2 V full, 48 V cut-off, 19.4–20 Ah, **1.15 kWh**, 5.9 kg budget. Each pack carries its own BMS and FET switch (B029), a 100 A Class T fuse (B031) and the pack half of the blind-mate connector (B111), so a removed pack has **dead terminals**. OR-ing, hand-off and inrush limiting are on the robot side (KES-PPM, docs/05 §4). Build procedure: docs/10 P06; bay: P06.1. Build in a cleared area with a sand bin and a Class D (lithium) or large AVD/water-mist extinguisher, never alone.

9. Inspection plan

Part class	Inspection	Accept
Bearing seats, gear and dowel bores	bore gauge / CMM, 100 %	per §2 fits
Gears	supplier report (profile, lead, runout) + incoming over-pin check	ISO 1328-1 grade 6
Cycloid discs	CMM or optical comparator (≥ 72 points per lobe), over-pins with two Ø5 (Ø4) gauge pins in opposite lobes, bore-to-profile concentricity, pair match	profile ±0.005 mm; concentricity ≤ 0.005 mm; pair over-pins within 0.003 mm
Cycloid ring	pin-groove circle by CMM	Ø84.000 / Ø68.000 ±0.005, pitch error ≤ 0.005 mm
Liquid jackets	proof 3 bar water 10 min, then 2 bar air decay	no leak; ≤ 0.01 bar drop in 15 min
Flatness-critical plates	surface plate + dial indicator	0.1 mm
Ti parts	dye-penetrant crack inspection after machining	no indications
Hardened pins and cams	file test + Rockwell on a witness piece per batch	≥ 58 HRC
Bonded tube joints	proof load 1.5 × structural-case axial + torsion on a coupon from each batch	no failure
Shells	fit check on the assembly before painting	gaps 1.0 ± 0.3 mm
Cells	100 % mass, OCV and 1 kHz IR (R3)	reject outliers

10. BOM cross-reference and BOM issues

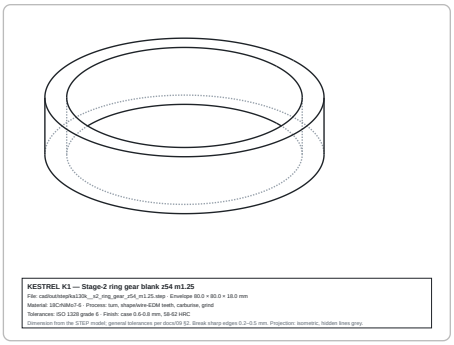
`data/bom.csv` / generated/bom.md now carries the Mk II bill (Mk II additions from B101). The tutorial uses those IDs. Items where the BOM and this package differ, to be reconciled by the BOM owner:

BOM line	BOM says	This package / reason
B007	KA-130 17:1 with "s2 m1.5 × 14 mm"	Plan A stage 2 is m1.25 × 18 mm (docs/04 §5.2, `structure_checks.md`, existing drawing); m1.5 is the ISO 6336 fallback
B013	crossed roller "~ID90/OD120"	no THK RB size; RB9016 (90 × 130 × 16) and a CAD seat change (H1)
B014	KA-110 / KA-90 crossed roller ~ID60/OD90	THK RB6013; confirm the PCD 100 KA-110P output hub overhang in CAD
B031	100 A Class T fuse	agreed (docs/05 §4.6); verify ≥ 125 V DC and ≥ 20 kA DC interrupting on the datasheet
B084	dyno "rated for 500 N·m"	the KA-130K peak is 493 N·m: absorber ≥ 600 N·m and a 1,000 N·m transducer (P01)
B106	KW-L2 ball screw Ø8 × 2 ("1 mm lead preferred if rated")	**1 mm lead** (H10): KSS Ø8 × 1, or Ø6 × 1 if its Ca ≥ 1.5 kN

BOM line	BOM says	This package / reason
B122	insulative PC coolant	40 % propylene glycol with aluminium-safe inhibitor (docs/03 §9.4) unless B122's freeze point and Al compatibility are confirmed
B016	rotor bearings "68xx / 61xx" × 52	sizes per actuator: 61808 (KA-130, KH-C25), 61806 (KA-90P, KH-C20), per drawing (KA-110P, KH-14); plus cycloid journals 61803 / 61802
—	not listed	hardened 100Cr6 bearing pins (planets, ankle cross), INA needle cages, cycloid needle cages, ISO 3096 ring-pin rollers, eccentric sleeves (in B010/B011 cost?), HeliCoil Plus inserts, DIN 580 M12 lifting eyes

11. Design holds: resolve before cutting metal

#	Hold	Why	Owner
H1	KA-130 crossed-roller seat is ID90/**OD120** in the CAD, docs/03 §5 and B013; no THK RB size matches. Use RB9016 (90 × 130 × 16) and re-rate it for 1.1 × 493 N·m plus the knee push-rod load (radial ≤ 12 kN with the knee torque limit of docs/10 P07 §F)	bearing not purchasable as modelled	CAD + BOM
H2	Through-bores: docs/03 §11 and docs/05 §9 pass 10 AWG leg power and EtherCAT through Ø20 / Ø15 actuator bores; the Mk II sun gears (§5.1 Plan A, docs/04) leave none, Plan B leaves Ø14 / Ø10 / Ø8 for signal only. Choose the plan and re-route power externally before gears are ordered	geometry impossible as written	docs/03 + docs/04 + docs/05
H3	`ka130k__*.step` and the drawings are K1 geometry (finned housing, z20/z50 stage 1). Mk II jacket, two-start helix, HC2 bay and stage-1 parts are not yet modelled	parts not drawn	CAD
H4	`detail__thigh_plate` : only 3 of the 12 holes of the PCD 105 hip flange lie inside the plate outline → slip ≈ 234 N·m against 364 N·m hip-pitch momentary and 406 N·m actuator peak. Needs a full-circle hip boss or a separate 12-hole adapter ring	joint would slip	CAD
H5	`detail__knee_hub` : (a) the 6× M5 PCD 52 crank and shank joints slip at ≈ 232 N·m against 535 N·m knee momentary; make crank and hub one piece, or ≥ 12× M6 12.9 on PCD ≥ 90; (b) the Ø42 × 9 bearing seats fit only 71905-size angular-contact bearings, whose static rating is below the X-case knee load (push-rod 8.8 kN + GRF 2.8 kN). Re-select (crossed roller or a 7206-size pair) and resize the hub	joint would slip; bearings overloaded	CAD + calc
H6	Cycloid disc bore Ø22 in `cycloid_profile.py` (comment: 6802 ball bearing) → Ø26 G6 for the C25 needle cage; C20 keeps Ø22 with a Ø16 cam	eccentric bearing capacity	CAD
H7	Knee push-rod force near the range ends. At full actuator torque: 18.9 kN at 0°, 23.1 kN at 130°. docs/04 §5.3's limit τ_act ≤ 535 / G(θ) still allows 15.5 kN at 0° and 21.2 kN at 130°, against the rod-end static rating ≈ 30 kN (unverified) and buckling 38.6 kN. Add the rod-force term of docs/10 P07 §F; `structure.py` should check rod force over the whole range at actuator capability	rod-end overload	calc + docs/04
H8	HC2: no purchasable 150 A pk, 100 V-class EtherCAT drive; custom (B025). EPC9186HC-class stages are rated ≤ 76 V input: OV trip at 74 V, chopper 70 / 68 V. HC2's 75 A continuous is below the motor's 84 A thermal rating, so the usable continuous is ≈ 263 / 217 N·m (docs/05 §6.1), not 295 / 243. **Its STO is uncertified:** through commissioning the safety function for the knee and hip is the external safety relay plus a DC contactor in each leg-zone feed (P02, docs/11 §2)	key part custom; ratings lower than the generated family; stop function unproven	electrical + calc
H9	Hip-pitch / thigh twin-plate layout: the KA-130H output is on one side only, and the 124 mm KA-130K does not fit between plates 80 mm apart in the envelope CAD	assembly order cannot be fixed until detailed	CAD
H10	KW-L2 screw lead: with the K1/R10 **2 mm** lead the ECX FLAT 32 L (98.5 mN·m continuous) gives only 272 N continuous against ≈ 318 N held for the 15 kg one-hand case. **Mk II uses a 1 mm lead** (5.53 N per mN·m): 545 N continuous, 754 N repeated at 136 mN·m, 972 N momentary at 176 mN·m (stall 540 mN·m), ≈ 164 mm/s at the 216 N speed corner (≥ 132 required). docs/13 §4.4 and BOM B106 still say Ø6/Ø8 × 2. Screw: KSS Ø8 × 1 (or Ø6 × 1 if its Ca ≥ 1.5 kN)	wrist would overheat in long holds with 2 mm	docs/04 + docs/13 + BOM
H11	Owner disagreements: loop order (docs/03 §9.2 knee jacket first; docs/05 §13 hip-pitch jacket first); coolant (docs/03 40 % propylene glycol; BOM B122 insulative PC coolant); reservoir (docs/03 sealed compensator; BOM B123 60 mL reservoir); EtherCAT slaves (docs/05 §7: 28, KW-L2 on the hand controller; docs/08 and BOM B107: 30, KW-L2 board with its own F28P65x slave). This package follows docs/03 and docs/05	inconsistent instructions	docs/03 / docs/05 / docs/08
H12	Pack envelope: 2 × 4 × 8 P50B cells fill the 150 × 98 × 185 mm box in `params.py` with ≈ 2 mm to spare; the BMS, fuse, handle and connector half (docs/03 §10.1: rear end cap) need ≈ 25 mm more (≈ 175 × 98 × 185 mm). docs/05 §3.1 also flags the 1.36 kg left for everything but cells	packs will not close as budgeted	CAD + params
H13	Bearing-pin hardness: the K1 package put needle rollers on Ti-6Al-4V (ankle cross pins, `structure_checks.md` row) and 17-4PH H900 (planet pins). Needle raceways need ≥ 58 HRC: use 100Cr6 60–64 HRC ground pins (§3). The strength rows stay conservative	raceways would brinell and wear	calc (label only)



DRAWING ka130k / s2 ring gear z54 m1.25

App. C Optional: scratch-built motors and actuators

This track covers the parts you make yourself: wound copper stators, magnet rotors, liquid jackets, cut gears, machined cycloids, housings and drives. In Mk II much of it is **no longer optional**. No commercial part meets these requirements, so the baseline (docs/04) builds:

- **KM-130x50-LC** (knee, hip pitch) with its liquid jacket;
- **KM-115x32-N50** (shoulders) and **KM-85x26-N50-4t** (shoulder yaw, elbow);
- KM-50x20 (wrist yaw, neck);
- the **C25 and C20 machined two-disc cycloids**;
- the **KW-L2** wrist linear actuators, which use a bought motor and screw in your own housing.

The optional part of the track replaces the remaining bought motors: RI115-PH by KM-115x32-N50, RI85-PH by KM-85x26-N50-3t, and the bought drives.

Design data: generated/actuator_family.md (Mk II motor table), research/R10 §5 and §8 (derivations of every Mk II motor), and generated/km_scratch_designs.md / generated/km_motor_designs.md for the K1 N42SH designs and the 36s/42p coil table. The two generated KM sheets still show the K1 N42SH variants; the Mk II N50UH numbers below come from R10 and `actuators.py`. Tags: **[E]** estimate (model or arithmetic shown), **[VT]** validation target, **[V]** datasheet.

1. What can and cannot be made: feasibility matrix

Part	Make at home	Make with a job shop	Buy only	Why / closest buildable alternative
Stator **laminations** (0.20 mm NO20)	× (laser/EDM + insulation-coated electrical steel)	✓ laser-cut or wire-EDM stacks to your DXF		Bonding is done at home. Keep 0.20 mm NO20 for Mk II: iron loss at maximum knee speed ≈ 85 W (R10 §3.2.5)
Segmented stator teeth + rectangular-wire bobbin coils (KA-150X only)	×	△ wire-EDM segments ✓; layer-wound rectangular coils need a bobbin winder		Not in the Mk II baseline (docs/04 §6)
Stator **winding** (round multi-strand)	✓	✓		Hand needle-winding with an indexing jig is practical for 36 slots
Magnets (sintered NdFeB, **N50UH**)	×	×	✓	Sintering, grinding and magnetising need an industrial process (hydrogen decrepitation, jet milling, aligned pressing, vacuum sintering at ~1,080 °C, pulse magnetisers). **Buy custom arcs, N50UH, NiCuNi coated, with grade certificates (Br, Hcj at temperature).** UH grades have longer lead times than SH; order early. Bonded NdFeB has about half the remanence and would cost ~40 % of the torque
Rotor back-iron, hubs, shafts	✓ (lathe)	✓		1018 ring, 7075 hub
Rotor **sleeve**	✓			wet-wound carbon tow on a mandrel
Liquid jacket (sleeve with helical channel)	✓ (CNC lathe threading cycle)	✓		§3.9. The two-start helix is a coarse thread cut with a grooving insert
Bearings	×	×	✓	precision-ground races and rollers
Planetary gears	△ (prototype external gears on a 4th axis)	✓ hob/shape, wire-EDM rings, carburise, grind		gear shop to drawing (docs/09 §5); printed gears for fit checks only
Cycloid discs, ring, pins (C25, C20)	△ only if you own a wire EDM	✓ wire-EDM discs, nitriding	pins ✓ (62 HRC dowels)	§4.2. **Baseline in Mk II**
Strain-wave flexspline	×	×	✓	thin-walled alloy-steel cup that flexes 2× per input turn; its fatigue life depends on steel, heat treatment, shot peening and tooth form. Only KH-14 still uses one (bought set)
Housings, carriers, flanges	✓ (mill/lathe)	✓		7075; Type III hard anodise outsourced

Part	Make at home	Make with a job shop	Buy only	Why / closest buildable alternative
Encoders	✓ magnetic (MT6835 on your PCB)		✓ AksIM-2 rings	the absolute ring/scale is not practical to make
Motor drives	△ (EV-class, ≤ 60 A)	✓ PCB assembly		§7. The **150 A GaN HC2** uses chip-scale GaN parts that need reflow and X-ray inspection: job-shop assembly only, never hand-soldered
MOSFETs, GaN, gate drivers, MCUs			✓	semiconductors
Cells	×	×	✓	pack assembly is yours (P06)
Flat BLDC motor for KW-L2 (maxon ECX FLAT)	×	×	✓	thin multi-pole outer-rotor windings with Halls; buy
Hand micro-motors (10 mm)	×	×	✓	coreless skew windings need dedicated machines; buy maxon/Faulhaber
Ball screws (Ø4–Ø8)	×	×	✓	ground thread + recirculating balls
Ti push-rods, crosses	✓ (care with Ti)	✓		

✓ feasible · △ possible with significant tooling/skill · × not realistically possible

2. The scratch-built motor family (Mk II)

All Mk II motors use **N50UH** magnets (R10 §3.2.4: Br +8.6 % over N42SH, Hcj ≥ 25 kOe for 4× rated current). Data from generated/actuator_family.md and `actuators.py` (R10 §8); resistances at 120 °C.

Motor	Use	Status	Die (OD, slots/poles)	Stack	Turns × paths	Kt (N·m/A rms)	R_LL 120 °C	Km	Cont / peak (N·m)	Mass
KM-130x50-LC	KA-130K knee, KA-130H hip pitch (4 + spare)	baseline	Ø130, 36/42 (K1 die)	50 mm	3 × 2	0.2242	20.26 mΩ	1.286	18.85 [VT, jacket] / 31.5 @150 A	2.615 kg
KM-115x32-N50	KH-25 shoulders (4)	baseline	Ø115, 36/40	32 mm	3 × 1	0.2522	70.2 mΩ	0.777	5.49 / 14.2 @60 A	1.355 kg
KM-85x26-N50-4t	KH-20 shoulder yaw, elbow (4)	baseline	Ø85, 36/42	26 mm	4 × 1	0.1856	146.1 mΩ	0.396	2.01 / 6.18	0.642 kg
KM-115x32-N50	KA-110P (replaces RI115-PH)	optional	as above							
KM-85x26-N50-3t	KA-90P (replaces RI85-PH)	optional	Ø85, 36/42	26 mm	3 × 1	0.1392	82.2 mΩ	0.396	2.01 / 6.18	0.642 kg
KM-85x17-N50-5t	KH-20 lighter variant	not used in Mk II	Ø85, 36/42	17 mm	5 × 1	0.1517	171.3 mΩ	0.299	1.31 / 4.04	0.434 kg
KM-50x20	KH-14 wrist yaw, neck (4)	baseline (unchanged)	Ø50, 12/14	20 mm	18 × 1	0.106	864.8 mΩ	0.093	0.28 / 0.9	0.23 kg

Why the KM-85x17 is not used: at the elbow its 60 s hold is 1.5 × 1.31 × 0.85 × 30 × 0.85 = 42.6 N·m against the 59 N·m one-handed-carry hold [E]. KM-85x12 (the K1 KH-25 scratch motor) is retired; the Mk II shoulder needs the Ø115.

The two dies. The Ø85 motors share one lamination die and one magnet arc (42 × 7.29° × 2.5 mm); only stack and turns change. The Ø130 die is the K1 die. KM-115x32 needs a

new Ø115 die (R10 §5.4).

Honest limit of the model (R10 §1.2 item 6). At the model's 7 A/mm² rule, the scratch Ø85 motors match the commercial kits on Km but not on the datasheet continuous torque (KM-85x26 1.96 vs ILM85x26 2.9 N·m). The sizing used the model's (lower) values, so measured continuous torque may come out higher. Never substitute a scratch motor for a datasheet motor (RI115-PH, RI85-PH) without re-running `calc/run_all.py` with the measured Kt, R and thermal rating.

3. Making a motor, step by step (KM-130x50-LC shown; the others are identical in method)

3.1 Verify the design in FEMM first (free, 2-D magnetostatic)

1. Draw one **symmetry section**. For 36s/42p the pattern repeats every 12 slots / 14 poles, so model 1/3 of the motor with anti-periodic/periodic boundaries.
2. Materials: NO20 BH curve, **N50UH at 80 °C (Br ≈ 1.29 T, μr 1.05)**, copper, air.
3. Check, at the continuous current (84 A rms → peak in phase A, half in B and C), stepping the rotor 0.5° over one pole pair: * open-circuit airgap fundamental Bg1 ≈ 1.31 T (R10 §5.1); * tooth and yoke peak flux ≤ 1.8 T at continuous current; torque linear to ≈ 130 A, then bending (T_sat ≈ 29.8 N·m, 31.5 N·m at 150 A); * Kt within ±10 % of 0.224 N·m/A rms; * **demagnetisation:** at 204 A and 120 °C magnet temperature, no magnet region below the N50UH knee (R10 §7.1); * cogging peak-to-peak < 1 % of the continuous torque (0.19 N·m).
4. **Teeth.** N50UH raises Bg from 0.97 to 1.06 T, so the teeth widen to stay at 1.6 T: ≈ 5.4 → 5.9 mm (scaled by Bg) [E]; slot area falls 59.2 → 52.2 mm² (R10 §3.2.4).
5. **Yoke.** The calculated yoke (2.1 mm) is magnetically sufficient but fragile. **Use 3.0 mm** (stator OD unchanged, slot depth −0.9 mm, about −6 % slot area). Confirm the flux densities and Km again.

3.2 Laminations

1. Export the lamination DXF from the FEMM geometry: 36 teeth, 1.2 mm deep shoes, 2.0 mm slot openings, 3 mm yoke, OD 130.00, 2 key notches on the OD, 2 bond notches.
2. Order a **prototype stack** from a lamination service: 0.20 mm NO20 with C-5 coating, **stack height 50.0 mm** (≈ 238–250 laminations depending on the stacking factor; specify the height, not the count), burr ≤ 0.02 mm, stacking factor ≥ 0.95.
3. If you receive loose laminations, bond them yourself: clean; stack on a hardened mandrel with key bars; press 1 MPa and wick Loctite 290 (or use self-bonding Backlack laminations); cure; check squareness ≤ 0.05 mm and that a slot gauge passes every slot.
4. Deburr the slot edges with a fibre brush. Check interlaminar insulation (> 1 kΩ across the stack, megger at 50 V).

3.3 Slot insulation

1. Cut **Nomex 410, 0.13 mm** liners to the slot perimeter plus 2 mm; crease and insert one per slot, folded over at the tooth tips.
2. Fit end-lamination insulators (printed PA12-CF or Nomex) on both faces.

3.4 Winding (the copper coil)

36 slots / 42 poles, coil k around tooth k ('+' clockwise seen from the lead end; table from generated/km_scratch_designs.md):

```
T01:+A T02:+B T03:-B T04:-C T05:+C T06:+A T07:-A T08:-B T09:+B T10:+C T11:-C T12:-A   (repeats ×3)
```

- **Conductor (KM-130x50-LC):** 3 turns × 2 parallel paths of **3.65 mm²** per turn: **10 strands of Ø0.67 mm grade-2 enamelled copper, Class H (200 °C)** in hand (10 × 0.353 = 3.53 mm²). Check: 2 half-coils × 3 turns × 3.65 mm² = 21.9 mm² copper in a 52.2 mm² slot = fill 0.42 [E], the model's value. Ten strands are stiff: wind with a tensioner and dress each turn with the nylon wedge.
- **Method (needle winding by hand):** mount the stator on a rotary indexing jig (10° detents); anchor the bundle at the lead end; wind 3 turns per tooth in the table direction, flat and tight against the liner; keep each coil in its half of the slot (double layer); lace with Kevlar at both faces; label every lead (A1s, A1f…).
- **Connection:** 12 coils per phase in **2 parallel paths of 6 coils in series** (alternating teeth of that phase in the sequence). A "−" coil connects finish-to-finish with its neighbour. Star point crimped and soldered, Class H sleeve.

- **Leads:** 3 phase leads, **8 AWG (or 2 × 12 AWG) 200 °C silicone** for 150 A peak / 84 A continuous; 2 NTCs (EPCOS B57861S) against the end turns at the hot spot near the lead exit.
- **Check before varnishing:** phase resistances equal within ±2 % (model **14.5 mΩ line-to-line at 20 °C** = 20.26 / (1 + 0.00393 × 100) [E]; use a 4-wire meter, leads matter at this level); insulation > 100 MΩ at 500 V; phase inductances equal within ±3 %.

3.5 Impregnation

Pre-heat to 80 °C, vacuum-dip in **Class H polyester-imide or epoxy varnish**, drain, bake per the varnish schedule (typically 2 h at 150 °C). Re-test insulation.

3.6 Rotor

1. **Back-iron ring:** 1018 steel, OD 89.6 mm (magnets add 2 × 3.0 for a 95.6 mm rotor OD), radial 3.6 mm, ground concentric ≤ 0.01 mm, pressed or bonded on the 7075 hub with its two bearing seats. **Ring length 50 mm.**
2. **Magnets:** 42 arcs **N50UH**, 3.0 mm thick, 7.29° at Ø95.6, **50 mm long, or 4 × 12.5 mm segmented axially** (lower eddy loss, easier to source); NiCuNi, radially magnetised, 10 % spares. Gaussmeter every magnet, mark the N face.
3. **Bonding fixture:** non-magnetic indexing ring with 42 pockets, alternating polarity; high-temperature structural epoxy (Loctite 9492 / 3M 2216 class); one magnet at a time, slid in from the side with a non-magnetic pusher. **Never put fingers between a magnet and steel.**
4. **Sleeve:** wet-wind carbon tow 0.3 mm over the magnets, cure, grind to the final OD (airgap 0.60 mm).
5. **Balance:** static on knife edges, then dynamic at 1,000 rpm, to G2.5.

3.7 Assembly (rotor into stator): magnet safety

The radial pull of a 42-pole rotor entering a stator is several hundred newtons and acts suddenly; the N50UH, 50 mm rotor pulls harder than K1's. 1. Mount the jacketed stator in its housing vertically on the press table. 2. Fit a **Ø20 mm hardened guide rod** through the housing bearing seat and the rotor hub bore; a printed PTFE sleeve on the rotor keeps it concentric. 3. Lower the rotor with the press quill (never by hand) until the bearings seat. 4. Remove the sleeve, fit the second bearing and the wave spring, rotate by hand: smooth, faint cogging only.

3.8 Bearings and encoder

Rotor bearings (68xx thin-section) j5/H6, wave spring 30–60 N. Motor encoder: **AksIM-2 ring** on the hub (the hollow rotor rules out a shaft-end MT6835).

3.9 Liquid jacket (baseline on KM-130x50-LC)

Placement in the thigh and hip: docs/03 §3.1–§3.2; the loop it connects to: docs/03 §9.

Item	Specification
Sleeve	6061-T6, ID 130 mm minus 0.06–0.08 mm interference, 2 mm web under the channel, length = stack + end turns (≈ 66 mm)
Channel	**two-start helix, each start 6 mm wide × 3 mm deep, lead 18 mm** (a channel every 9 mm); each start runs ≈ 3 turns (≈ 1.3 m) over the active length; plenum grooves at both ends join the starts to the ports
Seal	2 EPDM O-rings at each end on the sleeve OD (3.53 mm section, AS568 -2xx size for the ≈ Ø140 bore); leak-check port between each pair
Housing	7075 bore H7 over the sleeve OD; housing wall 2.5 mm at the jacket → OD ≈ 145 mm [E]
Ports	1/8 BSPP (or M8 × 1) with barbed fittings, **inlet at the bottom, outlet at the top** when standing (air purges upward)
Potting	after the shrink fit, pot the end turns into the sleeve with thermally conductive epoxy (≥ 1.5 W/m·K) so end-turn heat also reaches the coolant
Mass	≈ +0.15 kg net per actuator over the finned K1 housing (R10 §4 rule 6)

Making it: 1. Turn the sleeve blank; bore to the shrink diameter. 2. Cut the channel on a CNC lathe as a **2-start "thread" with an 18 mm lead using a 6 mm grooving insert**, in several depth passes to 3.0 mm. Cut the end plenums and O-ring grooves. Deburr; anodise the wetted faces. 3. Heat the sleeve to 150 °C (Al expansion 23.6×10^{-6} × 130 mm × 125 K = 0.38 mm, ample clearance) and drop it over the impregnated stator on a locating mandrel. Do not exceed the varnish's temperature class. 4. Pot the end turns. Fit O-rings (EPDM only, lubricated with a glycol-compatible silicone grease) and press into the

housing. 5. **Leak test:** 2 bar air, under water, 10 min, no bubbles; then 2 bar pressure decay ≤ 5 mbar in 10 min (R10 §5.1).

First-article thermal test (gate before building four). Build the first jacket on a spare stack with a dummy winding (or on a K1 KM-130x45 stator, R10 §7.2). Feed 0.5 L/min of 40 % glycol at 25 °C, inject a known copper loss (DC through two phases in series), log winding (NTC + resistance rise) and coolant in/out to steady state. R_th = (T_winding − T_coolant,mean) / P. **Pass: ≤ 0.30 K/W.** Above that, F_cool falls below 2.0 and the continuous rating scales with √F_cool (R10 §7.2); re-run `calc/run_all.py` with the measured value.

Fallback (R10 §5.1 variants): air-only (no jacket) gives 0.64 × the jacketed continuous torque; the leg margins still close at ≈ 1.13 (knee) (docs/03 §9.6).

3.10 Motor acceptance tests (before building it into an actuator)

Test	Equipment	Pass (KM-130x50-LC model)
Phase resistance (4-wire) at 20 °C	milliohm meter	**14.5 mΩ L-L ±10 %**, phases within ±2 %
Inductance	LCR meter, 1 kHz	phases within ±3 %; record Ld/Lq by rotor position
Hi-pot, winding → stack	1,000 V AC (or 1,500 V DC) 60 s	no breakdown, leakage < 1 mA
Insulation resistance	500 V DC	> 100 MΩ
Back-EMF constant	spin at 500 rpm with a second motor, scope L-L	**Ke L-L peak 0.183 V·s/rad ±7 %** (≈ 9.6 V peak at 500 rpm; 0.818 × Kt as in the generated sheets) [E], THD < 5 %
Torque constant	locked rotor on a lever + load cell, 10–80 A, then 2 s pulses to 150 A on the HC2	**Kt 0.224 N·m/A rms ±8 %**; linear to ≈ 130 A; **31.5 N·m ±10 % at 150 A** [E]
Cogging	rotate slowly with a torque arm	< 0.19 N·m peak-to-peak
Jacket leak	2 bar air	§3.9
Thermal (in the jacket, 0.5 L/min, 25 °C coolant)	84 A rms (18.85 N·m equivalent), steady state	winding ≤ 25 + 0.30 × 215 ≈ 90 °C → R_th ≤ 0.30 K/W [VT]

If Kt, Km or R_th is more than 10 % off, update `calc/kestrel/actuators.py` and re-run `calc/run_all.py` before building more.

3.11 Mk II variants: KM-115x32-N50 and the KM-85 family

Same method as §3.1–§3.10; the differences are below. Conductor sizes are [E]: slot area scaled by the Ø130 N50UH reduction (52.2 / 59.2) at fill 0.42. Each one reproduces the model resistance within ≈ 1 % (e.g. KM-115x32-N50: K1 4-turn 109.4 mΩ × 36/48 × 2.49/2.90 = 70.5 mΩ vs 70.2 mΩ).

Item	KM-115x32-N50	KM-85x26-N50-4t	KM-85x26-N50-3t	KM-85x17-N50-5t
Die	**new Ø115**, 36 slots / **40 poles** (kw1 0.945), airgap Ø84.5, gap 0.60	Ø85 K1 die, 36/42 (kw1 0.933), airgap Ø59.5	same	same
Teeth	≈ 4.7 → 5.1 mm for N50UH [E]; 3.0 mm mechanical yoke check	per FEMM		
Stack	32 mm	26 mm	26 mm	17 mm
Magnets	40 arcs N50UH, 2.8 mm, 7.65° (0.85 of the 9° pole pitch) at rotor OD 83.9, 32 mm long	42 arcs N50UH, 2.5 mm, 7.29°, 26 mm	same	17 mm
Turns × paths (N_ph)	3 × 1 (36)	4 × 1 (48)	3 × 1 (36)	5 × 1 (60)
Conductor per turn	≈ 2.9 mm² ≈ **8 × Ø0.67**	≈ 1.43 mm² ≈ **4 × Ø0.67**	≈ 1.9 mm² ≈ 7 × Ø0.59	≈ 1.15 mm² ≈ 3 × Ø0.70
R_LL 20 °C (from 120 °C ÷ 1.393)	50.4 mΩ	104.9 mΩ	59.0 mΩ	123.0 mΩ
Ke L-L peak (0.818 × Kt)	0.206 V·s/rad	0.152	0.114	0.124
No-load @ 52 V	189 rad/s (1,805 rpm)	257 rad/s (2,458 rpm)	343 rad/s	315 rad/s
Thermal acceptance (in its housing, derate 0.85)	4.67 N·m ≈ 18.5 A rms → winding ≤ 120 °C	1.71 N·m ≈ 9.2 A rms	1.71 N·m ≈ 12.3 A rms	1.11 N·m ≈ 7.3 A rms

Item	KM-115x32-N50	KM-85x26-N50-4t	KM-85x26-N50-3t	KM-85x17-N50-5t
Rotor ID (gear / cable bore)	74.6 mm	51.5 mm	51.5 mm	51.5 mm

36 slots / 40 poles coil table (KM-115x32-N50) [E: balanced double-layer allocation from the slot-EMF star, kd 0.960 × kp 0.985 = kw 0.945, matching the generated KM-115x32 sheet]. It repeats every 9 teeth, 4 times:

```
T01:+A T02:-A T03:-B T04:+B T05:-B T06:-C T07:+C T08:-C T09:-A   (T10–T18, T19–T27, T28–T36 repeat)
```

Each phase forms groups of three adjacent teeth with the middle coil reversed (e.g. T36 –A, T01 +A, T02 –A). Same sign convention as §3.4. **Check phase sequence and back-EMF balance (three phases within ±2 %) on the first stator before winding the rest.**

4. Making the transmissions

4.1 Planetary (KA family)

- **Recommended:** a gear shop machines to the docs/09 §5 table (hob/shape, carburise, grind, crown, tip relief). Mk II tooth plans are in docs/04 §5.2 and §7: KA-130K 17:1 (z16/z18/z52 × 4 planets, then z18/z18/z54 m1.25 × 18 mm), KA-130H 14:1 (z24/z18/z60 × 3, same stage 2), KA-110P 29:1 (6.0 × 4.83), KA-90P 28:1 (5.6 × 5.0).
- **Prototype path at home:** print the gears to check meshing and assembly inside the rotor; CNC-hob external gears in 17-4PH on a 4th axis for low-load tests; wire-EDM the rings from pre-hardened 42CrMo4 (job shop). Heat treatment is always outsourced.
- Inspection: over-pin measurement for tooth thickness; backlash 0.03–0.06 mm at m1.25.

4.2 Machined two-disc cycloid C25 / C20 (KH-25, KH-20) — baseline in Mk II

Item	C25 (shoulder pitch, roll)	C20 (shoulder yaw, elbow)
Ratio	30:1 (31 ring pins, 30-lobe discs)	30:1
Ring-pin circle	Ø84 mm (R 42)	Ø68 mm (R 34)
Ring pins	Ø5 needle rollers (ISO 3096 G2) or Ø5 m6 dowels (60 HRC) in a 42CrMo4 ring	Ø4
Eccentricity E	0.9 mm (short-width coefficient K1 = E·31/R = 0.66; min epitrochoid curvature radius 4.9 mm vs pin radius 2.5)	0.8 mm (K1 0.73; 3.7 vs 2.0 mm)
Discs	2 at 180°, **8 mm** each, stack-cut as a matched pair	2 at 180°, **6 mm** each in hardened 100Cr6 (**7 mm** if the nitrided 42CrMo4 route is used; see the screening below)
Disc material and process	**Route A (preferred, docs/09 §5.3): 100Cr6 hardened to 60–62 HRC, then wire EDM** (rough + 3 skims, bore and output holes in the same setup), lap the 2–5 µm recast layer. Route B: 42CrMo4 Q&T, wire EDM, gas nitride, and a pressed 100Cr6 sleeve in the bore as the needle raceway (a nitrided case is too soft for needles)	same
Profile	standard cycloid equations (below) with a 0.02 mm equidistant clearance	same
Output	**8 solid pins Ø8** (≥ 58 HRC) on PCD 52, supported at both ends (output disc + output ring); disc holes = 8 + 2E = **Ø9.80**	**6 solid pins Ø6** on PCD 40; holes Ø7.60
Input	two 100Cr6 eccentric sleeves (Ø20 h5, 60–62 HRC) keyed 180° apart on a hollow input shaft (Ø12 bore); one needle cage Fw 20 / Ew 26 per disc running directly on the cam and the hardened disc bore (docs/09 §5.3)	sleeves Ø16, cages Fw 16 / Ew 22, shaft bore Ø10
Ratings rated / repeated / avg / momentary	80 / 160 / 100 / 250 N·m **[VT]** (`actuators.py` `CYC`, R10 §5.6)	52 / 105 / 65 / 160 N·m **[VT]**
Mass (gear)	0.57 kg	0.43 kg
Expected backlash / efficiency	1–7 arcmin; 0.85 assumed (UCLA machined cycloid: 7.0 arcmin, 82.3 % [V, R10 S9])	same

Profile equations (`cad/cycloid_profile.py`, outputs `cad/out/cycloid_KH-C25.dxf` and `cycloid_KH-C20.dxf`), with pin-circle radius R, pin radius Rr, eccentricity E and N ring pins:

```
psi(t) = atan2( sin((1−N)t),  R/(E·N) − cos((1−N)t) )
x(t)   =  R·cos t − Rr·cos(t + psi) − E·cos(N·t)
y(t)   = −R·sin t + Rr·sin(t + psi) + E·sin(N·t)
```

The script checks E·N < R and Rr < 0.8 × the minimum curvature radius of the pin-centre epitrochoid (no undercut), and removes 0.02 mm for running clearance. Verify meshing in CAD before cutting.

Load screening [E] (assumptions stated; a profile-resolved contact analysis replaces this before the discs are cut):

Check	Formula / assumption	C25 at 160 / 250 N·m	C20 at 105 / 160 N·m
Torque per disc	T_d = 0.55 × T_out (2 discs, 1.1 load share)	88 / 138 N·m	58 / 88 N·m
Max ring-pin force	F_max ≈ 4·T_d / (K1·R·31)	407 / 636 N	301 / 458 N
Worst-case Hertz (max force at the minimum lobe curvature, ρ_lobe = ρ_epi,min − Rr)	p = √(F·E/(π·L·ρ_eff)), *E* 115 GPa; C25 ρ_eff = 1/(1/2.5 + 1/2.4) = 1.22 mm, L 8; C20 ρ_eff 0.92 mm	**1,240 / 1,550 MPa**	6 mm: 1,415 / 1,750 MPa; **7 mm: 1,310 / 1,620 MPa**
Output-pin bending (solid pin, simply supported, two disc loads)	F_op ≈ 4·T_d/(n·R_op), M ≈ F·span/4 (spans 19 / 15 mm)	≈ 160 / 250 MPa (Ø8, R 26)	≈ 340 / 520 MPa (Ø6, R 20)
Eccentric bearing load per disc	≥ T_d / (E·30)	≥ 3.3 kN	≥ 2.4 kN

Reading, with typical repeated-contact allowables (confirm for the chosen steel): ≈ 1,500 MPa for through-hardened 100Cr6 (route A), ≈ 1,300 MPa for nitrided 42CrMo4 (route B). The C25 passes on either route even with the worst-case geometry combination. The C20 at 6 mm passes on route A only (1,415 MPa); on route B use **7 mm discs** (+ ≈ 0.02 kg) or show ρ_eff ≥ 1.1 mm at the most-loaded pin with a profile-resolved check. Momentary values are static loads that hardened contacts carry. The pins are far below the yield of hardened dowels. The eccentric needle cages (C0 ≥ 8 kN on the C25) cover the ≥ 3.3 kN bearing load.

Assembly: ring pins in, discs on the eccentric needle cages 180° apart, solid output pins through both discs, output ring, crossed-roller output (THK RB5013 on C25, RB4010 on C20, docs/09 §5.3). Grease: RV-reducer grease (docs/09 §5.3) or lithium-complex EP, NLGI 1–2. Set the drive's peak-current limit in non-volatile parameters so the gear never sees more than its repeated rating (docs/09 §5.3: 21 A rms on KH-25, 18.8 A rms on KH-20, then trim on the dyno). Run in 2 h at 30 % load, both directions, then change the grease.

Tests (R10 §7.6), on the first unit of each size: backlash (input locked, ±2 N·m at the output, dial at 100 mm); torsional stiffness (torque–twist to the repeated rating, output encoder); efficiency vs speed at 25 / 50 / 100 % of rated (replaces η 0.85); **static overload to the momentary rating** (250 / 160 N·m) with no measurable permanent backlash change; 10^5-cycle endurance at the repeated rating before ordering the other units.

4.3 Strain-wave (KH-14 only)

Bought Leaderdrive CSG-14-50 sets. Not recommended to make (§1).

4.4 KW-L2 wrist linear actuator (4 units)

What you make: the housing, push tube, end fittings and the driver board's enclosure. What you buy: the motor, the screw, bearings and the driver electronics (docs/04 §9 for the requirement and the motor choice).

Item	Specification
Motor	maxon **ECX FLAT 32 L** high-torque, 24 V winding (Option A), screw coupled directly: 98.5 mN·m nominal, 540 mN·m stall, 10,600 rpm no-load, 71 g [V, maxon product page cited in generated/bom.md B024]
Screw	Ø6 × 2 mm ground ball screw with flange nut **if** C_a ≥ 2 kN and C_0a ≥ 3 kN; else **Ø8 × 2** (same force per mN·m: 2.765 N) [E]
Fixed end	angular-contact pair, back-to-back, preloaded, in a 7075 bearing block bolted to the motor flange
Coupling	rigid clamp hub from the motor shaft to the screw journal (no flexible coupling: it adds compliance and length)
Anti-rotation	nut carried in a push tube with a flat running in a POM-lined slot (or a miniature rail where space allows)
Push tube / rod end	Ti tube, M5 PTFE-lined rod end to the wrist lever (r_p 22 mm, d 16 mm)

Item	Specification
Stroke	±30 mm working (22 sin 70° + 16 sin 35°) + 5 mm each end to elastomer stops [E]
Driver	per forearm, 2-axis: F28P65x + DRV8353-class gate driver + 6 MOSFETs per axis, **24 V rail**, Hall + encoder inputs, current sensing for force estimation
Sealing	forearm is IP65 zone: O-ring housing joints, PTFE wiper at the rod exit, ePTFE vent

Build steps: machine the housing and bearing block → press and preload the bearing pair on the screw journal → fit the clamp hub to the motor shaft → check screw run-out ≤ 0.02 mm at the nut → fit the nut and push tube, set the anti-rotation flat → fit the rod end at mid-stroke → wire to the driver → home against the soft stops.

Tests (R10 §7.7): force–speed curve at 24 V: **≥ 754 N for 2 s** (generated combined-corner requirement, short-time stall), **≥ 272 N continuous for 10 min** with the winding ≤ 100 °C, **≥ 318 N for 60 s** (AO hold), **≥ 132 mm/s at 216 N**; backlash < 0.02 mm; 10^5 cycles at ±25 mm and 300 N. If the 32 L cannot make 754 N short-time, use Option B (ECX FLAT 22 L + 3.9:1) and accept a ≈ 0.94 speed margin, or lengthen the levers with a longer stroke and re-run the wrist requirement.

5. Housings and structure

Everything in docs/09 §3 is CNC work you can do with a capable mill and lathe (7075, Ti with care). New in Mk II: the jacket sleeve (§3.9), the KA-110P output hub for 12 × M5 on PCD 100 (docs/03 §5), the cycloid carrier rings and the V-band limb flanges (docs/03 §13). Type III hard anodise is outsourced; Type II is possible at home but softer.

6. Bearings, magnets, cells, semiconductors, bought motors

Buy them (§1). Authorised distributors only for cells (counterfeits are common, R3), magnets (request N50UH grade certificates with Br/Hcj at temperature) and maxon motors.

7. Building the joint drives yourself (cost-down option)

- **EV-class drives (≤ 60 A, 22 joints), TI TIDA-010979 pattern [R2]:** AM2612 or F28P650DK MCU (on-chip EtherCAT slave); 3 × DRV7167 100 V GaN half-bridges (60 A rms) or DRV8353RS + 6 × ISC022N10NM6; isolated current sensing; 2 × BiSS-C; EtherCAT PHYs; Ø70 mm annular board with a 15 mm bore.
- **HC2 (150 A rms, knee and hip pitch):** buy an EPC9186HC-class power stage (150 A rms, ≤ 76 V, EPC2361 100 V GaN [R, R10 S26]) and add your controller, or lay out a 6-layer board with paralleled GaN (R10 §5.1). Chip-scale GaN: reflow and X-ray at a PCB assembler. Heat path: board bonded to the rear cap, which is thermally tied to the jacket. Test 150 A for 2 s and 99 A for 10 s into a locked rotor (R10 §7.3). **Firmware is the long pole.**
- **Firmware:** FOC (SVPWM 20–40 kHz; ≥ 40 kHz on HC2 for the 712 Hz electrical frequency), current/velocity/position loops, field weakening, CiA-402 over EtherCAT, encoder calibration, thermal model with magnet-temperature Kt correction, **phase-short on fault** (R10 §3.4), fault handling.
- **STO:** two independent hardware channels removing gate-driver supply/enable; cannot be overridden by firmware. Test the response time.
- **Effort:** 6–12 months for a robust EV-class drive; longer for HC2. Buy modules first; design your own as a later cost-down.

8. Cost and time comparison (per actuator class, rough)

Item	Buy route	Scratch route	Notes
Ø85 frameless motor	A$450–1,300 (CubeMars/TQ)	A$180–350 (laminations, magnets, wire) + ≈ 15 h	K1 figures; N50UH arcs cost more than N42SH (re-cost in docs/12)
Ø115 frameless motor	RI115-PH: see BOM	new die + parts + ≈ 18 h	not re-costed for Mk II
KM-130 motor	n/a	K1 KM-130x45: A$1,050–1,100 + ≈ 25 h	the 50 mm N50UH version costs more in magnets and laminations; re-cost in docs/12
Liquid jacket	n/a	6061 bar + O-rings + ≈ 6 h lathe work	plus ≈ 3 h leak and thermal test for the first article
Strain-wave set (size 20/25)	A$450–640 (Leaderdrive)	cycloid A$250–400 (EDM + pins) + ≈ 10 h	K1 figures; Mk II adds nitriding and a second skim pass

Item	Buy route	Scratch route	Notes
Drive (EV class)	A$945–1,600	A$150–250 parts + months of development	after the robot works
Encoder (motor side)	A$456 (AksIM-2)	A$18 (MT6835)	

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